

Three Nulls, One Verdict: Calibrating the `eforensics` Election-Fraud Model Against Two Decades of Korean Elections

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Abstract

Bayesian finite-mixture election-forensics models in the tradition of Mebane [2016] classify vote-counting units as clean or fraudulent from aggregate turnout and vote-share data alone, without independent evidence of manipulation. We evaluate the `eforensics` quasi-binomial-logistic (`qbl`) model against 20 years of Korean elections (2002–2025), fitting 576 separate model runs spanning general and presidential elections, two aggregation levels, up to five vote-counting channels, both directions of leader coding, and, critically, two independently constructed *fraud-free synthetic controls* calibrated to match each real dataset’s turnout and vote-share marginals without any fraud-generating mechanism. We find that the model reports substantial “fraud” (5–29% of units, into the millions of “fraudulent” votes) on data that is fraud-free by construction; that real-data fraud-share estimates correlate at $r \approx 0.96$ with their fraud-free null counterparts across 58 paired cells under each leader coding; that the model’s largest real-data estimates fall on presidential elections held before Korea had early voting or any fraud allegations (2002–2012), exceeding what it reports for the actually contested 2020 election; and that recoding the model to test fraud in the opposite (conservative-favoring) direction reproduces the identical pathology across the complete 11-election grid, with the Democratic-coded estimate exceeding the conservative-coded estimate in most, but not all, cells, an asymmetry itself matched by fraud-free nulls rather than indicating real fraud in either direction. A direct comparison of short against long MCMC chains on a matched subsample shows convergence failure and fraud-share estimates are statistically indistinguishable between settings, indicating the non-convergence pervasive in this corpus is not an artifact of insufficient sampling. We conclude that the model’s output depends on the shape of the (turnout, vote-share) distribution, not on fraud, and that this failure is not fixable by post-hoc recalibration, since recalibrating for validity would remove essentially all detected signal in the corpus.

Keywords: election forensics; Bayesian finite mixture models; null calibration; `eforensics`; early voting; South Korea; election fraud claims

1 Introduction

Bayesian election-fraud forensics models classify aggregation units (precincts, wards, administrative neighborhoods) into latent classes representing clean voting and one or more fraud mechanisms, using only reported vote totals and turnout. One of the most widely applied models in this tradition is Walter Mebane’s `eforensics` framework [Mebane, 2016, Ferrari et al., 2019], a finite mixture of overdispersed binomial models in which a subset of units are permitted to exhibit jointly elevated turnout and leader vote share, interpreted as evidence of vote manufacturing (converting abstentions into votes) and vote stealing (transferring votes from an opponent).

Korea offers an unusually clean natural experiment for auditing such a model. Nationwide in-person early voting (*sajeon tupyoo*, “pre-vote”) was introduced by statute in 2013 and first used in the 2014 local

elections; the first general election held under the system was the 20th (2016) and the first presidential election was the 19th (2017). Before 2014, no election-fraud claim of the count-manipulation type this literature targets was raised about a Korean national election.¹ Organized fraud allegations first emerged after the 21st general election (April 2020), concentrated almost entirely on early-voting ballots, and Mebane himself applied `eforensics` to that election, reporting anomalies he characterized as strongly suggestive of manipulation, with roughly 28.7% of pre-vote (early-voting) units flagged as fraudulent [Mebane, 2020]. That analysis has since become a central citation for the Korean fraud-claim movement, which resurfaced around the 2024 general election and was cited in connection with the December 2024 martial-law declaration, followed by a snap presidential election in June 2025. A peer-reviewed Korean-language study, Min and Yoo [2021], independently shows that Mebane’s original data construction was invalid from the outset rather than merely a defensible alternative specification (Section 6).

This creates a sharp, falsifiable question. If `eforensics` is a genuine fraud detector, it should report approximately zero fraud on synthetic data constructed to share Korean elections’ turnout and vote-share distributions but containing no fraud-generating mechanism, and it should report approximately zero fraud on the pre-2014 Korean elections for which no mainstream fraud narrative exists. We test both predictions directly, run the identical analysis in both the historically alleged direction and its opposite, and find that the model fails both tests in a way that is fully explained by the shape of the (turnout, vote-share) distribution rather than by any fraud process.

2 The `eforensics` quasi-binomial-logistic model

The `eforensics` package’s `qbl` variant assigns each unit i , with eligible voters N_i , leader votes w_i , and abstentions a_i , to one of three latent classes $Z_i \in \{1, 2, 3\}$ (no fraud, incremental fraud, extreme fraud) with population mixture weights $\pi = (\pi_1, \pi_2, \pi_3)$. Conditional on class, w_i and a_i follow overdispersed binomial likelihoods with per-unit random effects on turnout (τ_i), leader share (ν_i), and, for fraudulent classes, fraud-magnitude parameters governing manufactured votes (converted from abstentions, via ι) and stolen votes (transferred from the opponent, via χ). Population-level location equations $w \sim \text{prov}$ and $a \sim \text{prov}$ allow province-level fixed effects. Estimation is by MCMC via JAGS. Throughout this study, every model fit calls the official `eforensics` R package [Ferrari et al., 2019] directly, installed from Walter Mebane’s own repository at the University of Michigan (`UMeforensics/eforensics_public` on GitHub, https://github.com/UMeforensics/eforensics_public), via its own `eforensics()` function; no part of the fraud model, the likelihood, the priors, or the MCMC sampler was reimplemented for this study. Our own code is limited to data preparation (constructing the unit-level tables in Section 3), the two null-data generators (Section 3.3), and post-processing of the package’s posterior output; the inferential machinery itself is exactly what a user obtains by installing and running the published package unmodified.

Two properties of this specification are essential to what follows. First, fraud is identified purely from the *shape* of the (turnout, vote-share) distribution: fraudulent units are those whose joint elevation of τ_i and ν_i is inconsistent with the fitted clean-class distribution. No information external to aggregate vote counts enters the model. Second, and less widely emphasized, fraud is **unidirectional**: both manufacturing and stealing mechanisms only ever add votes to whichever party or candidate is designated the “leader.” The model has no mixture component for fraud against the leader; votes lost by the leader are absorbed into the clean class regardless of their actual cause. Every published application of `eforensics` to a contested election therefore tests fraud in one direction only, chosen by the analyst.

¹One partial qualification concerns the 2012 presidential election: the 2017 documentary *The Plan* (Korean: *Deo Peullaen*) alleged that electronic ballot-sorting-machine misclassification (the “K=1.5” pattern) manipulated results in favor of the conservative winner, Park Geun-hye. This claim was statistically disputed at the time and never became a mass movement comparable to the post-2020 controversy.

3 Data and experimental design

3.1 Elections and aggregation grid

We assembled official National Election Commission results, obtained from the Korean government’s open-data portal,² for eleven Korean national elections: five general (legislative) elections (18th [2008], 19th [2012], 20th [2016], 21st [2020], 22nd [2024]) and six presidential elections (16th [2002], 17th [2007], 18th [2012], 19th [2017], 20th [2022], 21st [2025]). The first five elections listed (18th/19th general, 16th/17th/18th presidential) predate early voting; all others were held under the early-voting system. Our primary grid (Section 3.2) covers all eleven elections completely, at both leader codings.

Each election was aggregated at two levels: *dong* (Korean administrative neighborhood, *beopjeong eupmyeondong*; 2,700–3,600 units after filtering) and *constituency* (*seongeogu*; 198–253 units). For early-voting-era elections, votes were further split by channel: at *dong* level, in-precinct early votes (*gwannae sajeon tupyoo*, “inside-precinct pre-vote”), same-day votes, and their pool; at constituency level, additionally out-of-precinct early votes (*gwano sajeon tupyoo*, “outside-precinct pre-vote,” observable only at this level) and combined totals. Pre-early-voting elections use a single “total, no early” channel. Units with fewer than 50 votes in any analyzed channel, or with zero leader votes across both early and same-day channels, were dropped; eligible-voter counts were reconstructed from registered-voter rows and floored at total votes cast. Because *eforensics* models turnout τ_i and leader share ν_i on the logit scale, units with either quantity closer than 0.001 to 0 or 1 were additionally excluded, since the logit transform is undefined at the boundary; this drops a small number of units with near-complete or near-zero turnout, or with an almost unanimous or entirely absent leader vote, before any model fitting takes place.

The Democratic-lineage party or candidate was coded as the model’s “leader” in the primary analysis (Section 4), regardless of whether that party won; Section 4.8 reports a parallel analysis with the conservative-lineage side coded as leader, completed for every election in the grid.

3.2 Two MCMC chain-length configurations

Model fits used JAGS with 4 chains throughout. Two configurations appear in this study, differing only in chain length. The *long-chain configuration* used 1,000 burn-in iterations, 1,000 adaptation iterations, and either 3,000 or 5,000 sampling iterations; fitting time under this configuration is substantial (roughly 30–170 minutes per cell at the *dong* level), which makes completing every cell of the full bidirectional grid impractical and leaves the conservative-coded side of that grid partial, concentrated on the highest-value elections (2012 presidential, 2020 general, 2024 general). The *short-chain configuration*, used for the primary analysis reported below, shortens sampling to 1,000 iterations and burn-in to 300 iterations, holding adaptation at 1,000 iterations; this reduces fitting time enough that the complete grid, 58 (election, level, channel) cells $\times$ 3 sources $\times$ 2 leader codings = 348 runs, can be completed without exception. The adaptation parameter (JAGS’s `n.adapt`, governing how many iterations the sampler uses to tune its proposal distribution before burn-in begins) defaults to 1,000 under both configurations and in `mebane_crosscheck.R` generally; it is exposed as a command-line argument to both the R script and the Python scripts (`r_n_adapt / R_N_ADAPT`) and was held fixed at its default throughout this study, so it is not a source of variation between the two configurations.

This design lets us ask directly whether the extensive non-convergence found under the long-chain configuration (Section 4.6) is an artifact of insufficient sampling that more iterations would resolve. Matching the 174 *dong*/constituency cells common to both configurations under Democratic-favoring coding, convergence status and fraud-share estimates are close to unchanged: 89.7% of long-chain runs and 90.8%

²General (legislative) election results: <https://www.data.go.kr/data/15025527/fileData.do>. Presidential election results: <https://www.data.go.kr/data/15025528/fileData.do>.

of short-chain runs fail $\hat{R} \leq 1.05$; median $\max\hat{R}$ is 1.64 (long chains) versus 1.65 (short chains); and fraud-share point estimates correlate at $r = 0.95$ across the two configurations despite a 3–5 $\times$ reduction in sampling iterations and a 3.3 $\times$ reduction in burn-in (Figure 1). Increasing `n.iter` toward 3,000 or `burn.in` toward 1,000 therefore would not be expected to materially change the convergence diagnostics reported throughout this paper; the non-convergence is a property of the posterior geometry the `qbl` model induces on this data, not of an under-sampled chain. All results in Sections 4–4.8 use the short-chain configuration, since it is the only one with complete coverage; where the long-chain configuration is shown separately (Figure 3), it is explicitly labeled.

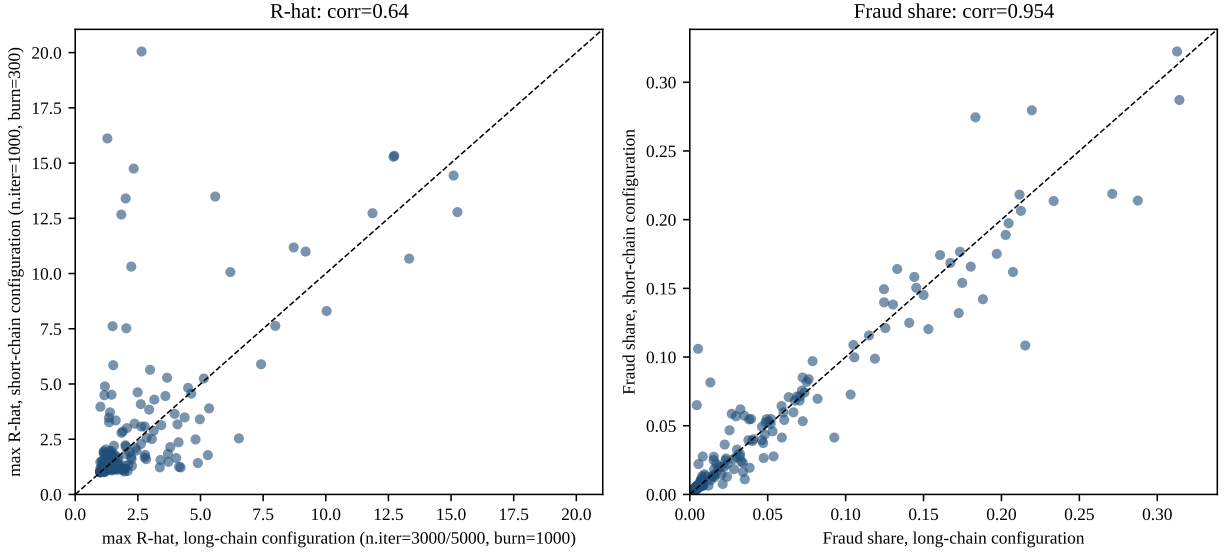


Figure 1: Chain-length robustness check: long-chain configuration (n.iter=3,000/5,000, burn.in=1,000) versus short-chain configuration (n.iter=1,000, burn.in=300) results on the same 174 Democratic-coded cells. Left: maximum $\hat{R}$ on the mixture weights. Right: fraud-share point estimates. Both correlate strongly across a 3–5 $\times$ change in sampling iterations, indicating the shorter chains used for the primary analysis are not the source of the paper’s non-convergence findings.

3.3 Three fraud-free controls: two ad hoc nulls and one model-consistent self-null

The central methodological device of this study is a set of synthetic datasets constructed to match each real dataset’s turnout and vote-share distributions while containing no fraud-generating process whatsoever. Two of these nulls are ad hoc, in the sense that the noise process used to simulate them is chosen by us rather than by the fitted model; the third instead draws directly from `eforensics`’ own fitted no-fraud generative equations, and is reported in Section 4.3.

Marginal null. For each unit i with real turnout τ_i and leader share ν_i , independently draw $\tau_i^* \sim \text{Beta}(k\tau_i, k(1 - \tau_i))$ and $\nu_i^* \sim \text{Beta}(k\nu_i, k(1 - \nu_i))$ with concentration $k = 200$, then reconstruct vote counts from τ_i^*, ν_i^* , and the unit’s true eligible-voter count N_i . This preserves every unit’s marginal turnout and share location, and hence all cross-unit heterogeneity in these two moments, while replacing whatever process actually generated the data with independent, clean Beta noise.

Joint null. Compute province-level means of (logit τ , logit ν) and the pooled 2×2 residual covariance across all units; simulate every unit as its province mean plus homogeneous bivariate Gaussian noise. This retains only province-level location and one global covariance structure; any unit-specific anomaly or fraud mechanism is destroyed by construction.

Model-consistent self-null. Unlike the two nulls above, this third null generator does not choose its own noise process. Instead, for a given cell it first fits `qbl` to the real data and harvests the posterior-median hyperparameters of the fitted model’s own class-1 (“no fraud”) data-generating equations (the province-level location terms and per-unit random-effect variances on turnout and leader share); it then simulates a synthetic dataset directly from those equations with every unit forced into class 1, i.e. population mixture weights fixed at $\pi = (1, 0, 0)$ by construction rather than merely estimated near it. This is fit through the identical `eforensics` workflow used elsewhere in the paper. Because it uses the model’s own fitted clean-class process rather than an externally chosen one, a positive finding here is the most direct possible demonstration that the false-positive problem originates in the model’s clean-class specification itself, rather than in some property of the marginal or joint null generators. Section 4.3 reports this analysis for the complete Democratic-coded grid under the short-chain configuration.

All three nulls are fit through the identical `eforensics` workflow, with identical filters and MCMC settings, as their paired real-data cell. The logic is straightforward: if `eforensics` reports substantial fraud on data that is fraud-free by construction, that is a measured false-positive rate; if it reports fraud on real data indistinguishable from what it reports on the paired null, its real-data output carries no information beyond the shape of the (turnout, share) marginal distribution that the null also reproduces.

Of 174 possible (election, level, channel, source) dem-coded cells, all 174 were fit to completion. A second grid with the conservative-lineage side coded as leader covers 54 cells: complete coverage of the 2012 presidential election, the 2020 general election, and the 2024 general election (Section 4.8).

4 Results

4.1 Massive false-positive fraud on data with no fraud

Across 120 completed null runs (marginal and joint nulls pooled, short-chain configuration), 46 (38.3%) report a fraud share (the posterior median of $\pi_2 + \pi_3$) above 5%, and 15 report above 15%. The largest false positives reach 28.7% of units (constituency-level total votes, 2025 presidential election, marginal null) with an estimated 7.3 million fraudulent votes. Table 1 lists selected examples.

Table 1: Selected fraud-free null runs with large false-positive fraud shares (short-chain configuration).

Cell (fraud-free null)	Fraud share	Fraudulent votes (F_w)	Units flagged
2025 pres., constituency, total, marginal	28.7%	7,340,668	25/242
2022 pres., constituency, total, joint	28.0%	7,349,496	27/248
2025 pres., constituency, total, joint	27.5%	7,598,564	26/242
2022 pres., constituency, total, marginal	21.9%	6,010,504	21/248
2025 pres., dong, pooled, joint	21.8%	3,602,274	379/3553
2017 pres., constituency, total, joint	21.4%	3,908,730	2/250

For calibration: Mebane’s 2020 analysis headlined 28.7% of early-voting units as fraudulent [Mebane, 2020]. Fraud-free synthetic data of the same country’s elections, matched only on turnout and share marginals, routinely produces flagged-unit rates and fraud shares of the same order of magnitude, and in several cells exceeds it.

4.2 Real data is statistically indistinguishable from fraud-free data

Pairing each of the 58 real-data cells with its two null counterparts under the short-chain configuration, real-data fraud share correlates at $r = 0.963$ with the marginal null and $r = 0.936$ with the joint null (Figure 2).

Restricted to the early-voting channels at the center of the Korean controversy, the mean paired difference (real minus null) is under 0.6 percentage points in either direction. Real data exceeds *both* of its nulls in only 12 of 58 cells (21%), roughly what sign noise alone would produce, and in every such case the excess is small relative to cross-null disagreement and occurs under severe non-convergence (Section 4.6).

The identical comparison under the long-chain configuration (Figure 3; 58 dem-coded cells with all three sources completed under that configuration, plus 18 conservative-coded cells) gives closely comparable correlations, $r = 0.984$ (marginal) and $r = 0.946$ (joint), confirming that the real-vs-null indistinguishability documented here does not depend on which chain-length configuration is used.

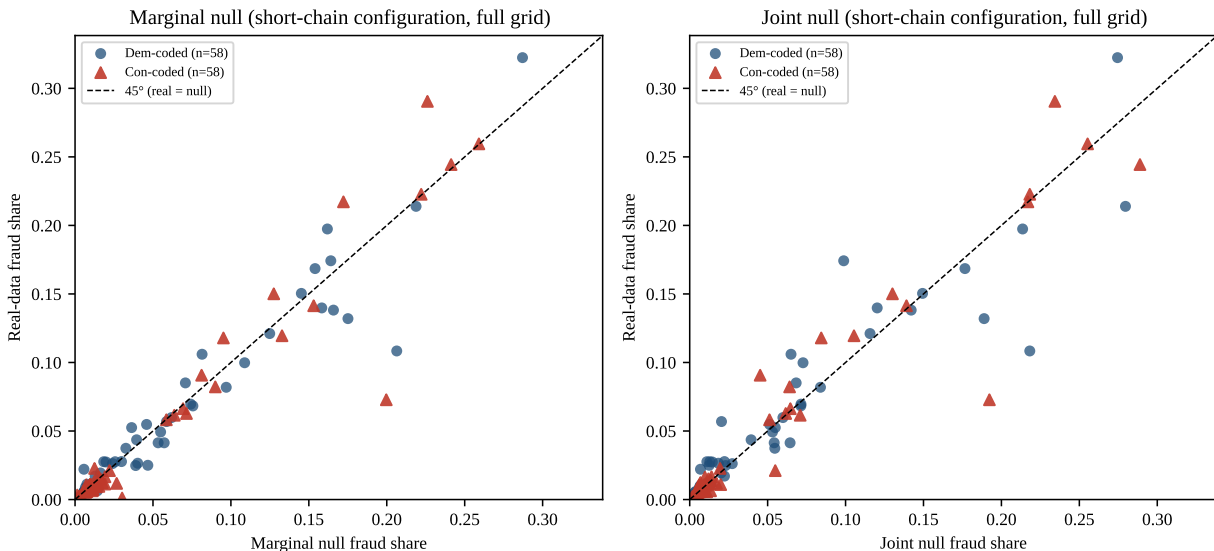


Figure 2: Real-data fraud share vs. fraud-free null fraud share, short-chain configuration ($n.iter=1,000$, $burn.in=300$), across the complete grid: 58 dem-coded cells (blue circles) and 58 conservative-coded cells (red triangles), all with real, marginal-null, and joint-null sources completed. Points falling near the 45° line indicate that the model’s real-data output is statistically indistinguishable from what it reports on data constructed to contain no fraud.

4.3 Confirmation with the model’s own no-fraud process: the self-null

The marginal and joint nulls of Section 3.3 are, by design, generated from noise processes chosen by us rather than by `eforensics` itself; a skeptic could in principle argue that their agreement with real data reflects some property of our chosen Beta or Gaussian noise rather than of the model’s clean-class specification. The model-consistent self-null closes this gap directly, since it does not choose a noise process at all: it uses `qbl`’s own posterior-median class-1 hyperparameters, fit to the real data, with population mixture weights fixed at $\pi = (1, 0, 0)$ by construction, so that every simulated unit is drawn from exactly the process the model itself treats as fraud-free.

We ran this analysis on the complete Democratic-coded grid under the short-chain configuration (58 election/level/channel cells). One cell (`pres21`, `dong`, `early`) failed with an invalid simulated draw, an isolated numerical edge case in the random-effect simulation step rather than a systematic problem, leaving 57 completed cells. Even under this maximally conservative construction, `eforensics` reports substantial fraud: the median fraud share across the 57 cells is 1.9% and the mean is 4.6%, with 13 cells (22.8%) exceeding a 5% fraud share and 5 cells (8.8%) exceeding 15%. The largest self-null estimates are concen-

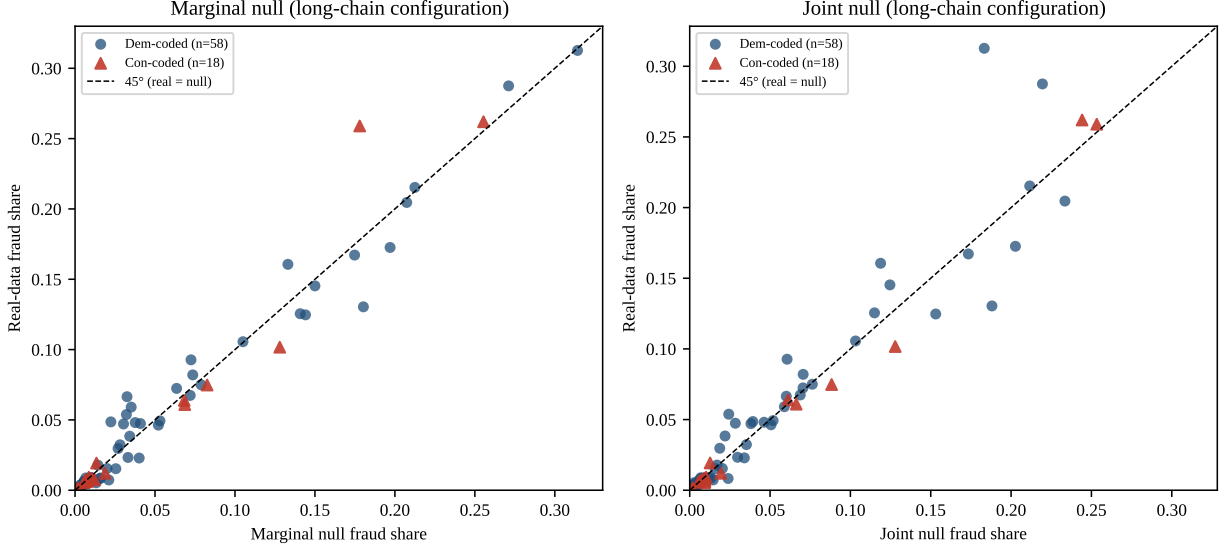


Figure 3: The same comparison under the long-chain configuration ($n_{\text{iter}}=3,000$ or $5,000$, $\text{burn.in}=1,000$): 58 dem-coded cells and 18 conservative-coded cells with all three sources completed under that configuration. Shown separately from Figure 2 because coverage differs between the two configurations (Section 3.2); the pattern is the same either way.

trated in exactly the same channels flagged elsewhere in this paper: constituency-level and pooled totals in the pre-early-voting and highly polarized presidential elections (Table 2).

Table 2: Selected model-consistent self-null results (π forced to $(1, 0, 0)$; short-chain configuration, Democratic-coded leader). π_1 is the model’s own posterior estimate of the no-fraud mixture weight on data drawn entirely from its class-1 process.

Cell (self-null, π forced to $(1, 0, 0)$)	π_1 (no fraud)	Fraud share	Units flagged
2025 pres., constituency, total	75.6%	24.5%	15/242
2025 pres., dong, pooled	80.2%	19.8%	302/3553
2017 pres., constituency, total	82.2%	17.8%	5/250
2022 pres., dong, pooled	83.1%	17.0%	230/3509
2002 pres., dong, total (no early)	84.5%	15.5%	229/3515
2002 pres., constituency, total (no early)	85.0%	14.9%	15/244

Pairing each self-null cell with its own real-data counterpart under the identical (election, level, channel) key, the two are strongly correlated ($r = 0.916$, $n = 57$); real data exceeds its self-null counterpart in 35 of 57 cells (61%), with a mean paired difference (real minus self-null) of under one percentage point (0.83pp). This is closely comparable in both magnitude and pattern to the $r = 0.963$ and $r = 0.936$ correlations already reported for the marginal and joint nulls (Section 4.2), and confirms that those results are not an artifact of the specific ad hoc noise processes used to build them: data generated entirely from `qbl`’s own fitted clean-class equations is, on this model’s own terms, no less “fraudulent” than the real elections it is compared against. Convergence fares no better here than elsewhere in the corpus: 51 of 57 self-null runs (89.5%) fail the $\hat{R} \leq 1.05$ criterion, matching the 87.8–90.8% failure rates found for real data and the other two nulls (Sections 3.2, 4.6). Figure 4 shows the full comparison; the complete 57-cell listing is in Appendix A.3.

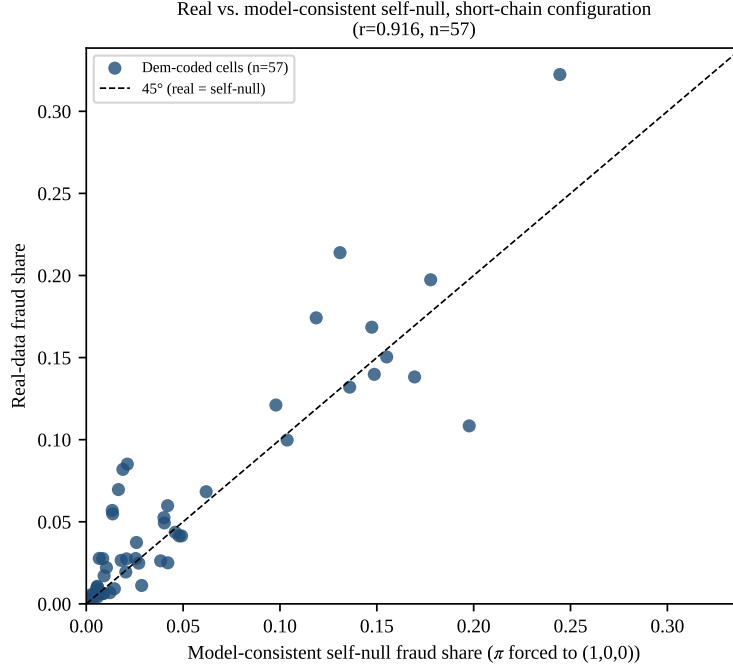


Figure 4: Real-data fraud share vs. model-consistent self-null fraud share (π forced to $(1, 0, 0)$), short-chain configuration, Democratic-coded leader, 57 completed cells. As in Figures 2–3, points near the 45° line indicate that real-data output carries no information beyond what the model’s own no-fraud process already produces.

This result also speaks directly to the recalibration question of Section 5: because the self-null is generated from the fitted model’s own class-1 process rather than an externally chosen approximation to it, its false-positive behavior cannot be attributed to the marginal or joint null generators being poorly matched to the real data. The false-positive problem is intrinsic to how `qbl` partitions its own posterior into clean and fraudulent classes, not to any feature of the two ad hoc nulls used elsewhere in this paper.

4.4 The pre-early-voting natural experiment

Table 3 reports real-data results for the five elections held before early voting existed, short-chain configuration. Two patterns are decisive. First, the presidential elections of 2002, 2007, and 2012, for which no early-voting-type fraud allegation has ever been raised in Korea, receive fraud shares of 7–17%, with the 2012 presidential election (13.2–16.9%) exceeding every early-voting-specific channel of the actually contested April 2020 election (0.1–6.0%; Section 4.7). If `eforensics`’ output were taken as evidence in the way it has been used for 2020, it would imply that 2002–2012 Korea was pervasively fraudulent and that 2020 was comparatively clean, the reverse of the claim structure the model has been used to support. Second, the general elections of the same era (2008, 2012) score near zero (0.5–0.7%), a roughly 25-fold swing driven by race structure (a single national presidential race with regionally polarized vote shares vs. many district-level party races) rather than by any fraud mechanism, exactly as the geometric account in Section 4.5 predicts. Both patterns are reproduced closely by the paired fraud-free nulls of the same elections.

Figure 5 places these estimates on a timeline alongside the introduction of early voting and the onset of fraud claims: the model’s largest outputs fall precisely where no claim exists.

Table 3: Real-data results for elections held before early voting existed (short-chain configuration).

Election	Level	Fraud share	Fraudulent votes (F_w)	Units flagged
Pres. 2002	Constituency	14.0%	2,356,322	16/244
Pres. 2002	Dong	15.0%	2,270,446	338/3515
Pres. 2007	Constituency	7.0%	1,073,737	8/248
Pres. 2007	Dong	8.5%	1,040,736	182/3576
Pres. 2012	Constituency	13.2%	2,962,604	18/251
Pres. 2012	Dong	16.9%	3,669,420	277/3479
Gen. 2008	Constituency	0.7%	0	0/198
Gen. 2008	Dong	0.5%	19,150	5/2733
Gen. 2012	Constituency	0.6%	0	0/209
Gen. 2012	Dong	0.6%	28,128	6/2923

4.5 The model signal tracks distributional geometry, not fraud

Mean real-data fraud share varies sharply and systematically by channel (short-chain configuration): same-day-only channels average 0.04–1.3%; early-only channels average 3.2–3.9%; pooled and total channels average 7.8–17.3%. Every one of these patterns is reproduced by the fraud-free nulls of the same channel. This tracks a straightforward channel-level regularity. Per-channel turnout $\tau_i = \text{channel votes}_i / N_i$ is mechanically deflated in same-day-only channels (roughly 45–55%, since a substantial share of voters vote early) and even more deflated in early-only channels (25–35%). Pooling restores true turnout (65–80%) while simultaneously mixing two subpopulations with different (τ, ν) profiles, inflating overdispersion; the mixture model absorbs this excess overdispersion into π_2 (incremental fraud, which dominates every large estimate in this corpus; extreme fraud π_3 is $\leq 0.3\%$ almost everywhere). The pre-2014 presidential elections combine genuinely high turnout (70–76%) with extreme regional polarization of vote share (ranging from roughly 10% to over 90% across provinces), placing their single channel in the same regime. Because both null generators preserve exactly this turnout-and-share geometry and destroy everything else, they reproduce the real estimates almost perfectly. The quantity `eforensics` recovers in this setting empirically depends, smoothly and predictably, on where the (τ, ν) cloud sits and how fat its tails are, computable with certainty from data containing no fraud.

4.6 Convergence and the model’s own multimodality diagnostic

Of 180 dem-coded runs under the short-chain configuration, 158 (87.8%) fail the standard Gelman–Rubin criterion ($\hat{R} \leq 1.05$) on the mixture weights; the median maximum $\hat{R}$ is 1.60, and 25 runs exceed $\hat{R} = 5$. As Section 3.2 showed directly, this is not attributable to the shorter chains used here: the long-chain configuration shows the same 89.7% failure rate on the overlapping cells. Mebane [2023] proposes a specific diagnostic for posterior multimodality in this model class, $M(\pi_j)$, the largest cross-chain difference in chain-specific posterior means, combined with a Hartigan dip-test p -value on pooled draws, arguing that its presence signals substantive model misspecification (“lost votes”) rather than a mere numerical artifact. This flag fires on 19/60 (31.7%) of real-data runs and 28/120 (23.3%) of fraud-free null runs: a diagnostic that fires at a comparable rate on data with and without fraud carries little evidential weight about fraud, whichever interpretation of its cause is correct.

4.7 The contested election itself: April 2020

For the election that launched the Korean controversy and that Mebane [2020] analyzed directly, short-chain configuration: dong-level in-precinct early votes score a real fraud share of 5.69% (26/3480 units

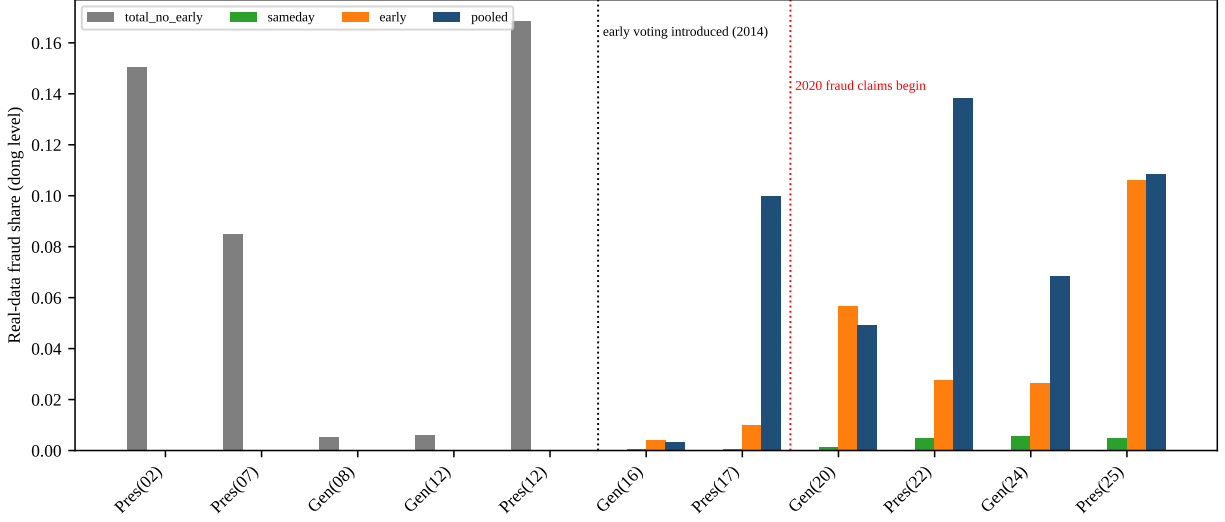


Figure 5: Real-data fraud share by election and channel (dong level, short-chain configuration), in chronological order. Dashed lines mark the introduction of early voting (2014) and the onset of organized fraud claims (2020). The largest estimates fall in the era with no claims; same-day-only channels remain near zero throughout.

flagged, $\hat{R} = 20.05$, severely not converged), against fraud-free nulls of 5.87% and 2.04%. Constituency-level early votes score 5.98% against nulls of 6.19% and 5.98%, the real value sitting almost exactly on top of its joint null and below its marginal null, no longer one of the cells where real exceeds both nulls (Section 4.2); under the long-chain configuration this same cell was one of the few where real exceeded both nulls, so the anomaly itself is unstable across chain length and should not be read as a robust signal either way. Out-of-precinct early votes, the specific channel most heavily targeted by fraud claims, score 2.76%, modestly above both its fraud-free controls (2.57%, 1.10%), a small excess of the same kind and magnitude as the 12 other cells in Section 4.2 where real data exceeds both nulls, and no larger than what cross-null disagreement alone would produce. Nothing in the 2020 early-vote channels stands out against the model’s own false-positive baseline, and every 2020 estimate is dwarfed by what the model incorrectly reports for the uncontested 2002–2012 elections or for fraud-free pooled data generally.

4.8 Bidirectional confirmation: fraud for whichever side is coded as leader

Because `eforensics` fraud mechanisms are unidirectional (Section 2), every result above tests only Democratic-favoring fraud. Under the short-chain configuration we reran the identical analysis, using the same data, the same filters, and the same null generators, with the conservative-lineage candidate or party coded as leader instead, for all 58 (election, level, channel) cells, all 11 elections, both leader codings, no exceptions; the long-chain configuration, by contrast, covers the conservative-coded side only for the 2012 presidential election and the 2020 and 2024 general elections (54 cells; Section 3.2). Across all 58 real-data (dem, con) pairs, the Democratic-coded estimate exceeds the conservative-coded estimate in 41 cells and the reverse holds in 17, an asymmetry consistent with the direction of the historical allegations but, as the cell-by-cell evidence below shows, fully explained by the same null-matched geometry as every other finding in this paper.

4.8.1 The 2012 presidential election

A natural objection to the bidirectional design is that the model, or our implementation of it, might simply have some built-in tendency to implicate the Democratic side regardless of which direction is actually tested, in which case the dem-coded estimates being frequently larger than the con-coded ones elsewhere in this paper would reflect that tendency rather than anything about the data. The 2012 presidential election is a useful check on this, because it is a pre-early-voting, regionally polarized, high-turnout race broadly similar in structure to the other pre-2014 elections. Table 4 and Figure 6 report both codings side by side, short-chain configuration.

Table 4: 2012 presidential election, dem- vs. conservative-coded leader (short-chain configuration).

Level	Leader coding	Source	Fraud share	$\hat{R}$
Dong	Dem	Real	16.9%	4.82
Dong	Dem	Marginal null	15.4%	7.52
Dong	Dem	Joint null	17.7%	1.97
Dong	Con	Real	26.0%	5.21
Dong	Con	Marginal null	25.9%	1.51
Dong	Con	Joint null	25.5%	3.52
Constituency	Dem	Real	13.2%	2.79
Constituency	Dem	Marginal null	17.5%	3.20
Constituency	Dem	Joint null	18.9%	1.19
Constituency	Con	Real	21.7%	5.03
Constituency	Con	Marginal null	17.2%	2.14
Constituency	Con	Joint null	21.7%	3.90

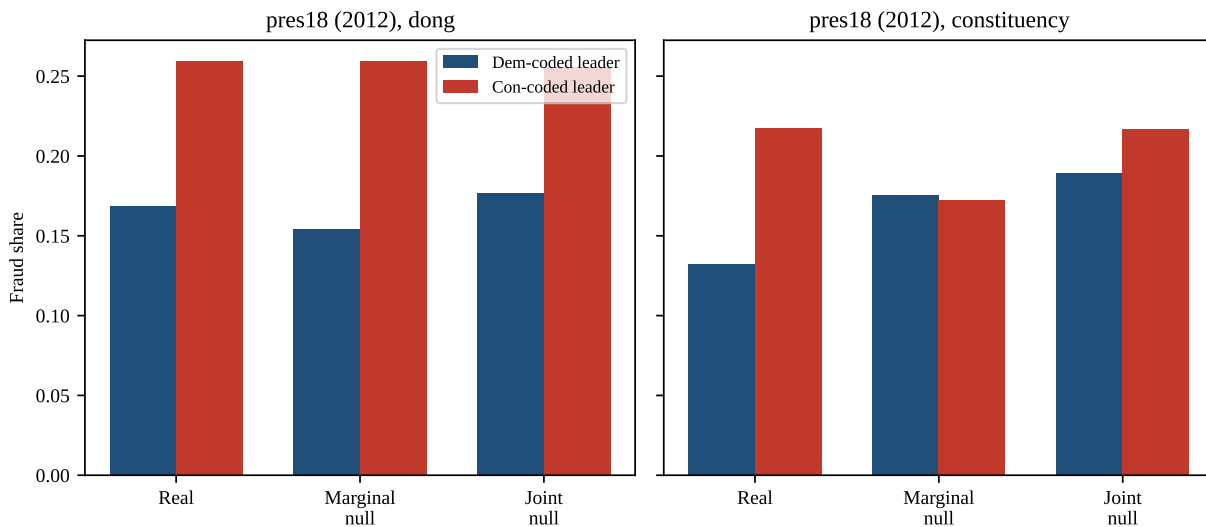


Figure 6: 2012 presidential election fraud share under Democratic-coded (blue) and conservative-coded (red) leader, real data and both fraud-free nulls, at dong and constituency level, short-chain configuration. The conservative-coded estimate is larger than the Democratic-coded estimate under every source, and both are closely tracked by their respective fraud-free nulls.

The model reports *more* fraud favoring the conservative winner ($\sim 26\%$ dong, $\sim 22\%$ constituency) than fraud favoring the Democratic loser ($\sim 17\%$, $\sim 13\%$) in the identical election, answering the objection above

directly: the model does not simply default to implicating the Democratic side regardless of the data, since here it implicates the conservative side more strongly. The conservative-coded estimate is closely matched by its own nulls at both levels (constituency: real 21.7% essentially identical to the joint null's 21.7%; dong: real 26.0% close to both its marginal null, 25.9%, and joint null, 25.5%), so neither estimate constitutes evidence of fraud in either direction. This reproduces, under substantially shorter MCMC chains and a different burn-in, the identical qualitative and near-identical quantitative pattern found under the long-chain configuration, itself evidence that the finding is not an artifact of either chain length. Pointed at either side of a regionally polarized, high-turnout, pre-early-voting election, *eforensics* reports large “fraud” favoring *that* side, matched almost exactly by fraud-free data with the same shape, regardless of which side is coded as leader.

4.8.2 The remaining elections, both directions

The complete grid lets us extend the same check to every other pre-2020 election. Two patterns emerge. First, among the pre-2014 elections, the two presidential elections not already discussed replicate the *pres18* pattern of large, closely null-matched estimates under both codings: the 2002 election scores 15.0% (dem) vs. 14.1% (con) at dong level and 14.0% vs. 12.0% at constituency level; the 2007 election scores 8.5% vs. 6.6% (dong) and 7.0% vs. 5.8% (constituency). The two pre-2014 general elections, by contrast, remain near zero under both codings (2008: 0.5% dem vs. 0.2% con at dong level; 2012: 0.6% vs. 1.7%), consistent with the race-structure account in Section 4.5: a single national presidential race is fraud-share-inflating regardless of which side is coded leader, while district-level party races are not. Second, the 2016 general election, the first held under early voting and hence the first available comparison point for how the dem/con asymmetry behaves right at the system's introduction, shows small, mixed-sign differences across channels (e.g., dong pooled: 0.3% dem vs. 1.5% con, con *larger*; constituency total: 0.6% vs. 1.1%, also con *larger*), unlike the mostly dem-larger pattern of 2020 and 2024. This is itself informative: if the dem/con asymmetry in later elections reflected a stable partisan skew in who chooses to vote early, the same skew should already be visible in 2016; that it is not, and that the sign even reverses in some 2016 channels, is more consistent with the asymmetry being driven by election-specific turnout composition (Section 4.7) than by anything structural to early voting itself.

4.8.3 The 2020 election, both directions

Conservative-coded estimates for the 2020 general election are smaller than Democratic-coded estimates in six of eight compared channels under the short-chain configuration (e.g., constituency total: 11.6% vs. 12.1%; dong early: 0.2% vs. 5.7%), directionally consistent with what the allegations claimed, though the constituency-level early and early-out channels show the conservative-coded value fractionally *larger* (5.98% vs. 5.98% and 2.76% vs. 2.76%, effectively tied at this precision) rather than smaller. This asymmetry, where it appears, is exactly what the geometric mechanism predicts even under zero fraud: 2020 early voting, introduced amid the COVID-19 pandemic, skewed toward younger and more urban voters and hence toward the Democratic side on both real and null data alike, and Section 4.8.2 shows this skew was largely absent in 2016, consistent with a COVID-specific rather than early-voting-general explanation. Consistent with a non-fraud explanation, the conservative-coded real values remain closely matched by their own conservative-coded nulls in every one of these channels, and the constituency-early “double exceedance” anomaly flagged under the long-chain configuration (Section 4.7) does not reproduce under the short-chain configuration for either coding, indicating it was itself a settings-sensitive artifact rather than a robust finding in need of a substantive explanation.

4.8.4 The 2024 election, both directions

The 2024 general election, the most recent Korean election with an active fraud narrative, shows conservative-coded real-data estimates smaller than Democratic-coded estimates in its two largest-signal channels (Figure 7): constituency total (11.8% vs. 17.4%) and dong pooled (6.3% vs. 6.8%). Both conservative-coded real values remain closely bracketed by their own nulls (constituency total: real 11.8% against nulls of 8.4% and 9.5%; dong pooled: real 6.3% against nulls of 6.2% and 7.2%), and the dong-pooled channel again shows the closest cross-coding agreement of any channel in either the 2020 or 2024 comparison.

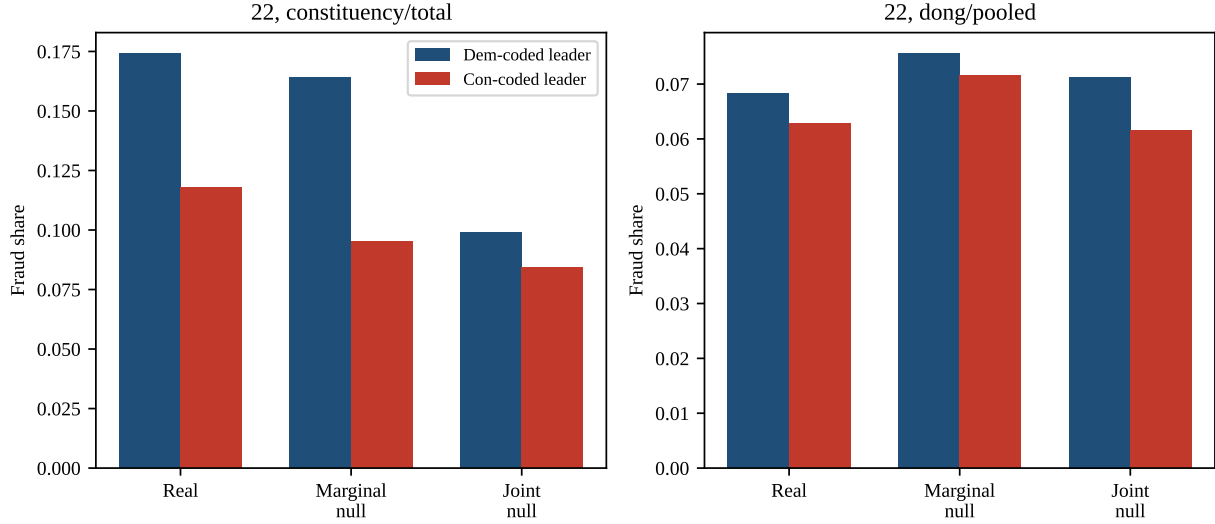


Figure 7: 2024 general election fraud share under Democratic-coded (blue) and conservative-coded (red) leader, real data and both fraud-free nulls, for the two largest-signal channels, short-chain configuration.

4.8.5 The 2025 election, both directions

The 21st presidential election (June 2025), held in the aftermath of the December 2024 martial-law declaration, is among the most politically charged elections in the corpus and shows the sharpest dem/con asymmetry in the entire grid: dong-level early votes score 10.6% under Democratic coding against only 0.2% under conservative coding, and constituency-level early-total scores 8.2% against 0.5%. Every conservative-coded value in this election remains small and closely tracked by its own nulls (e.g., dong early: real 0.2% against nulls of 0.3% and 0.2%), so despite showing the largest raw asymmetry in the study, this election provides no more evidence of fraud than any other: the Democratic-coded estimates are themselves matched by their own nulls throughout (Table 7 in Appendix A), and the asymmetry is consistent with 2025's unusually high and unusually partisan-charged turnout following the martial-law crisis, which is exactly the kind of extreme, non-fraud turnout event the geometry mechanism predicts will inflate estimates on whichever side that turnout leans toward.

4.8.6 Grid-wide convergence

Convergence and the Mebane multimodality flag replicate across the complete conservative-coded grid: 152/180 (84.4%) fail $\hat{R} \leq 1.05$, essentially the same rate as the 87.8% found under Democratic coding

(Section 4.6), and the multimodality flag fires on 29/180 (16.1%) of conservative-coded runs, again comparable to the Democratic-coded rate. Non-convergence and posterior multimodality are properties of this data and this model, not of which side is coded as leader.

5 Discussion: is eforensics credible as a fraud detector?

The evidence supports five independent, mutually reinforcing conclusions against treating eforensics’s positive outputs as evidence of fraud in this setting.

It has no specificity. It assigns 5–29% fraud shares and millions of fraudulent votes to data that is fraud-free by construction, under three distinct clean data-generating processes, including one generated entirely from the model’s own fitted no-fraud equations (Section 4.3), across both MCMC chain-length configurations and both leader codings. A detector whose false-positive output matches its real-data output in magnitude detects nothing.

It fails its historical control. The Korean fraud narrative is inseparable from early voting (introduced 2014; claims from 2020). The model’s largest real-data estimates fall on pre-2014 presidential elections with no mainstream fraud narrative, exceeding its estimates for the actually contested 2020 channels. The temporal pattern of its output is uncorrelated with, and in places anti-correlated with, the temporal pattern of the allegations it has been used to support.

Its output is predictable from fraud-irrelevant geometry. Fraud share rises monotonically with channel-level turnout and cross-unit (τ, ν) heterogeneity, a pattern fully reproduced under both nulls (Section 4.5). The estimand the model recovers in practice is overdispersion-plus-corner-mass, not fraud.

Its inferential machinery is unreliable in exactly this application. 84–91% of runs fail conventional convergence diagnostics across both leader codings and both chain-length configurations (Sections 3.2, 4.6), and the model’s own advanced multimodality diagnostic fires at comparable rates on fraud-free and real data.

It is not directionally selective. Coding the opposite party as leader reproduces the identical pathology across the complete grid (Section 4.8): large, null-matched “fraud” estimates appear for whichever side is coded as leader, and the model does not simply default to implicating the Democratic side regardless of the data. In the 2012 presidential election, for instance, the conservative-coded estimate is larger than the Democratic-coded one, both matched by their own fraud-free nulls, so the asymmetry seen in other elections is a property of each election’s data, not a built-in tendency of the model or our implementation of it.

5.1 Is the model salvageable by recalibration?

It is useful to locate this failure precisely rather than dismiss the model class wholesale. Failures of this kind fall on a three-level ladder: (i) *miscalibration*, fixable by recalibrating thresholds against a reference null distribution; (ii) *lack of power after calibration*, where size can be corrected but no signal remains; (iii) *non-identification of the estimand*, where the quantity recovered is not the one of interest regardless of calibration. Our evidence places eforensics-as-deployed at (iii) shading into (ii): fraud classes are identified purely through parametric shape assumptions on the clean class, so realistic clean-data heterogeneity (fat tails, high-turnout polarized units, multi-mode pooled voting) is absorbed into the fraud components, which is precisely why fraud-free nulls reproduce the real-data estimates. The model-consistent self-null (Section 4.3) rules out the remaining objection that this reflects only a poor match between our chosen null generators and the real data: fraud shares of the same order appear even on data drawn entirely from the model’s own fitted class-1 process, so the false-positive behavior is intrinsic to how qbl partitions its own posterior, not an artifact of the marginal or joint null construction. Recalibrating thresholds against an empirical null distribution would repair validity, since the false-positive rate could in principle be controlled, but because real and null outputs are already statistically indistinguishable, a validity-correct recalibrated test would report approximately no

detectable fraud anywhere in this 20-year corpus: calibration fixes size, not power, and would empty every positive finding the model has previously been used to support. Genuine repair would require redesign: a flexible clean-class distribution plus injection-verified power demonstrating that realistic fraud magnitudes remain detectable above a corrected false-positive floor, with the live possibility that a sufficiently realistic clean class leaves nothing detectable, implying that aggregate-count-only mixture forensics of this type is underpowered in principle for high-turnout, regionally polarized electorates such as Korea’s, rather than merely misapplied in this instance.

We note explicitly that Mebane’s own work cautions that `eforensics` “frauds” may reflect strategic behavior rather than malfeasance [Mebane, 2020]. Our findings sharpen this caveat considerably: the estimates do not merely admit benign explanations, they are quantitatively reproduced by explicitly fraud-free processes matched only on distributional shape, in both the alleged and the unalleged direction. Taken seriously, this caveat implies the model’s outputs should not have been treated as evidence in the Korean controversy.

6 Limitations

Several limitations qualify the conclusions above. First, the marginal and joint nulls are single-seed draws; the tightness of real-null agreement across 60 independent cells under each leader coding, and its replication by the model-consistent self-null (Section 4.3), makes seed noise an implausible sole driver, but a multi-seed ensemble would still be needed to attach formally calibrated p -values to the paired-difference comparisons of Section 4.2. Second, our aggregation choices (dong/constituency level, province fixed effects) differ from Mebane [2020]’s precinct-level units and covariates, but this should not be read as a mere difference between two equally defensible specifications: Min and Yoo [2021] independently show that Mebane’s original units were not administratively valid for Korean election data. His “district, election-day” and “voting-post, election-day” units duplicate each other, and, more consequentially, his early-vote turnout was computed against a “registered voters” denominator that does not exist as a fixed population for Korean in-precinct early voting, which has no pre-registration, unlike the U.S. case his software assumes. This produces measured early-vote turnout clustering near 100% by construction, which mechanically inflates `eforensics`’ manufactured-vote channel specifically, since that channel is triggered by turnout appearing implausibly high; correcting this error, Min and Yoo [2021] replicate the identical `eforensics-qbl` model and find Mebane’s headline fraud estimate falls by roughly half to two-thirds. Third, the conservative-coded grid is complete under the short-chain configuration (Section 4.8) but only partial under the long-chain configuration, so a long-chain, settings-matched replication of every bidirectional cell is not yet available. Fourth, convergence failure is pervasive (84–91% depending on coding and chain-length configuration), but, as Section 3.2 shows directly rather than by inference, this is not resolved by longer chains: matching 174 cells across a $3\text{--}5\times$ change in sampling iterations changes convergence rates by about one percentage point and leaves fraud-share estimates correlated at $r = 0.95$. None of these limitations bears on the paper’s central finding, which holds across three independent null constructions, two chain-length configurations, and both leader codings.

7 Further work

Two extensions would most directly sharpen the conclusions above. A *null ensemble with calibrated p -values*, refitting each null generator with multiple seeds per cell, would convert the paired-difference comparisons of Sections 4.2 and 4.3 into formal hypothesis tests; because a single dong-level fit already takes on the order of an hour of wall-clock time, this is best pursued first on the cells already flagged as exceeding all three nulls, or at the constituency level, where smaller unit counts make each fit substantially faster. An

injection/recovery power study, adding known fraud at realistic magnitudes to null data and measuring recovery above the false-positive floor established here, would establish whether any positive `eforensics` finding can be considered interpretable once the false-positive rate is accounted for, in this application or elsewhere. Beyond these two, extending the conservative-coded grid to the long-chain configuration and applying the identical procedure to placebo elections from a country without an active fraud discourse would each provide a further robustness check, though neither is expected to alter the present findings.

8 Conclusion

Across more than 600 model fits spanning two decades of Korean elections, two independent MCMC chain-length configurations, two independent aggregation levels, up to five vote-counting channels, three independently constructed fraud-free controls, and, in a complete grid, both directions of leader coding for every election studied, the `eforensics` quasi-binomial-logistic model’s fraud estimates are statistically indistinguishable from what it reports on data engineered to contain no fraud, are largest precisely where no mainstream fraud allegation has ever existed, and reproduce identically across the complete grid when the model is pointed at the opposite side of a contested election, including elections with no historical allegation in either direction. The third of these controls, drawn entirely from the fitted model’s own no-fraud generative equations with π forced to $(1, 0, 0)$, closes the remaining gap in this argument by showing that the false positives are not an artifact of any particular choice of null-generating process. A direct chain-length comparison further shows this is not an artifact of insufficient MCMC sampling: a 3–5 $\times$ reduction in iterations changes neither the convergence failure rate nor, to a close approximation, the fraud-share estimates themselves. The model’s output is well explained, in every instance we examined, by the shape of the turnout-and-vote-share distribution alone. We do not conclude that Korean elections have been free of any irregularity; we conclude only that this model, as deployed, provides no evidence on the question either way, and that its positive findings in the public record should not be treated as evidence of fraud.

Data availability

The raw election-result CSV files used throughout this study are official National Election Commission returns published by the Korean government’s open-data portal, <https://www.data.go.kr>. General (legislative) election results are available at <https://www.data.go.kr/data/15025527/fileData.do>, and presidential election results at <https://www.data.go.kr/data/15025528/fileData.do>; both pages provide per-election, precinct/dong-level result files in the format our data-loading code expects (Appendix B).

The processed unit-level tables, all model outputs, and the full analysis and figure-generation code are provided in https://mkimacad.github.io/south_korean_election_forensics.

Given its length, the codebase is not reproduced in full in this manuscript; Appendix B documents every script’s purpose, software requirements, and the exact commands needed to reproduce each stage of the analysis, from raw election CSVs through to the tables and figures presented above.

A Complete results listing

Tables 5–8 list every completed model run referenced in this paper: the long-chain configuration (Democratic- and conservative-coded, Tables 5–6) and the short-chain configuration (Democratic- and conservative-coded, Tables 7–8), 576 rows in total. The 95% credible interval reported alongside each fraud-share estimate is the sum of the independent per-unit-class HPD bounds for the incremental-

and extreme-fraud mixture weights (π_2, π_3), following the same conservative-sum convention used for F_w throughout the paper (Section 2); this is a deliberately conservative outer bound rather than a true joint interval, since the installed `eforensics` package does not currently expose the raw per-draw, per-unit fraud-proportion matrix needed to compute one directly. Rows with $\hat{R}$ marked * fail the $\hat{R} \leq 1.05$ convergence criterion. Election codes follow National Election Commission numbering: Pres02/07/12/17/22/25 are the 16th/17th/18th/19th/20th/21st presidential elections; Gen08/12/16/20/24 are the 18th/19th/20th/21st/22nd general elections.

Table 5: Long-chain configuration, Democratic-coded leader (n.iter=3000 or 5000, burn.in=1000, n.adapt=1000). $\hat{R}$ marked * fails the $\hat{R} \leq 1.05$ convergence criterion.

Elec.	Level	Channel	Source	N	Fraud share % [95% CI]	F_w	$\hat{R}$
Gen08	const.	total-no-early	real	198	0.69 [0.0,3.0]	0	1.00
Gen08	const.	total-no-early	marg.null	198	0.71 [0.0,3.1]	0	1.00
Gen08	const.	total-no-early	joint.null	198	0.70 [0.0,3.0]	0	1.00
Gen08	dong	total-no-early	real	2733	0.46 [0.2,0.8]	17,914	1.04
Gen08	dong	total-no-early	marg.null	2733	0.76 [0.4,1.2]	31,118	1.38*
Gen08	dong	total-no-early	joint.null	2733	0.40 [0.1,0.8]	38,051	1.24*
Gen12	const.	total-no-early	real	209	0.66 [0.0,2.8]	0	1.00
Gen12	const.	total-no-early	marg.null	209	0.67 [0.0,2.8]	0	1.00
Gen12	const.	total-no-early	joint.null	209	0.66 [0.0,2.8]	0	1.00
Gen12	dong	total-no-early	real	2923	0.54 [0.1,1.0]	22,770	2.23*
Gen12	dong	total-no-early	marg.null	2923	0.71 [0.4,1.1]	47,290	1.23*
Gen12	dong	total-no-early	joint.null	2923	0.28 [0.0,0.6]	24,787	1.97*
Pres02	const.	total-no-early	real	244	12.47 [7.4,18.5]	1,960,598	1.45*
Pres02	const.	total-no-early	marg.null	244	14.41 [7.0,23.2]	2,982,200	2.79*
Pres02	const.	total-no-early	joint.null	244	15.31 [10.1,21.5]	2,350,835	1.32*
Pres02	dong	total-no-early	real	3515	14.53 [11.2,17.0]	2,188,462	2.98*
Pres02	dong	total-no-early	marg.null	3515	15.00 [13.3,16.6]	2,492,242	1.52*
Pres02	dong	total-no-early	joint.null	3515	12.46 [10.7,14.5]	2,013,904	1.73*
Pres07	const.	total-no-early	real	248	8.20 [3.9,13.8]	1,218,100	1.64*
Pres07	const.	total-no-early	marg.null	248	7.36 [1.8,14.0]	973,233	2.64*
Pres07	const.	total-no-early	joint.null	248	7.06 [2.0,17.1]	930,913	3.80*

Table 5 – continued

Elec.	Level	Channel	Source	N	Fraud share % [95% CI]	F_w	$\hat{R}$
Pres07	dong	total-no-early	real	3576	7.24 [6.1,9.1]	882,177	2.49*
Pres07	dong	total-no-early	marg.null	3576	6.35 [5.2,12.2]	965,803	7.99*
Pres07	dong	total-no-early	joint.null	3576	7.05 [5.9,8.4]	925,051	1.93*
Pres12	const.	total-no-early	real	251	17.26 [10.5,24.8]	3,602,441	1.84*
Pres12	const.	total-no-early	marg.null	251	19.70 [11.9,28.7]	4,346,162	2.37*
Pres12	const.	total-no-early	joint.null	251	20.27 [14.1,28.1]	4,171,828	1.62*
Pres12	dong	total-no-early	real	3479	16.72 [12.0,18.5]	3,236,794	4.50*
Pres12	dong	total-no-early	marg.null	3479	17.48 [15.6,19.7]	3,739,725	2.04*
Pres12	dong	total-no-early	joint.null	3479	17.34 [15.9,18.9]	3,082,362	1.44*
Gen16	const.	early	real	234	0.73 [0.0,3.7]	35,169	1.91*
Gen16	const.	early	marg.null	234	2.11 [0.0,6.2]	189,116	2.17*
Gen16	const.	early	joint.null	234	1.46 [0.0,8.9]	189,027	4.03*
Gen16	const.	early-out	real	234	0.87 [0.0,3.4]	38,242	1.40*
Gen16	const.	early-out	marg.null	234	0.67 [0.0,2.8]	11,843	1.19*
Gen16	const.	early-out	joint.null	234	0.68 [0.0,2.8]	12,506	1.14*
Gen16	const.	early-total	real	234	3.84 [0.8,7.9]	366,447	1.49*
Gen16	const.	early-total	marg.null	234	3.42 [0.2,8.7]	325,867	2.40*
Gen16	const.	early-total	joint.null	234	2.20 [0.2,5.8]	175,006	1.62*
Gen16	const.	sameday	real	234	0.60 [0.0,2.5]	0	1.01
Gen16	const.	sameday	marg.null	234	0.60 [0.0,2.5]	0	1.01
Gen16	const.	sameday	joint.null	234	0.59 [0.0,2.5]	0	1.00
Gen16	const.	total	real	234	0.87 [0.0,3.3]	86,844	1.36*
Gen16	const.	total	marg.null	234	1.59 [0.0,5.3]	236,962	1.95*
Gen16	const.	total	joint.null	234	0.75 [0.0,3.2]	60,640	1.44*
Gen16	dong	early	real	3135	0.39 [0.1,1.7]	81,830	5.29*
Gen16	dong	early	marg.null	3135	0.37 [0.1,0.7]	46,687	1.45*
Gen16	dong	early	joint.null	3135	0.26 [0.0,1.3]	51,041	5.34*
Gen16	dong	pooled	real	3135	0.43 [0.1,0.8]	71,112	1.67*
Gen16	dong	pooled	marg.null	3135	0.55 [0.2,0.9]	106,763	1.40*
Gen16	dong	pooled	joint.null	3135	0.31 [0.1,0.6]	49,495	1.31*
Gen16	dong	sameday	real	3135	0.05 [0.0,0.2]	154	1.15*
Gen16	dong	sameday	marg.null	3135	0.07 [0.0,0.2]	13,593	1.21*
Gen16	dong	sameday	joint.null	3135	0.04 [0.0,0.2]	0	1.00
Pres17	const.	early	real	250	2.29 [0.4,5.7]	278,427	1.40*
Pres17	const.	early	marg.null	250	4.01 [1.5,7.8]	487,973	1.21*
Pres17	const.	early	joint.null	250	3.40 [0.0,10.7]	308,068	4.13*
Pres17	const.	early-out	real	250	4.71 [0.2,10.6]	401,782	3.06*
Pres17	const.	early-out	marg.null	250	3.03 [0.2,7.1]	338,010	2.02*
Pres17	const.	early-out	joint.null	250	3.81 [0.1,8.8]	349,003	2.85*
Pres17	const.	early-total	real	250	0.56 [0.0,2.4]	0	1.00
Pres17	const.	early-total	marg.null	250	0.55 [0.0,2.4]	0	1.00
Pres17	const.	early-total	joint.null	250	0.64 [0.0,2.6]	25,307	1.16*
Pres17	const.	sameday	real	250	0.57 [0.0,2.4]	0	1.00
Pres17	const.	sameday	marg.null	250	0.56 [0.0,2.5]	0	1.00
Pres17	const.	sameday	joint.null	250	0.56 [0.0,2.4]	0	1.00
Pres17	const.	total	real	250	20.46 [8.1,27.6]	4,221,885	3.16*

Table 5 – continued

Elec.	Level	Channel	Source	N	Fraud share % [95% CI]	F_w	$\hat{R}$
Pres17	const.	total	marg.null	250	20.74 [15.1,27.1]	4,538,349	1.16*
Pres17	const.	total	joint.null	250	23.35 [4.6,31.2]	4,224,649	5.13*
Pres17	dong	early	real	3491	0.71 [0.3,1.8]	157,068	3.67*
Pres17	dong	early	marg.null	3491	0.80 [0.3,1.9]	168,051	4.08*
Pres17	dong	early	joint.null	3491	0.59 [0.3,1.0]	94,931	1.44*
Pres17	dong	pooled	real	3491	10.56 [9.2,11.9]	1,959,121	1.49*
Pres17	dong	pooled	marg.null	3491	10.50 [5.8,12.0]	1,859,353	6.19*
Pres17	dong	pooled	joint.null	3491	10.32 [4.8,11.6]	1,377,730	7.42*
Pres17	dong	sameday	real	3491	0.06 [0.0,0.2]	17,022	1.45*
Pres17	dong	sameday	marg.null	3491	0.13 [0.0,0.3]	52,605	1.04
Pres17	dong	sameday	joint.null	3491	0.12 [0.0,0.3]	8,770	1.39*
Gen20	const.	early	real	253	6.65 [2.7,11.7]	940,273	1.69*
Gen20	const.	early	marg.null	253	3.25 [0.6,7.7]	444,747	1.97*
Gen20	const.	early	joint.null	253	6.01 [1.2,11.4]	806,296	2.45*
Gen20	const.	early-out	real	253	3.23 [0.8,7.0]	389,512	1.51*
Gen20	const.	early-out	marg.null	253	2.81 [0.7,6.1]	325,941	1.15*
Gen20	const.	early-out	joint.null	253	3.53 [0.4,7.1]	363,398	1.63*
Gen20	const.	early-total	real	253	2.33 [0.3,5.4]	309,502	1.35*
Gen20	const.	early-total	marg.null	253	3.31 [1.0,6.7]	464,833	1.13*
Gen20	const.	early-total	joint.null	253	2.98 [0.8,6.3]	394,520	1.23*
Gen20	const.	sameday	real	253	0.63 [0.0,2.6]	18,299	1.14*
Gen20	const.	sameday	marg.null	253	0.55 [0.0,2.4]	0	1.00
Gen20	const.	sameday	joint.null	253	0.82 [0.0,3.1]	53,496	1.35*
Gen20	const.	total	real	253	12.55 [6.4,18.6]	2,618,790	1.84*
Gen20	const.	total	marg.null	253	14.08 [8.9,20.8]	3,189,324	1.55*
Gen20	const.	total	joint.null	253	11.50 [3.2,17.4]	2,353,421	2.77*
Gen20	dong	early	real	3480	2.97 [1.9,4.1]	570,495	2.65*
Gen20	dong	early	marg.null	3480	2.69 [2.1,3.4]	662,028	1.28*
Gen20	dong	early	joint.null	3480	1.87 [1.1,6.9]	473,974	9.19*
Gen20	dong	pooled	real	3480	4.63 [3.9,5.5]	826,623	1.19*
Gen20	dong	pooled	marg.null	3480	5.21 [4.4,6.1]	910,319	1.11*
Gen20	dong	pooled	joint.null	3480	5.06 [4.2,5.9]	889,595	1.27*
Gen20	dong	sameday	real	3480	0.19 [0.0,0.4]	27,541	1.45*
Gen20	dong	sameday	marg.null	3480	0.14 [0.0,0.3]	14,320	1.14*
Gen20	dong	sameday	joint.null	3480	0.22 [0.1,0.5]	31,975	1.39*
Pres22	const.	early	real	250	1.54 [0.1,4.3]	88,463	1.34*
Pres22	const.	early	marg.null	250	2.00 [0.3,5.1]	131,799	1.33*
Pres22	const.	early	joint.null	250	2.03 [0.4,4.8]	215,208	1.15*
Pres22	const.	early-out	real	250	4.86 [1.5,9.1]	556,235	1.53*
Pres22	const.	early-out	marg.null	250	2.24 [0.1,5.8]	325,930	1.90*
Pres22	const.	early-out	joint.null	250	3.93 [0.0,10.2]	475,003	3.43*
Pres22	const.	early-total	real	250	4.93 [1.8,9.4]	391,493	1.52*
Pres22	const.	early-total	marg.null	250	5.31 [2.1,9.4]	533,290	1.30*
Pres22	const.	early-total	joint.null	250	5.19 [1.9,11.5]	741,076	2.25*
Pres22	const.	sameday	real	250	1.78 [0.1,5.1]	112,827	1.78*
Pres22	const.	sameday	marg.null	250	1.43 [0.1,3.9]	46,579	1.06*
Pres22	const.	sameday	joint.null	250	1.70 [0.0,4.7]	167,280	1.56*
Pres22	const.	total	real	248	28.75 [8.7,41.7]	6,702,732	4.63*
Pres22	const.	total	marg.null	248	27.11 [7.5,38.0]	5,915,814	4.97*
Pres22	const.	total	joint.null	248	21.95 [10.5,39.6]	5,331,194	4.80*
Pres22	dong	early	real	3509	5.38 [2.9,12.8]	589,160	12.71*
Pres22	dong	early	marg.null	3509	3.21 [1.7,14.9]	567,172	15.25*
Pres22	dong	early	joint.null	3509	2.43 [1.6,3.2]	311,907	2.01*
Pres22	dong	pooled	real	3509	13.04 [10.0,21.1]	2,496,708	8.72*
Pres22	dong	pooled	marg.null	3509	18.03 [8.6,20.4]	2,900,939	10.03*

Table 5 – continued

Elec.	Level	Channel	Source	N	Fraud [95% CI]	share %	F_w	$\hat{R}$
Pres22	dong	pooled	joint.null	3509	18.81 [16.8,20.8]		3,116,286	1.84*
Pres22	dong	sameday	real	3509	0.35 [0.1,0.7]		36,739	1.38*
Pres22	dong	sameday	marg.null	3509	0.48 [0.2,0.9]		67,335	1.34*
Pres22	dong	sameday	joint.null	3509	0.51 [0.2,1.0]		96,278	2.04*
Gen24	const.	early	real	246	4.81 [2.1,9.0]		680,905	1.22*
Gen24	const.	early	marg.null	246	3.76 [1.1,9.2]		591,522	2.06*
Gen24	const.	early	joint.null	246	4.63 [1.6,10.6]		755,608	2.12*
Gen24	const.	early-out	real	246	1.53 [0.0,5.6]		163,197	2.62*
Gen24	const.	early-out	marg.null	246	2.55 [0.0,8.2]		331,995	3.13*
Gen24	const.	early-out	joint.null	246	1.58 [0.0,4.9]		152,565	2.09*
Gen24	const.	early-total	real	246	0.90 [0.0,6.7]		182,314	3.72*
Gen24	const.	early-total	marg.null	246	0.94 [0.0,3.2]		75,160	1.24*
Gen24	const.	early-total	joint.null	246	1.20 [0.0,6.7]		259,146	3.38*
Gen24	const.	sameday	real	246	0.87 [0.0,3.5]		56,225	1.69*
Gen24	const.	sameday	marg.null	246	1.01 [0.0,3.6]		91,854	1.49*
Gen24	const.	sameday	joint.null	246	1.34 [0.0,4.6]		155,986	1.99*
Gen24	const.	total	real	246	16.06 [5.3,24.1]		3,425,501	3.40*
Gen24	const.	total	marg.null	246	13.31 [3.9,19.4]		2,733,030	2.83*
Gen24	const.	total	joint.null	246	11.88 [4.1,22.0]		2,716,267	3.97*
Gen24	dong	early	real	2739	4.74 [2.7,7.2]		619,168	5.59*
Gen24	dong	early	marg.null	2739	4.09 [2.3,18.1]		803,556	15.10*
Gen24	dong	early	joint.null	2739	2.84 [1.6,4.6]		447,956	3.70*
Gen24	dong	pooled	real	2739	6.74 [5.6,8.2]		1,117,138	1.58*
Gen24	dong	pooled	marg.null	2739	7.20 [5.9,8.5]		1,168,700	1.53*
Gen24	dong	pooled	joint.null	2739	6.88 [5.7,8.6]		1,267,053	2.00*
Gen24	dong	sameday	real	2739	0.43 [0.2,0.9]		52,731	1.51*
Gen24	dong	sameday	marg.null	2739	0.58 [0.1,2.0]		102,276	4.89*
Gen24	dong	sameday	joint.null	2739	0.59 [0.3,1.0]		95,103	1.18*
Pres25	const.	early	real	253	9.27 [2.9,20.6]		1,456,140	4.21*
Pres25	const.	early	marg.null	253	7.25 [3.2,15.9]		1,223,597	2.77*
Pres25	const.	early	joint.null	253	6.06 [2.5,10.8]		1,002,167	1.63*
Pres25	const.	early-out	real	253	5.91 [0.1,10.2]		662,683	2.95*
Pres25	const.	early-out	marg.null	253	3.51 [0.0,11.3]		606,160	4.37*
Pres25	const.	early-out	joint.null	253	5.89 [0.7,14.0]		893,777	3.60*
Pres25	const.	early-total	real	253	7.50 [3.0,12.7]		604,576	1.73*
Pres25	const.	early-total	marg.null	253	7.88 [4.2,12.7]		748,911	1.25*
Pres25	const.	early-total	joint.null	253	7.63 [3.4,12.8]		1,122,497	1.59*
Pres25	const.	sameday	real	253	0.83 [0.0,3.5]		19,930	1.73*
Pres25	const.	sameday	marg.null	253	1.56 [0.1,4.5]		54,652	1.53*
Pres25	const.	sameday	joint.null	253	2.39 [0.3,7.2]		177,679	2.26*
Pres25	const.	total	real	242	31.27 [12.3,41.6]		7,513,377	4.14*
Pres25	const.	total	marg.null	242	31.42 [22.7,40.6]		8,663,934	1.19*
Pres25	const.	total	joint.null	242	18.33 [5.5,37.4]		5,723,107	6.53*
Pres25	dong	early	real	3553	0.53 [0.2,1.1]		117,094	2.24*
Pres25	dong	early	marg.null	3553	1.32 [0.8,2.2]		281,099	2.62*
Pres25	dong	early	joint.null	3553	0.45 [0.1,1.0]		108,067	2.33*
Pres25	dong	pooled	real	3553	21.53 [8.8,24.0]		3,308,778	12.73*
Pres25	dong	pooled	marg.null	3553	21.25 [9.0,23.2]		3,299,247	11.86*
Pres25	dong	pooled	joint.null	3553	21.16 [7.8,23.2]		3,049,986	13.33*
Pres25	dong	sameday	real	3553	0.38 [0.2,0.7]		36,380	1.24*
Pres25	dong	sameday	marg.null	3553	0.36 [0.1,0.6]		37,168	1.10*
Pres25	dong	sameday	joint.null	3553	0.46 [0.2,0.8]		87,547	1.06*

Table 6: Long-chain configuration, conservative-coded leader (partial; excludes the still-incomplete 2022 presidential slice).

Elec.	Level	Channel	Source	N	Fraud share [95% CI]	%	F_w	$\hat{R}$
Pres12	const.	total-no-early	real	251	26.20 [19.7,33.1]		5,295,088	1.24*
Pres12	const.	total-no-early	marg.null	251	25.54 [11.1,33.1]		5,212,909	3.50*
Pres12	const.	total-no-early	joint.null	251	24.42 [10.9,32.0]		5,176,784	3.24*
Pres12	dong	total-no-early	real	3479	25.90 [23.8,28.8]		4,657,560	2.35*
Pres12	dong	total-no-early	marg.null	3479	17.80 [15.4,27.5]		3,852,548	9.33*
Pres12	dong	total-no-early	joint.null	3479	25.33 [14.1,27.3]		4,530,868	9.82*
Gen20	const.	early	real	253	1.93 [0.0,6.1]		238,134	2.62*
Gen20	const.	early	marg.null	253	1.34 [0.0,6.1]		154,583	2.87*
Gen20	const.	early	joint.null	253	1.27 [0.0,4.0]		129,202	1.45*
Gen20	const.	early-out	real	253	0.86 [0.0,3.9]		64,974	2.00*
Gen20	const.	early-out	marg.null	253	1.08 [0.0,3.6]		84,193	1.45*
Gen20	const.	early-out	joint.null	253	0.90 [0.0,3.3]		46,425	1.48*
Gen20	const.	early-total	real	253	0.93 [0.0,3.2]		76,409	1.24*
Gen20	const.	early-total	marg.null	253	0.86 [0.0,3.5]		62,688	1.65*
Gen20	const.	early-total	joint.null	253	1.02 [0.0,3.3]		86,065	1.08*
Gen20	const.	sameday	real	253	0.61 [0.0,2.5]		13,561	1.17*
Gen20	const.	sameday	marg.null	253	0.63 [0.0,2.6]		15,299	1.16*
Gen20	const.	sameday	joint.null	253	0.56 [0.0,2.5]		0	1.02
Gen20	const.	total	real	253	7.48 [3.1,13.6]		1,785,730	1.99*
Gen20	const.	total	marg.null	253	8.26 [4.6,13.2]		1,953,568	1.26*
Gen20	const.	total	joint.null	253	8.83 [5.2,13.6]		2,071,848	1.06*
Gen20	dong	early	real	3480	0.54 [0.2,1.0]		94,990	1.69*
Gen20	dong	early	marg.null	3480	0.56 [0.3,1.0]		121,719	1.86*
Gen20	dong	early	joint.null	3480	0.93 [0.4,1.4]		200,713	1.78*
Gen20	dong	pooled	real	3480	6.09 [5.0,7.3]		808,907	1.86*
Gen20	dong	pooled	marg.null	3480	6.86 [5.9,8.2]		973,634	1.80*
Gen20	dong	pooled	joint.null	3480	6.63 [5.7,7.7]		1,485,049	1.32*
Gen20	dong	sameday	real	3480	0.08 [0.0,0.3]		2,417	1.43*
Gen20	dong	sameday	marg.null	3480	0.13 [0.0,0.4]		16,188	1.44*
Gen20	dong	sameday	joint.null	3480	0.18 [0.0,0.4]		29,213	1.14*
Gen24	const.	early	real	246	1.20 [0.0,4.4]		134,868	1.90*
Gen24	const.	early	marg.null	246	1.88 [0.3,4.8]		225,726	1.15*
Gen24	const.	early	joint.null	246	1.94 [0.2,4.9]		217,380	1.25*
Gen24	const.	early-out	real	246	0.75 [0.0,2.9]		34,451	1.16*
Gen24	const.	early-out	marg.null	246	1.21 [0.0,4.6]		86,961	2.16*
Gen24	const.	early-out	joint.null	246	0.94 [0.0,3.3]		40,666	1.21*
Gen24	const.	early-total	real	246	0.64 [0.0,2.6]		22,521	1.17*
Gen24	const.	early-total	marg.null	246	0.75 [0.0,2.9]		41,080	1.15*
Gen24	const.	early-total	joint.null	246	0.95 [0.0,3.3]		62,043	1.25*
Gen24	const.	sameday	real	246	0.73 [0.0,2.9]		33,286	1.17*
Gen24	const.	sameday	marg.null	246	0.65 [0.0,2.6]		16,376	1.17*
Gen24	const.	sameday	joint.null	246	0.74 [0.0,3.0]		33,862	1.17*
Gen24	const.	total	real	246	10.17 [3.4,17.6]		2,082,252	2.87*
Gen24	const.	total	marg.null	246	12.80 [7.4,19.7]		2,837,318	1.69*
Gen24	const.	total	joint.null	246	12.79 [7.2,19.9]		2,658,190	1.96*
Gen24	dong	early	real	2739	0.65 [0.2,2.1]		113,371	4.82*

Table 6 – continued

Elec.	Level	Channel	Source	N	Fraud share [95% CI]	F_w	$\hat{R}$
Gen24	dong	early	marg.null	2739	0.81 [0.3,2.4]	125,165	4.98*
Gen24	dong	early	joint.null	2739	0.65 [0.2,2.0]	134,756	4.34*
Gen24	dong	pooled	real	2739	6.39 [5.4,7.6]	874,482	1.28*
Gen24	dong	pooled	marg.null	2739	6.83 [5.4,8.3]	1,074,383	2.12*
Gen24	dong	pooled	joint.null	2739	6.12 [5.1,7.2]	1,231,961	1.27*
Gen24	dong	sameday	real	2739	0.16 [0.0,0.5]	10,504	1.91*
Gen24	dong	sameday	marg.null	2739	0.21 [0.0,0.5]	19,936	1.13*
Gen24	dong	sameday	joint.null	2739	0.19 [0.0,0.5]	25,514	1.40*

Table 7: Short-chain configuration, Democratic-coded leader (n.iter=1000, burn.in=300, n.adapt=1000). Full coverage, all 11 elections.

Elec.	Level	Channel	Source	N	Fraud share [95% CI]	%	F_w	$\hat{R}$
Gen08	const.	total-no-early	real	198	0.73 [0.0,3.0]		0	1.01
Gen08	const.	total-no-early	marg.null	198	0.72 [0.0,3.2]		0	1.01
Gen08	const.	total-no-early	joint.null	198	0.90 [0.0,3.7]		44,182	1.47*
Gen08	dong	total-no-early	real	2733	0.52 [0.3,0.9]		19,150	1.11*
Gen08	dong	total-no-early	marg.null	2733	0.66 [0.3,1.1]		27,474	1.54*
Gen08	dong	total-no-early	joint.null	2733	0.33 [0.1,0.7]		24,658	1.36*
Gen12	const.	total-no-early	real	209	0.63 [0.0,2.8]		0	1.03
Gen12	const.	total-no-early	marg.null	209	0.67 [0.0,2.9]		0	1.00
Gen12	const.	total-no-early	joint.null	209	0.65 [0.0,2.9]		0	1.00
Gen12	dong	total-no-early	real	2923	0.59 [0.3,1.1]		28,128	1.70*
Gen12	dong	total-no-early	marg.null	2923	0.64 [0.4,1.0]		47,601	1.10*
Gen12	dong	total-no-early	joint.null	2923	0.38 [0.1,0.7]		27,046	1.15*
Pres02	const.	total-no-early	real	244	13.98 [8.1,20.6]		2,356,322	1.72*
Pres02	const.	total-no-early	marg.null	244	15.83 [8.7,22.9]		2,724,799	1.79*
Pres02	const.	total-no-early	joint.null	244	12.03 [5.7,18.1]		1,771,908	1.87*
Pres02	dong	total-no-early	real	3515	15.04 [9.7,16.8]		2,270,446	5.64*
Pres02	dong	total-no-early	marg.null	3515	14.52 [9.5,16.8]		2,336,821	5.85*
Pres02	dong	total-no-early	joint.null	3515	14.94 [13.6,16.5]		2,370,727	1.34*
Pres07	const.	total-no-early	real	248	6.96 [3.3,11.6]		1,073,737	1.28*
Pres07	const.	total-no-early	marg.null	248	7.42 [2.4,16.2]		1,207,574	3.07*
Pres07	const.	total-no-early	joint.null	248	7.14 [2.7,13.3]		1,016,127	2.14*
Pres07	dong	total-no-early	real	3576	8.51 [6.0,11.1]		1,040,736	4.62*
Pres07	dong	total-no-early	marg.null	3576	7.08 [5.8,12.6]		1,041,303	7.63*
Pres07	dong	total-no-early	joint.null	3576	6.83 [5.9,7.8]		928,204	1.17*
Pres12	const.	total-no-early	real	251	13.20 [7.9,24.5]		2,962,604	2.79*
Pres12	const.	total-no-early	marg.null	251	17.51 [7.2,25.9]		4,011,570	3.20*

Table 7 – continued

Elec.	Level	Channel	Source	N	Fraud share % [95% CI]	F_w	$\hat{R}$
Pres12	const.	total-no-early	joint.null	251	18.89 [13.2,24.9]	4,303,202	1.19*
Pres12	dong	total-no-early	real	3479	16.85 [12.4,19.1]	3,669,420	4.82*
Pres12	dong	total-no-early	marg.null	3479	15.40 [10.8,19.3]	3,561,176	7.52*
Pres12	dong	total-no-early	joint.null	3479	17.66 [15.9,19.4]	3,239,645	1.97*
Gen16	const.	early	real	234	1.12 [0.0,3.8]	63,334	1.09*
Gen16	const.	early	marg.null	234	0.76 [0.0,3.5]	45,634	1.91*
Gen16	const.	early	joint.null	234	1.35 [0.0,4.4]	96,605	1.65*
Gen16	const.	early-out	real	234	0.68 [0.0,2.8]	11,580	1.14*
Gen16	const.	early-out	marg.null	234	0.74 [0.0,2.8]	24,807	1.19*
Gen16	const.	early-out	joint.null	234	0.58 [0.0,2.4]	0	1.01
Gen16	const.	early-total	real	234	1.94 [0.0,5.2]	134,436	1.73*
Gen16	const.	early-total	marg.null	234	1.63 [0.0,5.3]	136,883	2.09*
Gen16	const.	early-total	joint.null	234	2.01 [0.0,8.0]	254,492	3.35*
Gen16	const.	sameday	real	234	0.60 [0.0,2.5]	0	1.01
Gen16	const.	sameday	marg.null	234	0.59 [0.0,2.6]	0	1.01
Gen16	const.	sameday	joint.null	234	0.60 [0.0,2.6]	0	1.06*
Gen16	const.	total	real	234	0.64 [0.0,2.8]	34,560	1.18*
Gen16	const.	total	marg.null	234	1.43 [0.1,4.0]	202,723	1.17*
Gen16	const.	total	joint.null	234	0.75 [0.0,3.1]	35,514	1.43*
Gen16	dong	early	real	3135	0.42 [0.1,0.8]	79,592	1.78*
Gen16	dong	early	marg.null	3135	0.50 [0.2,1.7]	101,762	4.51*
Gen16	dong	early	joint.null	3135	0.40 [0.1,1.3]	95,765	3.89*
Gen16	dong	pooled	real	3135	0.34 [0.1,0.7]	58,867	1.39*
Gen16	dong	pooled	marg.null	3135	0.47 [0.2,0.9]	64,867	1.76*
Gen16	dong	pooled	joint.null	3135	0.26 [0.1,0.6]	43,132	1.43*
Gen16	dong	sameday	real	3135	0.04 [0.0,0.2]	0	1.02
Gen16	dong	sameday	marg.null	3135	0.05 [0.0,0.2]	4,247	1.17*
Gen16	dong	sameday	joint.null	3135	0.04 [0.0,0.2]	0	1.03
Pres17	const.	early	real	250	2.48 [0.5,5.8]	414,680	1.24*
Pres17	const.	early	marg.null	250	3.89 [0.4,7.9]	450,458	1.83*
Pres17	const.	early	joint.null	250	2.30 [0.0,6.3]	252,473	2.36*
Pres17	const.	early-out	real	250	3.74 [0.4,8.9]	569,634	2.51*
Pres17	const.	early-out	marg.null	250	3.26 [0.4,7.5]	370,573	2.17*
Pres17	const.	early-out	joint.null	250	5.46 [1.1,11.7]	649,612	2.56*
Pres17	const.	early-total	real	250	2.21 [0.0,8.5]	253,550	3.97*
Pres17	const.	early-total	marg.null	250	0.57 [0.0,2.7]	0	1.03
Pres17	const.	early-total	joint.null	250	0.69 [0.0,3.1]	31,566	1.45*
Pres17	const.	sameday	real	250	0.55 [0.0,2.4]	0	1.01
Pres17	const.	sameday	marg.null	250	0.55 [0.0,2.5]	0	1.02
Pres17	const.	sameday	joint.null	250	0.54 [0.0,2.4]	0	1.02
Pres17	const.	total	real	250	19.74 [4.3,26.9]	3,858,227	4.29*
Pres17	const.	total	marg.null	250	16.19 [5.6,26.9]	3,721,124	4.49*
Pres17	const.	total	joint.null	250	21.36 [2.6,27.6]	3,908,730	5.25*
Pres17	dong	early	real	3491	1.02 [0.5,3.0]	197,417	5.28*
Pres17	dong	early	marg.null	3491	1.33 [0.7,2.3]	189,440	3.17*
Pres17	dong	early	joint.null	3491	0.68 [0.4,1.1]	147,652	1.28*
Pres17	dong	pooled	real	3491	9.98 [4.4,11.2]	1,834,130	7.61*
Pres17	dong	pooled	marg.null	3491	10.88 [3.7,12.1]	2,024,332	10.07*
Pres17	dong	pooled	joint.null	3491	7.27 [6.0,11.4]	1,406,150	5.89*
Pres17	dong	sameday	real	3491	0.07 [0.0,0.2]	14,574	1.22*
Pres17	dong	sameday	marg.null	3491	0.09 [0.0,0.3]	25,414	1.08*

Table 7 – continued

Elec.	Level	Channel	Source	N	Fraud share [95% CI]	F_w	$\hat{R}$
Pres17	dong	sameday	joint.null	3491	0.09 [0.0,0.5]	26,718	3.72*
Gen20	const.	early	real	253	5.98 [2.4,10.7]	858,974	1.66*
Gen20	const.	early	marg.null	253	6.19 [3.0,10.3]	943,198	1.04
Gen20	const.	early	joint.null	253	5.98 [1.3,10.2]	811,257	1.99*
Gen20	const.	early-out	real	253	2.76 [0.1,6.3]	272,470	1.81*
Gen20	const.	early-out	marg.null	253	2.57 [0.1,6.0]	279,014	1.84*
Gen20	const.	early-out	joint.null	253	1.10 [0.0,3.8]	97,340	1.42*
Gen20	const.	early-total	real	253	2.62 [0.6,9.7]	507,062	3.27*
Gen20	const.	early-total	marg.null	253	2.40 [0.3,6.5]	346,243	1.90*
Gen20	const.	early-total	joint.null	253	2.73 [0.4,7.3]	400,417	2.04*
Gen20	const.	sameday	real	253	0.87 [0.0,2.9]	52,030	1.11*
Gen20	const.	sameday	marg.null	253	0.66 [0.0,2.7]	24,907	1.18*
Gen20	const.	sameday	joint.null	253	0.64 [0.0,2.7]	16,916	1.19*
Gen20	const.	total	real	253	12.11 [7.9,17.6]	2,846,787	1.14*
Gen20	const.	total	marg.null	253	12.49 [7.7,21.1]	3,189,486	2.20*
Gen20	const.	total	joint.null	253	11.57 [4.7,20.9]	2,728,151	3.07*
Gen20	dong	early	real	3480	5.69 [2.0,18.5]	1,124,745	20.05*
Gen20	dong	early	marg.null	3480	5.87 [1.9,17.3]	1,078,501	16.12*
Gen20	dong	early	joint.null	3480	2.04 [1.4,7.5]	639,191	10.99*
Gen20	dong	pooled	real	3480	4.93 [4.2,5.7]	888,479	1.03
Gen20	dong	pooled	marg.null	3480	5.49 [4.7,6.4]	1,098,328	1.19*
Gen20	dong	pooled	joint.null	3480	5.31 [4.4,6.2]	998,937	1.38*
Gen20	dong	sameday	real	3480	0.12 [0.0,0.3]	12,277	1.08*
Gen20	dong	sameday	marg.null	3480	0.14 [0.0,0.4]	12,980	1.95*
Gen20	dong	sameday	joint.null	3480	0.17 [0.0,0.4]	17,767	1.95*
Pres22	const.	early	real	250	2.74 [0.3,10.3]	389,318	3.47*
Pres22	const.	early	marg.null	250	1.99 [0.2,4.7]	94,948	1.12*
Pres22	const.	early	joint.null	250	1.45 [0.1,4.1]	164,094	1.21*
Pres22	const.	early-out	real	250	5.25 [1.6,9.5]	620,741	1.59*
Pres22	const.	early-out	marg.null	250	3.63 [0.1,8.6]	556,542	2.85*
Pres22	const.	early-out	joint.null	250	5.49 [0.0,10.5]	724,435	3.13*
Pres22	const.	early-total	real	250	5.48 [2.4,9.4]	378,381	1.13*
Pres22	const.	early-total	marg.null	250	4.59 [1.9,8.8]	508,667	1.28*
Pres22	const.	early-total	joint.null	250	5.13 [1.3,9.4]	649,470	1.60*
Pres22	const.	sameday	real	250	1.71 [0.2,4.4]	64,168	1.25*
Pres22	const.	sameday	marg.null	250	1.29 [0.0,4.1]	62,967	1.49*
Pres22	const.	sameday	joint.null	250	2.23 [0.5,5.3]	135,547	1.17*
Pres22	const.	total	real	248	21.39 [8.7,38.1]	5,306,355	4.56*
Pres22	const.	total	marg.null	248	21.88 [10.8,35.9]	6,010,504	3.40*
Pres22	const.	total	joint.null	248	27.96 [17.0,38.5]	7,349,496	2.49*
Pres22	dong	early	real	3509	2.77 [1.9,11.0]	450,749	15.28*
Pres22	dong	early	marg.null	3509	2.98 [2.2,10.0]	501,169	12.78*
Pres22	dong	early	joint.null	3509	2.23 [1.5,9.1]	527,051	13.40*
Pres22	dong	pooled	real	3509	13.82 [9.5,20.5]	2,874,517	11.18*
Pres22	dong	pooled	marg.null	3509	16.58 [11.0,20.3]	3,245,128	8.30*
Pres22	dong	pooled	joint.null	3509	14.21 [7.4,19.8]	2,350,082	12.67*
Pres22	dong	sameday	real	3509	0.50 [0.2,0.8]	51,731	1.20*
Pres22	dong	sameday	marg.null	3509	0.65 [0.4,1.1]	64,089	1.42*
Pres22	dong	sameday	joint.null	3509	0.65 [0.3,1.1]	106,122	1.82*
Gen24	const.	early	real	246	4.36 [1.5,8.4]	642,413	1.39*
Gen24	const.	early	marg.null	246	3.95 [1.2,11.7]	675,939	3.00*
Gen24	const.	early	joint.null	246	3.94 [1.3,8.1]	606,590	1.33*
Gen24	const.	early-out	real	246	2.50 [0.0,10.3]	433,760	4.09*
Gen24	const.	early-out	marg.null	246	4.67 [0.2,10.0]	561,919	2.88*
Gen24	const.	early-out	joint.null	246	1.25 [0.1,3.5]	96,448	1.06*

Table 7 – continued

Elec.	Level	Channel	Source	N	Fraud share % [95% CI]	F_w	$\hat{R}$
Gen24	const.	early-total	real	246	0.92 [0.0,3.4]	81,441	1.47*
Gen24	const.	early-total	marg.null	246	1.47 [0.0,4.4]	185,961	1.78*
Gen24	const.	early-total	joint.null	246	0.64 [0.0,2.7]	23,464	1.22*
Gen24	const.	sameday	real	246	1.07 [0.0,3.3]	83,584	1.11*
Gen24	const.	sameday	marg.null	246	1.33 [0.0,4.2]	127,246	1.65*
Gen24	const.	sameday	joint.null	246	1.04 [0.0,3.5]	83,314	1.30*
Gen24	const.	total	real	246	17.42 [11.3,24.7]	4,203,065	1.56*
Gen24	const.	total	marg.null	246	16.41 [10.3,23.5]	3,854,885	1.60*
Gen24	const.	total	joint.null	246	9.88 [4.9,22.6]	2,636,042	3.65*
Gen24	dong	early	real	2739	2.65 [1.3,10.4]	578,511	13.49*
Gen24	dong	early	marg.null	2739	4.02 [1.3,11.5]	669,397	14.44*
Gen24	dong	early	joint.null	2739	1.84 [1.2,2.7]	344,182	1.84*
Gen24	dong	pooled	real	2739	6.83 [5.9,8.0]	1,150,452	1.13*
Gen24	dong	pooled	marg.null	2739	7.56 [6.4,9.0]	1,294,177	1.50*
Gen24	dong	pooled	joint.null	2739	7.13 [5.9,9.1]	1,299,479	2.23*
Gen24	dong	sameday	real	2739	0.55 [0.3,1.0]	48,479	1.18*
Gen24	dong	sameday	marg.null	2739	0.64 [0.2,1.1]	72,724	1.42*
Gen24	dong	sameday	joint.null	2739	0.72 [0.3,1.2]	79,666	1.55*
Pres25	const.	early	real	253	4.14 [1.5,7.7]	830,750	1.22*
Pres25	const.	early	marg.null	253	5.33 [1.5,10.0]	922,708	1.68*
Pres25	const.	early	joint.null	253	5.42 [2.5,9.4]	1,045,889	1.16*
Pres25	const.	early-out	real	253	4.14 [0.2,10.7]	532,457	3.84*
Pres25	const.	early-out	marg.null	253	5.72 [0.2,11.3]	725,987	3.49*
Pres25	const.	early-out	joint.null	253	6.44 [1.0,16.9]	1,104,769	4.46*
Pres25	const.	early-total	real	253	8.19 [3.7,13.1]	891,451	1.56*
Pres25	const.	early-total	marg.null	253	9.70 [5.4,15.0]	1,423,472	1.35*
Pres25	const.	early-total	joint.null	253	8.39 [3.8,14.0]	1,112,902	1.78*
Pres25	const.	sameday	real	253	2.76 [0.6,6.2]	133,808	1.26*
Pres25	const.	sameday	marg.null	253	1.84 [0.2,4.6]	87,877	1.15*
Pres25	const.	sameday	joint.null	253	1.29 [0.0,3.9]	106,973	1.29*
Pres25	const.	total	real	242	32.24 [24.1,41.9]	8,957,677	1.24*
Pres25	const.	total	marg.null	242	28.71 [8.1,40.3]	7,340,668	4.89*
Pres25	const.	total	joint.null	242	27.45 [16.5,39.5]	7,598,564	2.54*
Pres25	dong	early	real	3553	10.60 [5.1,16.5]	1,279,220	10.31*
Pres25	dong	early	marg.null	3553	8.15 [6.3,9.6]	982,100	2.28*
Pres25	dong	early	joint.null	3553	6.50 [3.6,16.9]	1,029,625	14.75*
Pres25	dong	pooled	real	3553	10.84 [7.8,27.8]	2,591,130	15.34*
Pres25	dong	pooled	marg.null	3553	20.64 [8.5,23.5]	3,583,441	12.73*
Pres25	dong	pooled	joint.null	3553	21.82 [11.1,23.8]	3,602,274	10.67*
Pres25	dong	sameday	real	3553	0.50 [0.2,0.8]	40,749	1.18*
Pres25	dong	sameday	marg.null	3553	0.55 [0.3,0.9]	50,457	1.55*
Pres25	dong	sameday	joint.null	3553	0.62 [0.3,1.0]	90,664	1.26*

Table 8: Short-chain configuration, conservative-coded leader. Full coverage, all 11 elections.

Elec.	Level	Channel	Source	N	Fraud share [95% CI]	F_w	$\hat{R}$
Gen08	const.	total-no-early	real	198	0.70 [0.0,3.0]	0	1.01
Gen08	const.	total-no-early	marg.null	198	0.69 [0.0,2.9]	0	1.02
Gen08	const.	total-no-early	joint.null	198	0.74 [0.0,3.1]	0	1.01
Gen08	dong	total-no-early	real	2733	0.23 [0.1,0.5]	8,727	1.09*
Gen08	dong	total-no-early	marg.null	2733	0.30 [0.1,0.6]	13,432	1.28*
Gen08	dong	total-no-early	joint.null	2733	0.18 [0.0,0.5]	24,098	1.87*
Gen12	const.	total-no-early	real	194	1.09 [0.0,4.0]	55,302	1.11*
Gen12	const.	total-no-early	marg.null	194	0.97 [0.0,3.6]	33,186	1.16*
Gen12	const.	total-no-early	joint.null	194	0.93 [0.0,5.6]	95,704	2.60*
Gen12	dong	total-no-early	real	2656	1.66 [1.0,2.4]	78,163	1.65*
Gen12	dong	total-no-early	marg.null	2656	1.92 [1.3,2.7]	112,911	1.46*
Gen12	dong	total-no-early	joint.null	2656	1.40 [0.9,2.0]	199,036	1.04
Pres02	const.	total-no-early	real	244	11.95 [6.8,18.3]	2,441,174	1.67*
Pres02	const.	total-no-early	marg.null	244	13.28 [9.0,18.7]	2,688,154	1.08*
Pres02	const.	total-no-early	joint.null	244	10.54 [5.3,17.2]	2,142,791	2.06*
Pres02	dong	total-no-early	real	3515	14.14 [12.8,15.5]	2,448,858	1.23*
Pres02	dong	total-no-early	marg.null	3515	15.31 [13.4,16.9]	2,838,928	2.11*
Pres02	dong	total-no-early	joint.null	3515	13.91 [12.6,15.5]	2,629,329	1.43*
Pres07	const.	total-no-early	real	248	5.82 [2.9,10.1]	768,866	1.11*
Pres07	const.	total-no-early	marg.null	248	5.86 [2.2,10.2]	1,002,539	1.38*
Pres07	const.	total-no-early	joint.null	248	5.12 [1.7,9.8]	781,383	1.67*
Pres07	dong	total-no-early	real	3576	6.63 [5.6,7.7]	1,092,182	1.57*
Pres07	dong	total-no-early	marg.null	3576	6.93 [5.9,8.0]	1,302,605	1.54*
Pres07	dong	total-no-early	joint.null	3576	6.46 [5.4,7.5]	1,237,504	1.50*
Pres12	const.	total-no-early	real	251	21.72 [8.8,34.0]	4,973,235	5.03*
Pres12	const.	total-no-early	marg.null	251	17.22 [10.6,27.0]	4,391,244	2.14*

Table 8 – continued

Elec.	Level	Channel	Source	N	Fraud share % [95% CI]	F_w	$\hat{R}$
Pres12	const.	total-no-early	joint.null	251	21.70 [7.7,29.9]	4,765,314	3.90*
Pres12	dong	total-no-early	real	3479	25.95 [19.8,28.0]	4,804,954	5.21*
Pres12	dong	total-no-early	marg.null	3479	25.91 [24.1,27.6]	5,215,066	1.51*
Pres12	dong	total-no-early	joint.null	3479	25.53 [22.9,29.3]	5,646,767	3.52*
Gen16	const.	early	real	229	1.24 [0.0,4.3]	106,257	1.72*
Gen16	const.	early	marg.null	229	1.75 [0.0,7.8]	215,175	3.09*
Gen16	const.	early	joint.null	229	0.69 [0.0,2.9]	15,516	1.17*
Gen16	const.	early-out	real	229	0.58 [0.0,2.5]	0	1.00
Gen16	const.	early-out	marg.null	229	0.59 [0.0,2.7]	0	1.03
Gen16	const.	early-out	joint.null	229	0.60 [0.0,2.6]	0	1.01
Gen16	const.	early-total	real	229	2.27 [0.0,5.5]	165,325	1.81*
Gen16	const.	early-total	marg.null	229	1.25 [0.0,4.3]	102,491	1.75*
Gen16	const.	early-total	joint.null	229	1.95 [0.1,5.5]	185,154	1.76*
Gen16	const.	sameday	real	229	0.61 [0.0,2.6]	0	1.00
Gen16	const.	sameday	marg.null	229	0.59 [0.0,2.6]	0	1.01
Gen16	const.	sameday	joint.null	229	0.62 [0.0,2.6]	0	1.00
Gen16	const.	total	real	229	1.07 [0.0,3.7]	110,307	1.18*
Gen16	const.	total	marg.null	229	0.72 [0.0,3.1]	30,744	1.09*
Gen16	const.	total	joint.null	229	0.60 [0.0,2.6]	0	1.01
Gen16	dong	early	real	3058	0.73 [0.2,1.5]	46,853	3.57*
Gen16	dong	early	marg.null	3058	1.16 [0.4,3.5]	106,326	6.75*
Gen16	dong	early	joint.null	3058	0.53 [0.2,1.0]	72,045	1.51*
Gen16	dong	pooled	real	3058	1.51 [1.0,2.1]	63,948	1.67*
Gen16	dong	pooled	marg.null	3058	1.63 [1.2,2.2]	87,938	1.21*
Gen16	dong	pooled	joint.null	3058	1.20 [0.6,1.9]	145,797	2.35*
Gen16	dong	sameday	real	3058	0.09 [0.0,0.3]	1,688	1.07*
Gen16	dong	sameday	marg.null	3058	0.23 [0.0,0.5]	5,510	1.82*
Gen16	dong	sameday	joint.null	3058	0.12 [0.0,0.4]	8,318	1.34*
Pres17	const.	early	real	250	1.05 [0.0,4.5]	26,346	2.21*
Pres17	const.	early	marg.null	250	0.92 [0.0,7.1]	93,624	3.74*
Pres17	const.	early	joint.null	250	1.06 [0.0,3.3]	41,471	1.27*
Pres17	const.	early-out	real	250	0.56 [0.0,2.5]	0	1.01
Pres17	const.	early-out	marg.null	250	0.72 [0.0,2.8]	12,340	1.17*
Pres17	const.	early-out	joint.null	250	1.09 [0.0,3.8]	20,931	1.73*
Pres17	const.	early-total	real	250	0.57 [0.0,2.4]	0	1.01
Pres17	const.	early-total	marg.null	250	0.71 [0.0,2.8]	6,597	1.17*
Pres17	const.	early-total	joint.null	250	0.69 [0.0,3.5]	19,512	1.92*
Pres17	const.	sameday	real	250	0.60 [0.0,2.5]	0	1.02
Pres17	const.	sameday	marg.null	250	0.71 [0.0,4.4]	120,430	2.61*
Pres17	const.	sameday	joint.null	250	0.56 [0.0,2.5]	0	1.01
Pres17	const.	total	real	250	15.01 [9.4,21.7]	2,758,946	1.62*
Pres17	const.	total	marg.null	250	12.75 [2.5,24.3]	2,301,644	4.76*
Pres17	const.	total	joint.null	250	13.01 [4.4,22.7]	2,858,246	3.62*
Pres17	dong	early	real	3491	0.33 [0.0,1.0]	22,437	4.13*
Pres17	dong	early	marg.null	3491	0.14 [0.0,1.7]	22,591	8.48*
Pres17	dong	early	joint.null	3491	0.53 [0.1,1.1]	55,245	3.31*
Pres17	dong	pooled	real	3491	9.07 [2.3,17.1]	1,125,726	16.71*
Pres17	dong	pooled	marg.null	3491	8.12 [3.9,11.6]	1,118,084	7.95*
Pres17	dong	pooled	joint.null	3491	4.52 [2.5,14.0]	1,154,015	16.06*
Pres17	dong	sameday	real	3491	0.11 [0.0,5.5]	141,004	18.12*
Pres17	dong	sameday	marg.null	3491	3.02 [0.0,5.2]	249,949	11.17*

Table 8 – continued

Elec.	Level	Channel	Source	N	Fraud share [95% CI]	F_w	$\hat{R}$
Pres17	dong	sameday	joint.null	3491	0.12 [0.0,2.7]	75,580	11.69*
Gen20	const.	early	real	253	1.18 [0.0,5.6]	167,831	2.89*
Gen20	const.	early	marg.null	253	2.67 [0.4,6.3]	314,211	1.59*
Gen20	const.	early	joint.null	253	1.72 [0.1,4.6]	170,204	1.37*
Gen20	const.	early-out	real	253	0.60 [0.0,2.5]	10,016	1.17*
Gen20	const.	early-out	marg.null	253	0.55 [0.0,2.4]	0	1.00
Gen20	const.	early-out	joint.null	253	0.94 [0.0,3.3]	48,500	1.26*
Gen20	const.	early-total	real	253	0.64 [0.0,2.7]	24,011	1.12*
Gen20	const.	early-total	marg.null	253	1.11 [0.0,3.5]	104,062	1.24*
Gen20	const.	early-total	joint.null	253	1.34 [0.0,3.9]	132,876	1.43*
Gen20	const.	sameday	real	253	0.55 [0.0,2.5]	0	1.01
Gen20	const.	sameday	marg.null	253	0.56 [0.0,2.4]	0	1.00
Gen20	const.	sameday	joint.null	253	0.57 [0.0,2.5]	0	1.00
Gen20	const.	total	real	253	8.22 [4.3,14.5]	2,134,650	1.79*
Gen20	const.	total	marg.null	253	9.00 [4.7,14.8]	2,081,116	1.55*
Gen20	const.	total	joint.null	253	6.41 [3.2,10.8]	1,459,665	1.23*
Gen20	dong	early	real	3480	1.39 [0.3,3.8]	126,636	9.10*
Gen20	dong	early	marg.null	3480	1.27 [0.2,4.2]	117,028	9.69*
Gen20	dong	early	joint.null	3480	1.35 [0.8,3.5]	279,526	5.89*
Gen20	dong	pooled	real	3480	6.15 [5.3,7.0]	851,223	1.02
Gen20	dong	pooled	marg.null	3480	6.37 [5.1,7.8]	937,443	2.28*
Gen20	dong	pooled	joint.null	3480	7.09 [6.2,8.0]	1,658,790	1.19*
Gen20	dong	sameday	real	3480	0.09 [0.0,0.3]	6,345	1.62*
Gen20	dong	sameday	marg.null	3480	0.15 [0.0,0.4]	14,585	1.44*
Gen20	dong	sameday	joint.null	3480	0.20 [0.0,0.4]	39,513	1.32*
Pres22	const.	early	real	250	1.63 [0.0,4.6]	211,797	1.73*
Pres22	const.	early	marg.null	250	1.51 [0.0,6.8]	228,330	2.93*
Pres22	const.	early	joint.null	250	0.98 [0.0,3.2]	105,586	1.11*
Pres22	const.	early-out	real	250	2.11 [0.1,5.9]	228,294	2.00*
Pres22	const.	early-out	marg.null	250	2.21 [0.0,5.8]	139,713	2.09*
Pres22	const.	early-out	joint.null	250	5.49 [0.2,10.3]	560,311	2.75*
Pres22	const.	early-total	real	250	0.98 [0.0,3.6]	13,227	1.57*
Pres22	const.	early-total	marg.null	250	1.27 [0.0,3.8]	34,410	1.50*
Pres22	const.	early-total	joint.null	250	0.63 [0.0,2.5]	19,412	1.17*
Pres22	const.	sameday	real	250	0.92 [0.0,3.2]	26,056	1.28*
Pres22	const.	sameday	marg.null	250	1.36 [0.0,3.8]	130,900	1.17*
Pres22	const.	sameday	joint.null	250	0.56 [0.0,2.6]	0	1.01
Pres22	const.	total	real	248	24.44 [9.8,40.1]	5,248,392	5.89*
Pres22	const.	total	marg.null	248	24.13 [10.0,37.7]	6,125,868	4.56*
Pres22	const.	total	joint.null	248	28.89 [20.9,39.7]	7,072,095	1.92*
Pres22	dong	early	real	3509	0.58 [0.3,1.0]	119,758	1.48*
Pres22	dong	early	marg.null	3509	0.69 [0.3,1.1]	118,375	1.70*
Pres22	dong	early	joint.null	3509	0.43 [0.2,0.8]	88,901	1.60*
Pres22	dong	pooled	real	3509	22.27 [20.4,23.9]	4,010,068	1.48*
Pres22	dong	pooled	marg.null	3509	22.21 [20.5,24.2]	4,201,595	1.57*
Pres22	dong	pooled	joint.null	3509	21.82 [13.3,23.8]	4,325,166	8.13*
Pres22	dong	sameday	real	3509	0.20 [0.1,0.4]	13,175	1.04
Pres22	dong	sameday	marg.null	3509	0.24 [0.0,0.5]	31,524	1.68*
Pres22	dong	sameday	joint.null	3509	0.41 [0.1,0.8]	101,112	1.92*
Gen24	const.	early	real	246	1.13 [0.1,3.5]	97,128	1.09*
Gen24	const.	early	marg.null	246	1.91 [0.0,7.9]	264,800	3.42*
Gen24	const.	early	joint.null	246	1.04 [0.0,3.7]	94,728	1.48*
Gen24	const.	early-out	real	246	1.09 [0.0,5.5]	102,493	2.77*
Gen24	const.	early-out	marg.null	246	1.03 [0.0,4.0]	59,210	1.91*
Gen24	const.	early-out	joint.null	246	1.99 [0.2,6.4]	197,829	2.24*

Table 8 – continued

Elec.	Level	Channel	Source	N	Fraud share [95% CI]	F_w	$\hat{R}$
Gen24	const.	early-total	real	246	0.72 [0.0,3.1]	41,697	1.59*
Gen24	const.	early-total	marg.null	246	0.68 [0.0,2.6]	20,395	1.12*
Gen24	const.	early-total	joint.null	246	0.66 [0.0,2.8]	20,138	1.18*
Gen24	const.	sameday	real	246	0.93 [0.0,4.6]	152,384	2.36*
Gen24	const.	sameday	marg.null	246	0.77 [0.0,3.1]	35,780	1.13*
Gen24	const.	sameday	joint.null	246	0.59 [0.0,2.5]	0	1.00
Gen24	const.	total	real	246	11.79 [4.9,18.3]	2,606,558	2.22*
Gen24	const.	total	marg.null	246	9.52 [5.6,14.5]	2,119,617	1.16*
Gen24	const.	total	joint.null	246	8.44 [3.8,18.2]	2,148,107	2.79*
Gen24	dong	early	real	2739	0.60 [0.3,1.0]	110,100	1.14*
Gen24	dong	early	marg.null	2739	0.77 [0.3,4.4]	190,614	9.79*
Gen24	dong	early	joint.null	2739	0.80 [0.2,5.4]	263,831	11.87*
Gen24	dong	pooled	real	2739	6.28 [5.2,7.5]	943,255	1.31*
Gen24	dong	pooled	marg.null	2739	7.15 [6.2,8.3]	1,222,421	1.14*
Gen24	dong	pooled	joint.null	2739	6.16 [5.1,7.4]	1,267,380	1.42*
Gen24	dong	sameday	real	2739	0.17 [0.0,0.4]	10,222	1.31*
Gen24	dong	sameday	marg.null	2739	0.27 [0.1,0.6]	23,370	1.49*
Gen24	dong	sameday	joint.null	2739	0.24 [0.0,0.5]	35,956	1.49*
Pres25	const.	early	real	253	0.96 [0.0,3.4]	74,687	1.51*
Pres25	const.	early	marg.null	253	1.53 [0.0,5.5]	108,920	2.53*
Pres25	const.	early	joint.null	253	0.75 [0.0,3.0]	14,512	1.43*
Pres25	const.	early-out	real	253	1.12 [0.0,3.7]	42,579	1.46*
Pres25	const.	early-out	marg.null	253	0.96 [0.0,3.5]	19,711	1.70*
Pres25	const.	early-out	joint.null	253	1.25 [0.1,3.6]	69,115	1.06*
Pres25	const.	early-total	real	253	0.53 [0.0,2.3]	0	1.01
Pres25	const.	early-total	marg.null	253	0.68 [0.0,4.7]	25,333	2.85*
Pres25	const.	early-total	joint.null	253	0.54 [0.0,2.3]	0	1.01
Pres25	const.	sameday	real	253	0.85 [0.0,3.1]	88,985	1.33*
Pres25	const.	sameday	marg.null	253	0.99 [0.0,4.0]	215,466	1.86*
Pres25	const.	sameday	joint.null	253	0.89 [0.0,3.2]	124,514	1.47*
Pres25	const.	total	real	242	29.05 [7.6,36.7]	5,378,922	5.23*
Pres25	const.	total	marg.null	242	22.61 [4.5,35.4]	4,542,425	5.65*
Pres25	const.	total	joint.null	242	23.43 [16.4,35.2]	5,926,735	2.40*
Pres25	dong	early	real	3553	0.24 [0.1,0.5]	40,390	1.60*
Pres25	dong	early	marg.null	3553	0.28 [0.0,1.0]	46,700	3.98*
Pres25	dong	early	joint.null	3553	0.22 [0.1,0.5]	33,564	1.41*
Pres25	dong	pooled	real	3553	7.28 [5.8,21.4]	1,742,480	18.19*
Pres25	dong	pooled	marg.null	3553	19.97 [10.2,21.7]	3,397,576	10.20*
Pres25	dong	pooled	joint.null	3553	19.23 [17.8,20.6]	3,912,520	1.05*
Pres25	dong	sameday	real	3553	0.72 [0.3,1.2]	209,630	1.95*
Pres25	dong	sameday	marg.null	3553	1.14 [0.7,1.6]	297,481	1.54*
Pres25	dong	sameday	joint.null	3553	0.82 [0.4,1.3]	188,867	1.81*

A.1 Mixture-component probabilities

Tables 9–12 report the three underlying mixture-component probabilities, no-fraud (π_1), incremental fraud (π_2), and extreme fraud (π_3), with their individual 95% highest posterior density intervals, for the same 576 runs listed in Tables 5–8. These three probabilities sum to 1 for every row up to rounding; the fraud share reported throughout the main text is $\pi_2 + \pi_3$, and the conservative combined interval given in Tables 5–8 is the sum of the π_2 and π_3 interval bounds shown here.

Table 9: Mixture-component probabilities, long-chain configuration, Democratic-coded leader. $p_{\text{no-fraud}} + p_{\text{incr.}} + p_{\text{extr.}} = 1$ for every row up to rounding; fraud share as reported elsewhere in this paper is $p_{\text{incr.}} + p_{\text{extr.}}$.

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen08	const.	total-no-early	real	99.17 [97.7,100.0]	0.35 [0.0,1.5]	0.35 [0.0,1.5]
Gen08	const.	total-no-early	marg.null	99.13 [97.6,100.0]	0.35 [0.0,1.5]	0.36 [0.0,1.5]
Gen08	const.	total-no-early	joint.null	99.15 [97.6,100.0]	0.35 [0.0,1.5]	0.35 [0.0,1.5]
Gen08	dong	total-no-early	real	99.53 [99.2,99.8]	0.43 [0.2,0.7]	0.03 [0.0,0.1]
Gen08	dong	total-no-early	marg.null	99.23 [98.8,99.6]	0.74 [0.4,1.1]	0.03 [0.0,0.1]
Gen08	dong	total-no-early	joint.null	99.59 [99.3,99.8]	0.37 [0.1,0.7]	0.03 [0.0,0.1]
Gen12	const.	total-no-early	real	99.21 [97.8,100.0]	0.33 [0.0,1.4]	0.33 [0.0,1.4]
Gen12	const.	total-no-early	marg.null	99.20 [97.7,100.0]	0.34 [0.0,1.4]	0.33 [0.0,1.4]
Gen12	const.	total-no-early	joint.null	99.21 [97.8,100.0]	0.32 [0.0,1.4]	0.33 [0.0,1.4]
Gen12	dong	total-no-early	real	99.45 [99.0,99.8]	0.51 [0.1,0.9]	0.02 [0.0,0.1]
Gen12	dong	total-no-early	marg.null	99.28 [98.9,99.6]	0.69 [0.4,1.0]	0.02 [0.0,0.1]
Gen12	dong	total-no-early	joint.null	99.70 [99.4,100.0]	0.26 [0.0,0.5]	0.02 [0.0,0.1]
Pres02	const.	total-no-early	real	87.41 [82.2,92.2]	12.19 [7.4,17.3]	0.28 [0.0,1.2]
Pres02	const.	total-no-early	marg.null	85.41 [77.7,92.5]	13.94 [7.0,21.5]	0.47 [0.0,1.7]
Pres02	const.	total-no-early	joint.null	84.55 [78.9,89.3]	15.03 [10.1,20.4]	0.28 [0.0,1.2]
Pres02	dong	total-no-early	real	85.48 [83.1,88.8]	14.46 [11.2,16.7]	0.07 [0.0,0.2]
Pres02	dong	total-no-early	marg.null	84.98 [83.5,86.5]	14.80 [13.3,16.2]	0.21 [0.1,0.4]
Pres02	dong	total-no-early	joint.null	87.53 [85.6,89.2]	12.40 [10.7,14.3]	0.06 [0.0,0.2]
Pres07	const.	total-no-early	real	91.67 [86.9,95.7]	7.92 [3.9,12.6]	0.28 [0.0,1.1]
Pres07	const.	total-no-early	marg.null	92.50 [86.8,97.9]	7.08 [1.8,12.8]	0.28 [0.0,1.2]
Pres07	const.	total-no-early	joint.null	92.78 [83.7,97.8]	6.78 [2.0,15.9]	0.28 [0.0,1.2]
Pres07	dong	total-no-early	real	92.76 [90.9,93.9]	7.22 [6.1,9.0]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	marg.null	93.64 [87.9,94.8]	6.33 [5.2,12.1]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	joint.null	92.94 [91.7,94.1]	7.03 [5.9,8.3]	0.02 [0.0,0.1]
Pres12	const.	total-no-early	real	82.53 [76.2,89.1]	16.80 [10.5,23.1]	0.46 [0.0,1.7]
Pres12	const.	total-no-early	marg.null	80.19 [72.0,87.6]	19.42 [11.9,27.5]	0.28 [0.0,1.2]

Table 9 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres12	const.	total-no-early	joint.null	79.59 [72.6,85.5]	19.99 [14.1,26.9]	0.29 [0.0,1.2]
Pres12	dong	total-no-early	real	83.28 [81.6,87.8]	16.58 [12.0,18.2]	0.14 [0.0,0.3]
Pres12	dong	total-no-early	marg.null	82.49 [80.6,84.3]	17.32 [15.6,19.3]	0.16 [0.0,0.4]
Pres12	dong	total-no-early	joint.null	82.65 [81.2,84.1]	17.28 [15.9,18.8]	0.06 [0.0,0.2]
Gen16	const.	early	real	99.06 [97.1,100.0]	0.44 [0.0,2.4]	0.30 [0.0,1.3]
Gen16	const.	early	marg.null	97.76 [94.5,99.8]	1.82 [0.0,5.0]	0.29 [0.0,1.2]
Gen16	const.	early	joint.null	98.35 [91.9,100.0]	1.16 [0.0,7.6]	0.30 [0.0,1.3]
Gen16	const.	early-out	real	98.97 [97.2,100.0]	0.57 [0.0,2.1]	0.30 [0.0,1.3]
Gen16	const.	early-out	marg.null	99.19 [97.8,100.0]	0.37 [0.0,1.5]	0.30 [0.0,1.3]
Gen16	const.	early-out	joint.null	99.19 [97.8,100.0]	0.39 [0.0,1.5]	0.29 [0.0,1.3]
Gen16	const.	early-total	real	96.03 [92.8,98.9]	3.53 [0.8,6.6]	0.31 [0.0,1.3]
Gen16	const.	early-total	marg.null	96.44 [92.1,99.6]	3.12 [0.2,7.4]	0.30 [0.0,1.3]
Gen16	const.	early-total	joint.null	97.66 [95.0,99.6]	1.90 [0.2,4.5]	0.30 [0.0,1.3]
Gen16	const.	sameday	real	99.28 [98.0,100.0]	0.30 [0.0,1.3]	0.30 [0.0,1.3]
Gen16	const.	sameday	marg.null	99.27 [98.0,100.0]	0.30 [0.0,1.2]	0.30 [0.0,1.3]
Gen16	const.	sameday	joint.null	99.30 [98.0,100.0]	0.29 [0.0,1.2]	0.29 [0.0,1.2]
Gen16	const.	total	real	98.98 [97.3,100.0]	0.57 [0.0,2.0]	0.30 [0.0,1.3]
Gen16	const.	total	marg.null	98.24 [95.4,99.8]	1.30 [0.0,4.0]	0.29 [0.0,1.3]
Gen16	const.	total	joint.null	99.11 [97.5,100.0]	0.45 [0.0,1.9]	0.30 [0.0,1.3]
Gen16	dong	early	real	99.60 [98.4,99.9]	0.37 [0.1,1.6]	0.02 [0.0,0.1]
Gen16	dong	early	marg.null	99.62 [99.3,99.9]	0.35 [0.1,0.6]	0.02 [0.0,0.1]
Gen16	dong	early	joint.null	99.73 [98.7,100.0]	0.23 [0.0,1.2]	0.02 [0.0,0.1]
Gen16	dong	pooled	real	99.56 [99.2,99.8]	0.41 [0.1,0.7]	0.02 [0.0,0.1]
Gen16	dong	pooled	marg.null	99.44 [99.1,99.7]	0.53 [0.2,0.8]	0.02 [0.0,0.1]
Gen16	dong	pooled	joint.null	99.68 [99.4,99.9]	0.29 [0.1,0.5]	0.02 [0.0,0.1]
Gen16	dong	sameday	real	99.94 [99.8,100.0]	0.03 [0.0,0.1]	0.02 [0.0,0.1]
Gen16	dong	sameday	marg.null	99.92 [99.8,100.0]	0.05 [0.0,0.2]	0.02 [0.0,0.1]
Gen16	dong	sameday	joint.null	99.95 [99.9,100.0]	0.02 [0.0,0.1]	0.02 [0.0,0.1]
Pres17	const.	early	real	97.57 [95.1,99.4]	2.00 [0.4,4.5]	0.28 [0.0,1.2]
Pres17	const.	early	marg.null	95.87 [92.9,98.3]	3.74 [1.5,6.6]	0.27 [0.0,1.2]
Pres17	const.	early	joint.null	96.46 [90.0,100.0]	3.11 [0.0,9.5]	0.28 [0.0,1.2]
Pres17	const.	early-out	real	95.18 [90.1,99.6]	4.43 [0.2,9.4]	0.27 [0.0,1.2]
Pres17	const.	early-out	marg.null	96.85 [93.7,99.5]	2.74 [0.2,5.8]	0.28 [0.0,1.2]
Pres17	const.	early-out	joint.null	96.09 [92.0,99.8]	3.53 [0.1,7.6]	0.28 [0.0,1.2]

Table 9 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres17	const.	early-total	real	99.34 [98.1,100.0]	0.29 [0.0,1.2]	0.27 [0.0,1.2]
Pres17	const.	early-total	marg.null	99.34 [98.1,100.0]	0.28 [0.0,1.2]	0.27 [0.0,1.2]
Pres17	const.	early-total	joint.null	99.24 [97.9,100.0]	0.36 [0.0,1.4]	0.28 [0.0,1.2]
Pres17	const.	sameday	real	99.31 [98.1,100.0]	0.28 [0.0,1.2]	0.29 [0.0,1.3]
Pres17	const.	sameday	marg.null	99.32 [98.0,100.0]	0.29 [0.0,1.3]	0.28 [0.0,1.2]
Pres17	const.	sameday	joint.null	99.32 [98.1,100.0]	0.28 [0.0,1.2]	0.28 [0.0,1.2]
Pres17	const.	total	real	79.44 [73.2,91.6]	20.18 [8.1,26.4]	0.28 [0.0,1.2]
Pres17	const.	total	marg.null	79.14 [73.5,84.3]	20.47 [15.1,26.0]	0.28 [0.0,1.2]
Pres17	const.	total	joint.null	76.54 [69.5,95.0]	23.08 [4.6,30.1]	0.27 [0.0,1.2]
Pres17	dong	early	real	99.28 [98.2,99.7]	0.69 [0.3,1.8]	0.02 [0.0,0.1]
Pres17	dong	early	marg.null	99.19 [98.1,99.7]	0.78 [0.3,1.8]	0.02 [0.0,0.1]
Pres17	dong	early	joint.null	99.40 [99.1,99.7]	0.57 [0.3,0.9]	0.02 [0.0,0.1]
Pres17	dong	pooled	real	89.44 [88.2,90.8]	10.54 [9.2,11.8]	0.02 [0.0,0.1]
Pres17	dong	pooled	marg.null	89.49 [88.0,94.2]	10.48 [5.8,11.9]	0.02 [0.0,0.1]
Pres17	dong	pooled	joint.null	89.67 [88.4,95.2]	10.30 [4.8,11.5]	0.02 [0.0,0.1]
Pres17	dong	sameday	real	99.93 [99.8,100.0]	0.04 [0.0,0.1]	0.02 [0.0,0.1]
Pres17	dong	sameday	marg.null	99.86 [99.7,100.0]	0.11 [0.0,0.2]	0.02 [0.0,0.1]
Pres17	dong	sameday	joint.null	99.87 [99.7,100.0]	0.10 [0.0,0.2]	0.02 [0.0,0.1]
Gen20	const.	early	real	93.23 [89.0,97.0]	6.37 [2.7,10.5]	0.28 [0.0,1.2]
Gen20	const.	early	marg.null	96.60 [92.9,99.1]	2.97 [0.6,6.5]	0.28 [0.0,1.2]
Gen20	const.	early	joint.null	93.86 [89.3,98.4]	5.74 [1.2,10.2]	0.28 [0.0,1.2]
Gen20	const.	early-out	real	96.63 [93.7,98.9]	2.95 [0.8,5.8]	0.28 [0.0,1.2]
Gen20	const.	early-out	marg.null	97.08 [94.6,99.0]	2.54 [0.7,4.9]	0.27 [0.0,1.2]
Gen20	const.	early-out	joint.null	96.37 [93.4,99.2]	3.25 [0.4,5.9]	0.27 [0.0,1.2]
Gen20	const.	early-total	real	97.55 [95.2,99.5]	2.05 [0.3,4.2]	0.27 [0.0,1.2]
Gen20	const.	early-total	marg.null	96.56 [93.9,98.6]	3.04 [1.0,5.5]	0.28 [0.0,1.2]
Gen20	const.	early-total	joint.null	96.91 [94.3,98.9]	2.72 [0.8,5.1]	0.26 [0.0,1.1]
Gen20	const.	sameday	real	99.25 [97.9,100.0]	0.35 [0.0,1.4]	0.27 [0.0,1.2]
Gen20	const.	sameday	marg.null	99.32 [98.1,100.0]	0.28 [0.0,1.3]	0.27 [0.0,1.2]
Gen20	const.	sameday	joint.null	99.05 [97.5,100.0]	0.54 [0.0,1.9]	0.29 [0.0,1.2]
Gen20	const.	total	real	87.33 [82.1,93.2]	12.27 [6.4,17.5]	0.28 [0.0,1.2]
Gen20	const.	total	marg.null	85.79 [80.1,90.8]	13.82 [8.9,19.6]	0.26 [0.0,1.2]
Gen20	const.	total	joint.null	88.40 [83.3,96.5]	11.22 [3.2,16.3]	0.28 [0.0,1.2]
Gen20	dong	early	real	97.02 [95.9,98.0]	2.95 [1.9,4.0]	0.02 [0.0,0.1]
Gen20	dong	early	marg.null	97.30 [96.6,97.9]	2.67 [2.1,3.4]	0.02 [0.0,0.1]
Gen20	dong	early	joint.null	98.12 [93.1,98.9]	1.85 [1.1,6.8]	0.02 [0.0,0.1]
Gen20	dong	pooled	real	95.36 [94.5,96.1]	4.61 [3.9,5.4]	0.02 [0.0,0.1]
Gen20	dong	pooled	marg.null	94.78 [93.9,95.5]	5.19 [4.4,6.0]	0.02 [0.0,0.1]
Gen20	dong	pooled	joint.null	94.93 [94.1,95.7]	5.04 [4.2,5.9]	0.02 [0.0,0.1]
Gen20	dong	sameday	real	99.80 [99.6,100.0]	0.17 [0.0,0.4]	0.02 [0.0,0.1]

Table 9 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen20	dong	sameday	marg.null	99.85 [99.7,100.0]	0.12 [0.0,0.3]	0.02 [0.0,0.1]
Gen20	dong	sameday	joint.null	99.77 [99.6,99.9]	0.20 [0.1,0.4]	0.02 [0.0,0.1]
Pres22	const.	early	real	98.34 [96.3,99.8]	1.26 [0.1,3.1]	0.28 [0.0,1.2]
Pres22	const.	early	marg.null	97.88 [95.5,99.5]	1.72 [0.3,3.9]	0.28 [0.0,1.2]
Pres22	const.	early	joint.null	97.85 [95.8,99.5]	1.75 [0.4,3.7]	0.27 [0.0,1.2]
Pres22	const.	early-out	real	95.02 [91.6,98.2]	4.58 [1.5,7.9]	0.28 [0.0,1.2]
Pres22	const.	early-out	marg.null	97.64 [94.8,99.8]	1.96 [0.1,4.5]	0.28 [0.0,1.3]
Pres22	const.	early-out	joint.null	95.92 [90.6,100.0]	3.65 [0.0,9.0]	0.29 [0.0,1.2]
Pres22	const.	early-total	real	94.94 [91.3,97.9]	4.65 [1.8,8.2]	0.29 [0.0,1.2]
Pres22	const.	early-total	marg.null	94.56 [91.1,97.4]	5.03 [2.1,8.3]	0.28 [0.0,1.2]
Pres22	const.	early-total	joint.null	94.67 [89.4,97.9]	4.91 [1.9,10.3]	0.28 [0.0,1.2]
Pres22	const.	sameday	real	98.07 [95.6,99.8]	1.50 [0.1,3.9]	0.28 [0.0,1.2]
Pres22	const.	sameday	marg.null	98.45 [96.6,99.7]	1.15 [0.1,2.7]	0.27 [0.0,1.2]
Pres22	const.	sameday	joint.null	98.17 [96.0,99.8]	1.42 [0.0,3.5]	0.28 [0.0,1.2]
Pres22	const.	total	real	70.91 [61.5,88.7]	23.59 [6.9,32.6]	5.16 [1.8,9.1]
Pres22	const.	total	marg.null	72.31 [64.0,91.9]	21.75 [7.5,28.9]	5.35 [0.1,9.0]
Pres22	const.	total	joint.null	77.80 [62.3,88.2]	18.32 [9.8,32.8]	3.63 [0.7,6.9]
Pres22	dong	early	real	94.54 [87.2,97.1]	5.36 [2.9,12.7]	0.02 [0.0,0.1]
Pres22	dong	early	marg.null	96.78 [85.2,98.3]	3.19 [1.7,14.8]	0.02 [0.0,0.1]
Pres22	dong	early	joint.null	97.56 [96.9,98.4]	2.41 [1.6,3.1]	0.02 [0.0,0.1]
Pres22	dong	pooled	real	86.85 [79.3,89.9]	12.92 [10.0,20.6]	0.12 [0.0,0.5]
Pres22	dong	pooled	marg.null	81.92 [79.7,91.4]	17.79 [8.6,20.0]	0.24 [0.0,0.4]
Pres22	dong	pooled	joint.null	81.17 [79.3,83.0]	18.69 [16.8,20.6]	0.12 [0.0,0.3]
Pres22	dong	sameday	real	99.64 [99.4,99.8]	0.33 [0.1,0.6]	0.02 [0.0,0.1]
Pres22	dong	sameday	marg.null	99.52 [99.2,99.8]	0.46 [0.2,0.8]	0.02 [0.0,0.1]
Pres22	dong	sameday	joint.null	99.48 [99.1,99.8]	0.49 [0.2,0.9]	0.02 [0.0,0.1]
Gen24	const.	early	real	95.05 [91.7,97.7]	4.53 [2.1,7.8]	0.28 [0.0,1.3]
Gen24	const.	early	marg.null	96.10 [91.5,98.6]	3.49 [1.1,8.0]	0.27 [0.0,1.2]
Gen24	const.	early	joint.null	95.22 [90.1,98.0]	4.35 [1.6,9.4]	0.28 [0.0,1.2]
Gen24	const.	early-out	real	98.27 [95.1,100.0]	1.24 [0.0,4.3]	0.29 [0.0,1.3]
Gen24	const.	early-out	marg.null	97.31 [92.6,100.0]	2.27 [0.0,6.9]	0.28 [0.0,1.3]
Gen24	const.	early-out	joint.null	98.27 [95.8,100.0]	1.30 [0.0,3.7]	0.28 [0.0,1.2]
Gen24	const.	early-total	real	98.91 [94.1,100.0]	0.62 [0.0,5.5]	0.28 [0.0,1.2]
Gen24	const.	early-total	marg.null	98.93 [97.3,99.9]	0.66 [0.0,2.1]	0.28 [0.0,1.2]
Gen24	const.	early-total	joint.null	98.60 [94.0,100.0]	0.91 [0.0,5.5]	0.28 [0.0,1.2]
Gen24	const.	sameday	real	98.99 [97.1,100.0]	0.58 [0.0,2.3]	0.29 [0.0,1.2]
Gen24	const.	sameday	marg.null	98.86 [97.0,100.0]	0.73 [0.0,2.3]	0.28 [0.0,1.2]
Gen24	const.	sameday	joint.null	98.51 [96.1,100.0]	1.06 [0.0,3.4]	0.28 [0.0,1.2]
Gen24	const.	total	real	83.84 [76.3,94.0]	15.78 [5.3,22.9]	0.28 [0.0,1.2]
Gen24	const.	total	marg.null	86.58 [81.2,95.6]	13.03 [3.9,18.2]	0.28 [0.0,1.2]
Gen24	const.	total	joint.null	88.05 [78.8,95.6]	11.59 [4.1,20.7]	0.29 [0.0,1.3]
Gen24	dong	early	real	95.26 [92.9,97.3]	4.71 [2.7,7.1]	0.02 [0.0,0.1]

Table 9 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen24	dong	early	marg.null	95.90 [82.0,97.7]	4.06 [2.3,18.0]	0.03 [0.0,0.1]
Gen24	dong	early	joint.null	97.15 [95.5,98.4]	2.81 [1.6,4.5]	0.03 [0.0,0.1]
Gen24	dong	pooled	real	93.25 [92.0,94.3]	6.71 [5.6,8.0]	0.03 [0.0,0.1]
Gen24	dong	pooled	marg.null	92.78 [91.6,94.1]	7.18 [5.9,8.4]	0.03 [0.0,0.1]
Gen24	dong	pooled	joint.null	93.11 [91.5,94.3]	6.86 [5.7,8.5]	0.02 [0.0,0.1]
Gen24	dong	sameday	real	99.56 [99.2,99.8]	0.41 [0.2,0.7]	0.03 [0.0,0.1]
Gen24	dong	sameday	marg.null	99.41 [98.1,99.8]	0.55 [0.1,1.9]	0.03 [0.0,0.1]
Gen24	dong	sameday	joint.null	99.40 [99.1,99.7]	0.56 [0.3,0.9]	0.02 [0.0,0.1]
Pres25	const.	early	real	90.54 [80.2,96.7]	8.99 [2.9,19.4]	0.28 [0.0,1.2]
Pres25	const.	early	marg.null	92.59 [84.7,96.4]	6.98 [3.2,14.7]	0.27 [0.0,1.2]
Pres25	const.	early	joint.null	93.81 [89.8,97.2]	5.79 [2.5,9.7]	0.27 [0.0,1.2]
Pres25	const.	early-out	real	93.98 [90.5,99.8]	5.62 [0.1,9.0]	0.29 [0.0,1.2]
Pres25	const.	early-out	marg.null	96.35 [89.4,100.0]	3.24 [0.0,10.2]	0.27 [0.0,1.1]
Pres25	const.	early-out	joint.null	93.99 [86.8,99.2]	5.61 [0.7,12.8]	0.28 [0.0,1.2]
Pres25	const.	early- total	real	92.38 [87.9,96.7]	7.14 [3.0,11.3]	0.36 [0.0,1.4]
Pres25	const.	early- total	marg.null	91.99 [88.1,95.5]	7.43 [4.2,11.1]	0.45 [0.0,1.6]
Pres25	const.	early- total	joint.null	92.24 [87.7,96.1]	7.35 [3.4,11.6]	0.27 [0.0,1.2]
Pres25	const.	sameday	real	99.01 [97.2,100.0]	0.55 [0.0,2.3]	0.28 [0.0,1.2]
Pres25	const.	sameday	marg.null	98.30 [96.1,99.8]	1.30 [0.1,3.3]	0.26 [0.0,1.2]
Pres25	const.	sameday	joint.null	97.45 [93.6,99.5]	2.11 [0.3,6.0]	0.29 [0.0,1.2]
Pres25	const.	total	real	67.94 [60.8,86.6]	27.10 [12.1,34.1]	4.17 [0.2,7.5]
Pres25	const.	total	marg.null	68.43 [62.2,74.5]	26.48 [20.3,32.6]	4.94 [2.4,8.0]
Pres25	const.	total	joint.null	81.83 [63.8,93.4]	14.93 [5.5,28.1]	3.40 [0.0,9.3]
Pres25	dong	early	real	99.46 [99.0,99.8]	0.51 [0.2,1.0]	0.02 [0.0,0.1]
Pres25	dong	early	marg.null	98.68 [97.8,99.2]	1.30 [0.8,2.1]	0.02 [0.0,0.1]
Pres25	dong	early	joint.null	99.54 [99.1,99.8]	0.43 [0.1,0.9]	0.02 [0.0,0.1]
Pres25	dong	pooled	real	78.44 [76.2,91.2]	21.25 [8.8,23.4]	0.28 [0.0,0.6]
Pres25	dong	pooled	marg.null	78.70 [77.1,91.0]	20.79 [8.9,22.5]	0.46 [0.1,0.7]
Pres25	dong	pooled	joint.null	78.76 [76.9,92.0]	20.81 [7.8,22.5]	0.35 [0.0,0.7]
Pres25	dong	sameday	real	99.61 [99.4,99.8]	0.37 [0.2,0.6]	0.02 [0.0,0.1]
Pres25	dong	sameday	marg.null	99.63 [99.4,99.8]	0.34 [0.1,0.5]	0.02 [0.0,0.1]
Pres25	dong	sameday	joint.null	99.53 [99.3,99.8]	0.44 [0.2,0.7]	0.02 [0.0,0.1]

Table 10: Mixture-component probabilities, long-chain configuration, conservative-coded leader (partial; excludes the still-incomplete 2022 presidential slice).

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres12	const.	total-no-early	real	73.68 [67.8,80.0]	25.93 [19.7,31.9]	0.27 [0.0,1.1]
Pres12	const.	total-no-early	marg.null	74.37 [67.7,88.6]	25.26 [11.1,32.0]	0.28 [0.0,1.2]
Pres12	const.	total-no-early	joint.null	75.46 [68.6,88.6]	24.14 [10.9,30.8]	0.28 [0.0,1.2]
Pres12	dong	total-no-early	real	74.10 [71.2,76.1]	25.85 [23.8,28.6]	0.05 [0.0,0.2]
Pres12	dong	total-no-early	marg.null	82.25 [72.6,84.5]	17.72 [15.4,27.3]	0.08 [0.0,0.3]
Pres12	dong	total-no-early	joint.null	74.65 [72.8,85.8]	25.25 [14.1,27.1]	0.08 [0.0,0.2]
Gen20	const.	early	real	97.93 [94.6,100.0]	1.64 [0.0,4.9]	0.28 [0.0,1.2]
Gen20	const.	early	marg.null	98.50 [94.5,99.9]	1.07 [0.0,4.9]	0.27 [0.0,1.2]
Gen20	const.	early	joint.null	98.60 [96.7,99.9]	0.99 [0.0,2.8]	0.27 [0.0,1.2]
Gen20	const.	early-out	real	98.96 [96.7,100.0]	0.60 [0.0,2.7]	0.27 [0.0,1.2]
Gen20	const.	early-out	marg.null	98.78 [97.0,100.0]	0.81 [0.0,2.4]	0.27 [0.0,1.2]
Gen20	const.	early-out	joint.null	98.96 [97.3,100.0]	0.61 [0.0,2.1]	0.29 [0.0,1.2]
Gen20	const.	early-total	real	98.94 [97.4,100.0]	0.65 [0.0,2.0]	0.29 [0.0,1.2]
Gen20	const.	early-total	marg.null	98.99 [97.2,100.0]	0.58 [0.0,2.3]	0.28 [0.0,1.2]
Gen20	const.	early-total	joint.null	98.85 [97.3,99.9]	0.76 [0.0,2.1]	0.26 [0.0,1.1]
Gen20	const.	sameday	real	99.26 [98.0,100.0]	0.35 [0.0,1.4]	0.27 [0.0,1.2]
Gen20	const.	sameday	marg.null	99.24 [97.9,100.0]	0.36 [0.0,1.4]	0.27 [0.0,1.2]
Gen20	const.	sameday	joint.null	99.32 [98.0,100.0]	0.28 [0.0,1.2]	0.28 [0.0,1.2]
Gen20	const.	total	real	92.37 [87.0,96.4]	7.20 [3.1,12.4]	0.28 [0.0,1.2]
Gen20	const.	total	marg.null	91.61 [87.6,95.1]	7.98 [4.6,12.0]	0.28 [0.0,1.2]
Gen20	const.	total	joint.null	91.04 [87.3,94.6]	8.56 [5.2,12.3]	0.28 [0.0,1.2]
Gen20	dong	early	real	99.45 [99.1,99.8]	0.52 [0.2,0.9]	0.02 [0.0,0.1]
Gen20	dong	early	marg.null	99.43 [99.0,99.7]	0.54 [0.3,0.9]	0.02 [0.0,0.1]
Gen20	dong	early	joint.null	99.06 [98.7,99.6]	0.91 [0.4,1.3]	0.02 [0.0,0.1]
Gen20	dong	pooled	real	93.90 [92.7,95.0]	6.07 [5.0,7.3]	0.02 [0.0,0.1]
Gen20	dong	pooled	marg.null	93.13 [91.8,94.0]	6.84 [5.9,8.2]	0.02 [0.0,0.1]
Gen20	dong	pooled	joint.null	93.36 [92.4,94.3]	6.61 [5.7,7.6]	0.02 [0.0,0.1]
Gen20	dong	sameday	real	99.91 [99.8,100.0]	0.06 [0.0,0.2]	0.02 [0.0,0.1]
Gen20	dong	sameday	marg.null	99.86 [99.7,100.0]	0.11 [0.0,0.3]	0.02 [0.0,0.1]
Gen20	dong	sameday	joint.null	99.81 [99.6,100.0]	0.16 [0.0,0.3]	0.02 [0.0,0.1]
Gen24	const.	early	real	98.63 [96.2,99.8]	0.91 [0.0,3.2]	0.29 [0.0,1.2]
Gen24	const.	early	marg.null	97.99 [95.9,99.5]	1.60 [0.3,3.5]	0.28 [0.0,1.2]
Gen24	const.	early	joint.null	97.93 [95.7,99.5]	1.66 [0.2,3.7]	0.28 [0.0,1.2]

Table 10 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen24	const.	early-out	real	99.12 [97.7,100.0]	0.47 [0.0,1.7]	0.28 [0.0,1.2]
Gen24	const.	early-out	marg.null	98.62 [96.1,100.0]	0.92 [0.0,3.4]	0.29 [0.0,1.2]
Gen24	const.	early-out	joint.null	98.93 [97.3,99.9]	0.67 [0.0,2.1]	0.28 [0.0,1.2]
Gen24	const.	early-total	real	99.24 [97.9,100.0]	0.36 [0.0,1.4]	0.28 [0.0,1.2]
Gen24	const.	early-total	marg.null	99.13 [97.7,100.0]	0.47 [0.0,1.7]	0.28 [0.0,1.2]
Gen24	const.	early-total	joint.null	98.91 [97.3,99.9]	0.67 [0.0,2.1]	0.28 [0.0,1.2]
Gen24	const.	sameday	real	99.14 [97.7,100.0]	0.46 [0.0,1.7]	0.27 [0.0,1.2]
Gen24	const.	sameday	marg.null	99.22 [97.9,100.0]	0.36 [0.0,1.5]	0.28 [0.0,1.2]
Gen24	const.	sameday	joint.null	99.13 [97.7,100.0]	0.46 [0.0,1.7]	0.28 [0.0,1.2]
Gen24	const.	total	real	89.71 [83.0,96.1]	9.88 [3.4,16.4]	0.29 [0.0,1.2]
Gen24	const.	total	marg.null	87.07 [81.2,92.4]	12.51 [7.4,18.5]	0.29 [0.0,1.2]
Gen24	const.	total	joint.null	87.09 [81.0,92.7]	12.52 [7.2,18.7]	0.27 [0.0,1.2]
Gen24	dong	early	real	99.33 [97.9,99.8]	0.62 [0.2,2.0]	0.03 [0.0,0.1]
Gen24	dong	early	marg.null	99.18 [97.7,99.7]	0.78 [0.3,2.3]	0.03 [0.0,0.1]
Gen24	dong	early	joint.null	99.33 [98.1,99.7]	0.63 [0.2,1.9]	0.03 [0.0,0.1]
Gen24	dong	pooled	real	93.59 [92.5,94.6]	6.37 [5.4,7.4]	0.02 [0.0,0.1]
Gen24	dong	pooled	marg.null	93.16 [91.8,94.7]	6.81 [5.4,8.2]	0.03 [0.0,0.1]
Gen24	dong	pooled	joint.null	93.87 [92.8,94.9]	6.09 [5.1,7.1]	0.02 [0.0,0.1]
Gen24	dong	sameday	real	99.83 [99.6,100.0]	0.13 [0.0,0.4]	0.03 [0.0,0.1]
Gen24	dong	sameday	marg.null	99.78 [99.6,99.9]	0.19 [0.0,0.4]	0.03 [0.0,0.1]
Gen24	dong	sameday	joint.null	99.80 [99.6,100.0]	0.16 [0.0,0.4]	0.02 [0.0,0.1]

Table 11: Mixture-component probabilities, short-chain configuration, Democratic-coded leader. Full coverage, all 11 elections.

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen08	const.	total-no-early	real	99.12 [97.5,99.9]	0.37 [0.0,1.5]	0.36 [0.0,1.5]
Gen08	const.	total-no-early	marg.null	99.13 [97.5,100.0]	0.34 [0.0,1.6]	0.38 [0.0,1.6]
Gen08	const.	total-no-early	joint.null	98.93 [97.0,99.9]	0.53 [0.0,2.2]	0.36 [0.0,1.4]
Gen08	dong	total-no-early	real	99.47 [99.2,99.7]	0.50 [0.3,0.8]	0.02 [0.0,0.1]
Gen08	dong	total-no-early	marg.null	99.33 [98.9,99.7]	0.63 [0.3,1.0]	0.03 [0.0,0.1]
Gen08	dong	total-no-early	joint.null	99.66 [99.4,99.9]	0.30 [0.1,0.6]	0.03 [0.0,0.1]
Gen12	const.	total-no-early	real	99.22 [97.8,100.0]	0.31 [0.0,1.3]	0.32 [0.0,1.5]
Gen12	const.	total-no-early	marg.null	99.19 [97.7,100.0]	0.32 [0.0,1.4]	0.36 [0.0,1.5]
Gen12	const.	total-no-early	joint.null	99.21 [97.8,100.0]	0.32 [0.0,1.4]	0.33 [0.0,1.4]
Gen12	dong	total-no-early	real	99.40 [99.0,99.7]	0.57 [0.3,1.0]	0.02 [0.0,0.1]
Gen12	dong	total-no-early	marg.null	99.35 [99.0,99.6]	0.61 [0.4,0.9]	0.03 [0.0,0.1]
Gen12	dong	total-no-early	joint.null	99.61 [99.4,99.8]	0.36 [0.1,0.6]	0.02 [0.0,0.1]
Pres02	const.	total-no-early	real	85.92 [80.4,91.8]	13.70 [8.1,19.4]	0.28 [0.0,1.2]
Pres02	const.	total-no-early	marg.null	83.93 [78.3,90.9]	15.29 [8.7,20.9]	0.54 [0.0,1.9]
Pres02	const.	total-no-early	joint.null	87.77 [82.5,93.8]	11.74 [5.7,16.8]	0.29 [0.0,1.3]
Pres02	dong	total-no-early	real	84.97 [83.4,90.1]	14.91 [9.7,16.5]	0.13 [0.0,0.3]
Pres02	dong	total-no-early	marg.null	85.45 [83.3,90.4]	14.41 [9.5,16.5]	0.11 [0.0,0.3]
Pres02	dong	total-no-early	joint.null	85.05 [83.7,86.4]	14.87 [13.6,16.3]	0.07 [0.0,0.2]
Pres07	const.	total-no-early	real	92.90 [89.0,96.2]	6.70 [3.3,10.3]	0.26 [0.0,1.2]
Pres07	const.	total-no-early	marg.null	92.41 [84.4,97.1]	7.16 [2.4,15.1]	0.26 [0.0,1.2]
Pres07	const.	total-no-early	joint.null	92.75 [87.4,96.9]	6.87 [2.7,12.1]	0.27 [0.0,1.2]
Pres07	dong	total-no-early	real	91.48 [89.0,94.0]	8.49 [6.0,11.0]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	marg.null	92.91 [87.4,94.2]	7.06 [5.8,12.6]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	joint.null	93.16 [92.2,94.0]	6.81 [5.9,7.7]	0.02 [0.0,0.1]
Pres12	const.	total-no-early	real	86.78 [76.3,91.9]	12.84 [7.9,23.1]	0.37 [0.0,1.5]
Pres12	const.	total-no-early	marg.null	82.31 [74.7,92.4]	17.15 [7.2,24.5]	0.36 [0.0,1.5]

Table 11 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres12	const.	total-no-early	joint.null	80.94 [75.6,86.0]	18.62 [13.2,23.7]	0.28 [0.0,1.3]
Pres12	dong	total-no-early	real	83.12 [81.1,87.6]	16.78 [12.4,18.9]	0.07 [0.0,0.2]
Pres12	dong	total-no-early	marg.null	84.58 [80.8,89.0]	15.27 [10.8,18.9]	0.13 [0.0,0.4]
Pres12	dong	total-no-early	joint.null	82.33 [80.8,83.8]	17.51 [15.9,18.9]	0.15 [0.0,0.4]
Gen16	const.	early	real	98.74 [96.9,99.8]	0.81 [0.0,2.4]	0.32 [0.0,1.4]
Gen16	const.	early	marg.null	99.08 [97.1,100.0]	0.47 [0.0,2.3]	0.29 [0.0,1.2]
Gen16	const.	early	joint.null	98.50 [96.3,99.9]	1.06 [0.0,3.1]	0.29 [0.0,1.3]
Gen16	const.	early-out	real	99.19 [97.7,100.0]	0.38 [0.0,1.5]	0.30 [0.0,1.3]
Gen16	const.	early-out	marg.null	99.13 [97.7,100.0]	0.48 [0.0,1.7]	0.27 [0.0,1.1]
Gen16	const.	early-out	joint.null	99.29 [98.0,100.0]	0.29 [0.0,1.2]	0.30 [0.0,1.2]
Gen16	const.	early-total	real	97.91 [95.7,100.0]	1.63 [0.0,3.9]	0.31 [0.0,1.4]
Gen16	const.	early-total	marg.null	98.14 [95.5,100.0]	1.34 [0.0,4.0]	0.29 [0.0,1.3]
Gen16	const.	early-total	joint.null	97.77 [92.7,100.0]	1.69 [0.0,6.8]	0.32 [0.0,1.3]
Gen16	const.	sameday	real	99.27 [98.0,100.0]	0.30 [0.0,1.2]	0.30 [0.0,1.3]
Gen16	const.	sameday	marg.null	99.30 [97.9,100.0]	0.29 [0.0,1.2]	0.30 [0.0,1.4]
Gen16	const.	sameday	joint.null	99.25 [97.9,100.0]	0.31 [0.0,1.3]	0.29 [0.0,1.3]
Gen16	const.	total	real	99.23 [97.8,100.0]	0.36 [0.0,1.6]	0.29 [0.0,1.2]
Gen16	const.	total	marg.null	98.43 [96.6,99.8]	1.14 [0.1,2.8]	0.29 [0.0,1.2]
Gen16	const.	total	joint.null	99.11 [97.5,100.0]	0.46 [0.0,1.9]	0.29 [0.0,1.2]
Gen16	dong	early	real	99.58 [99.3,99.9]	0.39 [0.1,0.7]	0.02 [0.0,0.1]
Gen16	dong	early	marg.null	99.49 [98.4,99.8]	0.47 [0.2,1.6]	0.02 [0.0,0.1]
Gen16	dong	early	joint.null	99.59 [98.8,99.9]	0.38 [0.1,1.2]	0.02 [0.0,0.1]
Gen16	dong	pooled	real	99.65 [99.4,99.9]	0.32 [0.1,0.6]	0.02 [0.0,0.1]
Gen16	dong	pooled	marg.null	99.52 [99.2,99.8]	0.45 [0.2,0.8]	0.02 [0.0,0.1]
Gen16	dong	pooled	joint.null	99.73 [99.5,99.9]	0.24 [0.1,0.5]	0.02 [0.0,0.1]
Gen16	dong	sameday	real	99.95 [99.9,100.0]	0.02 [0.0,0.1]	0.02 [0.0,0.1]
Gen16	dong	sameday	marg.null	99.94 [99.8,100.0]	0.03 [0.0,0.1]	0.02 [0.0,0.1]
Gen16	dong	sameday	joint.null	99.95 [99.8,100.0]	0.02 [0.0,0.1]	0.02 [0.0,0.1]
Pres17	const.	early	real	97.38 [95.0,99.3]	2.20 [0.5,4.6]	0.27 [0.0,1.2]
Pres17	const.	early	marg.null	96.02 [92.8,99.3]	3.62 [0.4,6.6]	0.27 [0.0,1.2]
Pres17	const.	early	joint.null	97.60 [94.4,100.0]	2.02 [0.0,5.2]	0.28 [0.0,1.1]
Pres17	const.	early-out	real	96.17 [91.8,99.5]	3.47 [0.4,7.8]	0.27 [0.0,1.1]
Pres17	const.	early-out	marg.null	96.58 [93.0,99.3]	2.99 [0.4,6.4]	0.28 [0.0,1.2]
Pres17	const.	early-out	joint.null	94.40 [89.4,98.9]	5.16 [1.1,10.4]	0.30 [0.0,1.3]
Pres17	const.	early-total	real	97.58 [92.2,100.0]	1.93 [0.0,7.4]	0.28 [0.0,1.1]

Table 11 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres17	const.	early-total	marg.null	99.30 [97.9,100.0]	0.30 [0.0,1.5]	0.27 [0.0,1.2]
Pres17	const.	early-total	joint.null	99.18 [97.5,100.0]	0.41 [0.0,1.9]	0.28 [0.0,1.2]
Pres17	const.	sameday	real	99.34 [98.1,100.0]	0.26 [0.0,1.2]	0.29 [0.0,1.2]
Pres17	const.	sameday	marg.null	99.33 [98.0,100.0]	0.29 [0.0,1.2]	0.27 [0.0,1.2]
Pres17	const.	sameday	joint.null	99.33 [98.1,100.0]	0.27 [0.0,1.2]	0.27 [0.0,1.2]
Pres17	const.	total	real	80.15 [73.7,95.1]	19.46 [4.3,25.7]	0.28 [0.0,1.2]
Pres17	const.	total	marg.null	83.74 [73.8,94.2]	15.91 [5.6,25.8]	0.27 [0.0,1.1]
Pres17	const.	total	joint.null	78.55 [72.9,96.8]	21.08 [2.6,26.4]	0.28 [0.0,1.2]
Pres17	dong	early	real	98.97 [97.1,99.5]	1.00 [0.5,2.9]	0.02 [0.0,0.1]
Pres17	dong	early	marg.null	98.66 [97.8,99.3]	1.31 [0.7,2.2]	0.02 [0.0,0.1]
Pres17	dong	early	joint.null	99.31 [99.0,99.6]	0.66 [0.4,1.0]	0.02 [0.0,0.1]
Pres17	dong	pooled	real	90.02 [88.9,95.5]	9.96 [4.4,11.1]	0.02 [0.0,0.1]
Pres17	dong	pooled	marg.null	89.12 [88.0,96.3]	10.86 [3.7,12.0]	0.02 [0.0,0.1]
Pres17	dong	pooled	joint.null	92.72 [88.6,94.0]	7.25 [6.0,11.3]	0.02 [0.0,0.1]
Pres17	dong	sameday	real	99.92 [99.8,100.0]	0.05 [0.0,0.2]	0.02 [0.0,0.1]
Pres17	dong	sameday	marg.null	99.90 [99.8,100.0]	0.07 [0.0,0.2]	0.02 [0.0,0.1]
Pres17	dong	sameday	joint.null	99.90 [99.5,100.0]	0.07 [0.0,0.4]	0.02 [0.0,0.1]
Gen20	const.	early	real	93.92 [90.0,97.4]	5.71 [2.4,9.6]	0.27 [0.0,1.2]
Gen20	const.	early	marg.null	93.68 [90.4,96.6]	5.93 [3.0,9.1]	0.26 [0.0,1.2]
Gen20	const.	early	joint.null	93.92 [90.5,98.4]	5.73 [1.3,9.1]	0.26 [0.0,1.2]
Gen20	const.	early-out	real	97.11 [94.5,99.8]	2.46 [0.1,4.9]	0.30 [0.0,1.3]
Gen20	const.	early-out	marg.null	97.34 [94.7,99.7]	2.32 [0.1,4.8]	0.25 [0.0,1.1]
Gen20	const.	early-out	joint.null	98.75 [97.0,99.9]	0.81 [0.0,2.4]	0.29 [0.0,1.3]
Gen20	const.	early-total	real	97.24 [90.9,99.2]	2.36 [0.6,8.6]	0.26 [0.0,1.1]
Gen20	const.	early-total	marg.null	97.43 [94.4,99.6]	2.12 [0.3,5.3]	0.28 [0.0,1.2]
Gen20	const.	early-total	joint.null	97.11 [93.5,99.3]	2.45 [0.4,6.0]	0.29 [0.0,1.2]
Gen20	const.	sameday	real	99.02 [97.6,99.9]	0.60 [0.0,1.8]	0.27 [0.0,1.2]
Gen20	const.	sameday	marg.null	99.21 [97.9,100.0]	0.36 [0.0,1.5]	0.30 [0.0,1.2]
Gen20	const.	sameday	joint.null	99.22 [97.9,100.0]	0.37 [0.0,1.5]	0.26 [0.0,1.2]
Gen20	const.	total	real	87.75 [83.3,91.9]	11.85 [7.9,16.4]	0.26 [0.0,1.2]
Gen20	const.	total	marg.null	87.41 [79.8,92.2]	12.22 [7.7,20.0]	0.27 [0.0,1.1]
Gen20	const.	total	joint.null	88.31 [80.1,95.1]	11.28 [4.7,19.7]	0.29 [0.0,1.2]
Gen20	dong	early	real	94.29 [81.6,98.0]	5.67 [2.0,18.4]	0.02 [0.0,0.1]
Gen20	dong	early	marg.null	94.12 [82.8,98.0]	5.85 [1.9,17.2]	0.02 [0.0,0.1]
Gen20	dong	early	joint.null	97.95 [92.5,98.6]	2.02 [1.4,7.4]	0.02 [0.0,0.1]
Gen20	dong	pooled	real	95.06 [94.3,95.8]	4.91 [4.2,5.7]	0.02 [0.0,0.1]
Gen20	dong	pooled	marg.null	94.50 [93.7,95.3]	5.47 [4.7,6.3]	0.02 [0.0,0.1]
Gen20	dong	pooled	joint.null	94.68 [93.8,95.6]	5.29 [4.4,6.1]	0.02 [0.0,0.1]
Gen20	dong	sameday	real	99.87 [99.7,100.0]	0.10 [0.0,0.2]	0.02 [0.0,0.1]
Gen20	dong	sameday	marg.null	99.85 [99.6,100.0]	0.12 [0.0,0.3]	0.02 [0.0,0.1]

Table 11 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen20	dong	sameday	joint.null	99.82 [99.6,100.0]	0.15 [0.0,0.3]	0.02 [0.0,0.1]
Pres22	const.	early	real	97.09 [90.4,99.3]	2.46 [0.3,9.0]	0.27 [0.0,1.3]
Pres22	const.	early	marg.null	97.88 [96.0,99.5]	1.70 [0.2,3.5]	0.29 [0.0,1.2]
Pres22	const.	early	joint.null	98.42 [96.6,99.8]	1.16 [0.1,2.8]	0.29 [0.0,1.3]
Pres22	const.	early-out	real	94.66 [91.0,98.0]	4.97 [1.6,8.4]	0.27 [0.0,1.1]
Pres22	const.	early-out	marg.null	96.23 [92.1,99.8]	3.35 [0.1,7.4]	0.28 [0.0,1.2]
Pres22	const.	early-out	joint.null	94.35 [90.4,100.0]	5.21 [0.0,9.2]	0.28 [0.0,1.3]
Pres22	const.	early- total	real	94.41 [91.3,97.2]	5.20 [2.4,8.2]	0.28 [0.0,1.2]
Pres22	const.	early- total	marg.null	95.30 [92.0,98.0]	4.34 [1.9,7.7]	0.26 [0.0,1.1]
Pres22	const.	early- total	joint.null	94.76 [91.4,98.4]	4.84 [1.3,8.2]	0.29 [0.0,1.2]
Pres22	const.	sameday	real	98.18 [96.2,99.7]	1.44 [0.2,3.2]	0.26 [0.0,1.2]
Pres22	const.	sameday	marg.null	98.59 [96.6,99.9]	1.01 [0.0,2.9]	0.27 [0.0,1.2]
Pres22	const.	sameday	joint.null	97.64 [95.4,99.3]	1.94 [0.5,4.0]	0.29 [0.0,1.3]
Pres22	const.	total	real	77.91 [64.6,89.9]	19.10 [8.7,31.7]	2.29 [0.0,6.4]
Pres22	const.	total	marg.null	76.87 [68.8,87.0]	17.79 [9.6,26.9]	4.09 [1.1,9.0]
Pres22	const.	total	joint.null	72.13 [64.8,80.1]	21.47 [16.1,27.9]	6.49 [0.9,10.5]
Pres22	dong	early	real	97.22 [89.0,98.0]	2.75 [1.9,11.0]	0.02 [0.0,0.1]
Pres22	dong	early	marg.null	97.01 [90.1,97.7]	2.96 [2.2,9.9]	0.02 [0.0,0.1]
Pres22	dong	early	joint.null	97.77 [91.0,98.5]	2.21 [1.5,9.0]	0.02 [0.0,0.1]
Pres22	dong	pooled	real	86.19 [79.6,90.5]	13.77 [9.5,20.3]	0.06 [0.0,0.2]
Pres22	dong	pooled	marg.null	83.43 [79.9,89.1]	16.39 [11.0,19.8]	0.18 [0.0,0.5]
Pres22	dong	pooled	joint.null	85.68 [80.3,92.6]	14.16 [7.4,19.6]	0.04 [0.0,0.2]
Pres22	dong	sameday	real	99.49 [99.2,99.7]	0.48 [0.2,0.7]	0.02 [0.0,0.1]
Pres22	dong	sameday	marg.null	99.34 [99.0,99.6]	0.63 [0.4,1.0]	0.02 [0.0,0.1]
Pres22	dong	sameday	joint.null	99.34 [99.0,99.7]	0.63 [0.3,1.0]	0.02 [0.0,0.1]
Gen24	const.	early	real	95.50 [92.3,98.2]	4.08 [1.5,7.3]	0.28 [0.0,1.1]
Gen24	const.	early	marg.null	95.89 [89.2,98.7]	3.65 [1.2,10.5]	0.30 [0.0,1.2]
Gen24	const.	early	joint.null	95.93 [92.8,98.5]	3.65 [1.3,6.8]	0.28 [0.0,1.3]
Gen24	const.	early-out	real	97.30 [90.5,99.9]	2.21 [0.0,9.1]	0.28 [0.0,1.2]
Gen24	const.	early-out	marg.null	95.19 [90.8,99.6]	4.37 [0.2,8.7]	0.29 [0.0,1.2]
Gen24	const.	early-out	joint.null	98.63 [97.0,99.8]	0.99 [0.1,2.4]	0.27 [0.0,1.1]
Gen24	const.	early- total	real	98.94 [97.2,100.0]	0.64 [0.0,2.2]	0.28 [0.0,1.2]
Gen24	const.	early- total	marg.null	98.38 [96.2,100.0]	1.21 [0.0,3.3]	0.26 [0.0,1.1]
Gen24	const.	early- total	joint.null	99.22 [97.9,100.0]	0.34 [0.0,1.5]	0.29 [0.0,1.2]
Gen24	const.	sameday	real	98.82 [97.2,99.9]	0.79 [0.0,2.2]	0.28 [0.0,1.2]
Gen24	const.	sameday	marg.null	98.55 [96.5,100.0]	1.02 [0.0,3.0]	0.31 [0.0,1.2]
Gen24	const.	sameday	joint.null	98.83 [97.2,99.9]	0.76 [0.0,2.2]	0.28 [0.0,1.3]
Gen24	const.	total	real	82.42 [76.5,88.4]	17.07 [11.3,23.2]	0.36 [0.0,1.5]
Gen24	const.	total	marg.null	83.50 [77.7,89.1]	16.03 [10.3,22.0]	0.38 [0.0,1.4]
Gen24	const.	total	joint.null	89.98 [77.9,94.4]	9.60 [4.9,21.4]	0.27 [0.0,1.2]
Gen24	dong	early	real	97.34 [89.7,98.7]	2.63 [1.3,10.3]	0.03 [0.0,0.1]
Gen24	dong	early	marg.null	95.97 [88.6,98.7]	3.99 [1.3,11.4]	0.03 [0.0,0.1]
Gen24	dong	early	joint.null	98.15 [97.4,98.8]	1.82 [1.2,2.6]	0.02 [0.0,0.1]
Gen24	dong	pooled	real	93.15 [92.2,94.2]	6.81 [5.9,7.9]	0.02 [0.0,0.1]
Gen24	dong	pooled	marg.null	92.42 [91.1,93.6]	7.54 [6.4,8.9]	0.03 [0.0,0.1]
Gen24	dong	pooled	joint.null	92.85 [91.0,94.1]	7.11 [5.9,9.0]	0.02 [0.0,0.1]
Gen24	dong	sameday	real	99.44 [99.1,99.7]	0.53 [0.3,0.9]	0.03 [0.0,0.1]

Table 11 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen24	dong	sameday	marg.null	99.34 [99.0,99.7]	0.62 [0.2,0.9]	0.03 [0.0,0.1]
Gen24	dong	sameday	joint.null	99.27 [98.9,99.7]	0.70 [0.3,1.1]	0.02 [0.0,0.1]
Pres25	const.	early	real	95.74 [92.9,98.2]	3.89 [1.5,6.6]	0.26 [0.0,1.1]
Pres25	const.	early	marg.null	94.50 [90.7,98.1]	5.06 [1.5,8.8]	0.27 [0.0,1.2]
Pres25	const.	early	joint.null	94.47 [91.3,97.3]	5.15 [2.5,8.2]	0.27 [0.0,1.2]
Pres25	const.	early-out	real	95.66 [90.0,99.6]	3.86 [0.2,9.6]	0.28 [0.0,1.1]
Pres25	const.	early-out	marg.null	94.19 [89.5,99.6]	5.46 [0.2,10.1]	0.26 [0.0,1.2]
Pres25	const.	early-out	joint.null	93.45 [83.4,98.4]	6.17 [1.0,15.8]	0.27 [0.0,1.2]
Pres25	const.	early-total	real	91.71 [87.4,95.7]	7.90 [3.7,11.9]	0.28 [0.0,1.2]
Pres25	const.	early-total	marg.null	90.14 [85.9,94.0]	9.34 [5.4,13.6]	0.37 [0.0,1.5]
Pres25	const.	early-total	joint.null	91.50 [86.5,95.7]	8.09 [3.8,12.7]	0.30 [0.0,1.3]
Pres25	const.	sameday	real	97.12 [94.4,99.0]	2.49 [0.6,5.0]	0.27 [0.0,1.2]
Pres25	const.	sameday	marg.null	98.02 [96.1,99.5]	1.54 [0.2,3.4]	0.30 [0.0,1.2]
Pres25	const.	sameday	joint.null	98.57 [96.7,99.8]	1.02 [0.0,2.8]	0.27 [0.0,1.1]
Pres25	const.	total	real	67.50 [61.2,74.0]	27.48 [22.0,34.0]	4.76 [2.1,7.9]
Pres25	const.	total	marg.null	70.43 [62.3,90.5]	24.21 [6.8,32.3]	4.50 [1.3,8.0]
Pres25	const.	total	joint.null	71.67 [64.4,81.6]	24.21 [15.7,31.1]	3.24 [0.8,8.5]
Pres25	dong	early	real	89.40 [83.6,94.9]	10.58 [5.1,16.4]	0.02 [0.0,0.1]
Pres25	dong	early	marg.null	91.85 [90.5,93.7]	8.13 [6.3,9.5]	0.02 [0.0,0.1]
Pres25	dong	early	joint.null	93.50 [83.2,96.4]	6.46 [3.6,16.7]	0.03 [0.0,0.2]
Pres25	dong	pooled	real	87.19 [76.5,92.2]	10.28 [7.8,23.0]	0.56 [0.0,4.8]
Pres25	dong	pooled	marg.null	79.29 [76.7,91.4]	20.20 [8.5,22.7]	0.44 [0.0,0.8]
Pres25	dong	pooled	joint.null	78.13 [76.3,88.8]	21.55 [11.1,23.3]	0.27 [0.0,0.5]
Pres25	dong	sameday	real	99.49 [99.2,99.7]	0.48 [0.2,0.8]	0.02 [0.0,0.1]
Pres25	dong	sameday	marg.null	99.44 [99.1,99.7]	0.53 [0.3,0.9]	0.02 [0.0,0.1]
Pres25	dong	sameday	joint.null	99.37 [99.0,99.6]	0.60 [0.3,0.9]	0.02 [0.0,0.1]

Table 12: Mixture-component probabilities, short-chain configuration, conservative-coded leader. Full coverage, all 11 elections.

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen08	const.	total-no-early	real	99.15 [97.6,100.0]	0.35 [0.0,1.5]	0.35 [0.0,1.5]
Gen08	const.	total-no-early	marg.null	99.17 [97.7,100.0]	0.34 [0.0,1.5]	0.35 [0.0,1.4]
Gen08	const.	total-no-early	joint.null	99.12 [97.5,100.0]	0.38 [0.0,1.6]	0.36 [0.0,1.5]
Gen08	dong	total-no-early	real	99.75 [99.6,99.9]	0.21 [0.1,0.4]	0.02 [0.0,0.1]
Gen08	dong	total-no-early	marg.null	99.69 [99.4,99.9]	0.27 [0.1,0.5]	0.02 [0.0,0.1]
Gen08	dong	total-no-early	joint.null	99.81 [99.5,100.0]	0.15 [0.0,0.4]	0.03 [0.0,0.1]
Gen12	const.	total-no-early	real	98.73 [96.7,99.9]	0.72 [0.0,2.4]	0.37 [0.0,1.6]
Gen12	const.	total-no-early	marg.null	98.89 [97.1,99.9]	0.60 [0.0,2.0]	0.36 [0.0,1.6]
Gen12	const.	total-no-early	joint.null	98.84 [95.3,100.0]	0.55 [0.0,4.0]	0.38 [0.0,1.6]
Gen12	dong	total-no-early	real	98.33 [97.7,98.9]	1.63 [1.0,2.3]	0.03 [0.0,0.1]
Gen12	dong	total-no-early	marg.null	98.07 [97.4,98.7]	1.89 [1.3,2.6]	0.02 [0.0,0.1]
Gen12	dong	total-no-early	joint.null	98.59 [98.1,99.1]	1.37 [0.9,1.9]	0.03 [0.0,0.1]
Pres02	const.	total-no-early	real	87.91 [82.5,92.8]	11.65 [6.8,17.0]	0.30 [0.0,1.3]
Pres02	const.	total-no-early	marg.null	86.58 [82.3,91.0]	12.99 [9.0,17.5]	0.28 [0.0,1.2]
Pres02	const.	total-no-early	joint.null	89.30 [83.6,94.3]	10.26 [5.3,16.0]	0.28 [0.0,1.2]
Pres02	dong	total-no-early	real	85.85 [84.6,87.1]	14.12 [12.8,15.4]	0.02 [0.0,0.1]
Pres02	dong	total-no-early	marg.null	84.68 [83.2,86.7]	15.29 [13.4,16.9]	0.02 [0.0,0.1]
Pres02	dong	total-no-early	joint.null	86.08 [84.6,87.4]	13.89 [12.6,15.4]	0.02 [0.0,0.1]
Pres07	const.	total-no-early	real	94.07 [90.7,96.8]	5.53 [2.9,8.8]	0.28 [0.0,1.3]
Pres07	const.	total-no-early	marg.null	94.03 [90.4,97.3]	5.57 [2.2,9.0]	0.29 [0.0,1.2]
Pres07	const.	total-no-early	joint.null	94.72 [91.2,98.1]	4.83 [1.7,8.5]	0.30 [0.0,1.3]
Pres07	dong	total-no-early	real	93.36 [92.3,94.4]	6.61 [5.6,7.6]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	marg.null	93.06 [92.0,94.1]	6.91 [5.9,7.9]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	joint.null	93.54 [92.5,94.5]	6.44 [5.4,7.4]	0.02 [0.0,0.1]
Pres12	const.	total-no-early	real	78.26 [66.8,91.1]	21.45 [8.8,32.9]	0.27 [0.0,1.1]
Pres12	const.	total-no-early	marg.null	82.46 [73.7,88.5]	16.65 [10.6,24.8]	0.57 [0.0,2.3]

Table 12 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres12	const.	total-no-early	joint.null	78.16 [71.0,92.0]	21.44 [7.7,28.6]	0.26 [0.0,1.3]
Pres12	dong	total-no-early	real	74.06 [72.2,80.2]	25.89 [19.8,27.8]	0.06 [0.0,0.2]
Pres12	dong	total-no-early	marg.null	74.09 [72.5,75.8]	25.81 [24.1,27.3]	0.09 [0.0,0.2]
Pres12	dong	total-no-early	joint.null	74.45 [70.8,77.1]	25.51 [22.9,29.2]	0.02 [0.0,0.1]
Gen16	const.	early	real	98.60 [96.4,100.0]	0.95 [0.0,3.0]	0.30 [0.0,1.2]
Gen16	const.	early	marg.null	98.08 [93.0,100.0]	1.44 [0.0,6.5]	0.31 [0.0,1.3]
Gen16	const.	early	joint.null	99.17 [97.7,100.0]	0.39 [0.0,1.6]	0.30 [0.0,1.3]
Gen16	const.	early-out	real	99.29 [98.0,100.0]	0.28 [0.0,1.3]	0.30 [0.0,1.2]
Gen16	const.	early-out	marg.null	99.27 [97.9,100.0]	0.30 [0.0,1.3]	0.30 [0.0,1.4]
Gen16	const.	early-out	joint.null	99.27 [97.9,100.0]	0.31 [0.0,1.3]	0.29 [0.0,1.2]
Gen16	const.	early-total	real	97.61 [95.3,100.0]	1.97 [0.0,4.2]	0.30 [0.0,1.3]
Gen16	const.	early-total	marg.null	98.60 [96.4,100.0]	0.96 [0.0,3.0]	0.29 [0.0,1.3]
Gen16	const.	early-total	joint.null	97.94 [95.3,99.8]	1.63 [0.1,4.1]	0.31 [0.0,1.3]
Gen16	const.	sameday	real	99.27 [98.0,100.0]	0.30 [0.0,1.3]	0.31 [0.0,1.3]
Gen16	const.	sameday	marg.null	99.28 [98.0,100.0]	0.28 [0.0,1.2]	0.31 [0.0,1.3]
Gen16	const.	sameday	joint.null	99.25 [98.0,100.0]	0.31 [0.0,1.3]	0.31 [0.0,1.3]
Gen16	const.	total	real	98.79 [96.9,99.9]	0.74 [0.0,2.3]	0.33 [0.0,1.4]
Gen16	const.	total	marg.null	99.14 [97.6,100.0]	0.40 [0.0,1.7]	0.32 [0.0,1.4]
Gen16	const.	total	joint.null	99.27 [97.9,100.0]	0.31 [0.0,1.3]	0.30 [0.0,1.3]
Gen16	dong	early	real	99.27 [98.6,99.8]	0.71 [0.2,1.4]	0.02 [0.0,0.1]
Gen16	dong	early	marg.null	98.83 [96.5,99.6]	1.14 [0.4,3.4]	0.02 [0.0,0.1]
Gen16	dong	early	joint.null	99.45 [99.1,99.8]	0.51 [0.2,0.9]	0.02 [0.0,0.1]
Gen16	dong	pooled	real	98.49 [97.9,99.0]	1.48 [1.0,2.0]	0.02 [0.0,0.1]
Gen16	dong	pooled	marg.null	98.36 [97.8,98.8]	1.61 [1.2,2.1]	0.02 [0.0,0.1]
Gen16	dong	pooled	joint.null	98.80 [98.1,99.3]	1.17 [0.6,1.8]	0.02 [0.0,0.1]
Gen16	dong	sameday	real	99.90 [99.8,100.0]	0.06 [0.0,0.2]	0.02 [0.0,0.1]
Gen16	dong	sameday	marg.null	99.76 [99.5,100.0]	0.21 [0.0,0.4]	0.02 [0.0,0.1]
Gen16	dong	sameday	joint.null	99.87 [99.7,100.0]	0.10 [0.0,0.2]	0.02 [0.0,0.1]
Pres17	const.	early	real	98.75 [96.3,100.0]	0.78 [0.0,3.3]	0.28 [0.0,1.2]
Pres17	const.	early	marg.null	98.90 [93.7,100.0]	0.65 [0.0,6.0]	0.28 [0.0,1.1]
Pres17	const.	early	joint.null	98.84 [97.2,99.9]	0.77 [0.0,2.1]	0.29 [0.0,1.2]
Pres17	const.	early-out	real	99.30 [98.1,100.0]	0.29 [0.0,1.3]	0.27 [0.0,1.2]

Table 12 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres17	const.	early-out	marg.null	99.16 [97.8,100.0]	0.45 [0.0,1.6]	0.27 [0.0,1.2]
Pres17	const.	early-out	joint.null	98.77 [96.9,100.0]	0.83 [0.0,2.7]	0.26 [0.0,1.1]
Pres17	const.	early-total	real	99.32 [98.1,100.0]	0.29 [0.0,1.2]	0.29 [0.0,1.2]
Pres17	const.	early-total	marg.null	99.16 [97.8,100.0]	0.46 [0.0,1.7]	0.25 [0.0,1.1]
Pres17	const.	early-total	joint.null	99.13 [97.2,100.0]	0.42 [0.0,2.3]	0.28 [0.0,1.2]
Pres17	const.	sameday	real	99.30 [98.0,100.0]	0.30 [0.0,1.2]	0.30 [0.0,1.2]
Pres17	const.	sameday	marg.null	99.10 [96.4,100.0]	0.44 [0.0,3.2]	0.27 [0.0,1.2]
Pres17	const.	sameday	joint.null	99.32 [98.0,100.0]	0.28 [0.0,1.3]	0.27 [0.0,1.2]
Pres17	const.	total	real	84.87 [78.7,90.0]	14.75 [9.4,20.5]	0.27 [0.0,1.2]
Pres17	const.	total	marg.null	87.14 [76.6,97.4]	12.48 [2.5,23.1]	0.27 [0.0,1.2]
Pres17	const.	total	joint.null	87.00 [78.0,95.0]	12.72 [4.4,21.4]	0.28 [0.0,1.3]
Pres17	dong	early	real	99.66 [99.1,100.0]	0.31 [0.0,0.9]	0.02 [0.0,0.1]
Pres17	dong	early	marg.null	99.85 [98.4,100.0]	0.12 [0.0,1.6]	0.02 [0.0,0.1]
Pres17	dong	early	joint.null	99.47 [99.0,99.9]	0.51 [0.1,1.0]	0.02 [0.0,0.1]
Pres17	dong	pooled	real	90.92 [83.0,97.7]	9.05 [2.3,17.0]	0.02 [0.0,0.1]
Pres17	dong	pooled	marg.null	91.87 [88.4,96.0]	8.10 [3.9,11.6]	0.02 [0.0,0.1]
Pres17	dong	pooled	joint.null	95.48 [86.1,97.5]	4.50 [2.5,13.9]	0.02 [0.0,0.1]
Pres17	dong	sameday	real	99.88 [94.6,100.0]	0.09 [0.0,5.4]	0.02 [0.0,0.1]
Pres17	dong	sameday	marg.null	96.99 [94.8,99.9]	3.00 [0.0,5.1]	0.02 [0.0,0.1]
Pres17	dong	sameday	joint.null	99.87 [97.3,100.0]	0.10 [0.0,2.6]	0.02 [0.0,0.1]
Gen20	const.	early	real	98.63 [95.0,100.0]	0.90 [0.0,4.5]	0.28 [0.0,1.1]
Gen20	const.	early	marg.null	97.19 [94.3,99.3]	2.39 [0.4,5.0]	0.28 [0.0,1.2]
Gen20	const.	early	joint.null	98.18 [96.0,99.8]	1.47 [0.1,3.5]	0.25 [0.0,1.1]
Gen20	const.	early-out	real	99.27 [98.0,100.0]	0.33 [0.0,1.4]	0.27 [0.0,1.1]
Gen20	const.	early-out	marg.null	99.33 [98.2,100.0]	0.28 [0.0,1.1]	0.27 [0.0,1.2]
Gen20	const.	early-out	joint.null	98.94 [97.3,99.9]	0.67 [0.0,2.1]	0.27 [0.0,1.1]
Gen20	const.	early-total	real	99.23 [97.9,100.0]	0.36 [0.0,1.4]	0.28 [0.0,1.2]
Gen20	const.	early-total	marg.null	98.76 [97.1,99.9]	0.83 [0.0,2.4]	0.29 [0.0,1.1]
Gen20	const.	early-total	joint.null	98.55 [96.8,100.0]	1.06 [0.0,2.7]	0.27 [0.0,1.2]
Gen20	const.	sameday	real	99.32 [98.1,100.0]	0.27 [0.0,1.2]	0.27 [0.0,1.3]
Gen20	const.	sameday	marg.null	99.32 [98.1,100.0]	0.27 [0.0,1.2]	0.29 [0.0,1.2]
Gen20	const.	sameday	joint.null	99.30 [98.1,100.0]	0.29 [0.0,1.3]	0.28 [0.0,1.2]
Gen20	const.	total	real	91.69 [86.5,95.6]	7.94 [4.3,13.3]	0.28 [0.0,1.1]
Gen20	const.	total	marg.null	90.93 [85.8,94.9]	8.73 [4.7,13.7]	0.26 [0.0,1.1]
Gen20	const.	total	joint.null	93.47 [89.9,96.5]	6.11 [3.2,9.7]	0.29 [0.0,1.2]

Table 12 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen20	dong	early	real	98.63 [96.3,99.7]	1.37 [0.3,3.7]	0.02 [0.0,0.1]
Gen20	dong	early	marg.null	98.72 [95.9,99.8]	1.25 [0.2,4.1]	0.02 [0.0,0.1]
Gen20	dong	early	joint.null	98.64 [96.6,99.2]	1.33 [0.8,3.4]	0.02 [0.0,0.1]
Gen20	dong	pooled	real	93.83 [93.0,94.7]	6.13 [5.3,6.9]	0.02 [0.0,0.1]
Gen20	dong	pooled	marg.null	93.62 [92.2,94.9]	6.35 [5.1,7.8]	0.02 [0.0,0.1]
Gen20	dong	pooled	joint.null	92.91 [92.0,93.8]	7.07 [6.2,8.0]	0.02 [0.0,0.1]
Gen20	dong	sameday	real	99.90 [99.8,100.0]	0.07 [0.0,0.2]	0.02 [0.0,0.1]
Gen20	dong	sameday	marg.null	99.84 [99.7,100.0]	0.13 [0.0,0.3]	0.02 [0.0,0.1]
Gen20	dong	sameday	joint.null	99.79 [99.6,99.9]	0.18 [0.0,0.4]	0.02 [0.0,0.1]
Pres22	const.	early	real	98.26 [96.0,99.9]	1.35 [0.0,3.5]	0.28 [0.0,1.2]
Pres22	const.	early	marg.null	98.27 [94.0,99.9]	1.23 [0.0,5.6]	0.28 [0.0,1.3]
Pres22	const.	early	joint.null	98.90 [97.4,99.9]	0.73 [0.0,2.1]	0.25 [0.0,1.1]
Pres22	const.	early-out	real	97.73 [94.8,99.9]	1.83 [0.1,4.7]	0.28 [0.0,1.2]
Pres22	const.	early-out	marg.null	97.70 [95.0,100.0]	1.92 [0.0,4.5]	0.29 [0.0,1.3]
Pres22	const.	early-out	joint.null	94.40 [90.4,99.6]	5.21 [0.2,9.2]	0.28 [0.0,1.1]
Pres22	const.	early-total	real	98.89 [97.0,100.0]	0.68 [0.0,2.4]	0.30 [0.0,1.2]
Pres22	const.	early-total	marg.null	98.64 [96.7,99.9]	0.99 [0.0,2.7]	0.28 [0.0,1.2]
Pres22	const.	early-total	joint.null	99.26 [98.0,100.0]	0.36 [0.0,1.4]	0.27 [0.0,1.1]
Pres22	const.	sameday	real	98.93 [97.3,100.0]	0.64 [0.0,2.0]	0.28 [0.0,1.2]
Pres22	const.	sameday	marg.null	98.53 [96.8,99.8]	1.08 [0.0,2.6]	0.27 [0.0,1.2]
Pres22	const.	sameday	joint.null	99.32 [98.0,100.0]	0.29 [0.0,1.4]	0.27 [0.0,1.2]
Pres22	const.	total	real	75.40 [61.0,89.4]	23.67 [9.8,37.9]	0.77 [0.0,2.2]
Pres22	const.	total	marg.null	75.59 [63.7,89.4]	23.45 [10.0,35.6]	0.68 [0.0,2.1]
Pres22	const.	total	joint.null	70.89 [63.2,77.7]	27.88 [20.9,36.7]	1.01 [0.0,3.0]
Pres22	dong	early	real	99.41 [99.1,99.7]	0.56 [0.3,0.9]	0.02 [0.0,0.1]
Pres22	dong	early	marg.null	99.30 [99.0,99.7]	0.67 [0.3,1.0]	0.02 [0.0,0.1]
Pres22	dong	early	joint.null	99.56 [99.3,99.8]	0.41 [0.2,0.7]	0.02 [0.0,0.1]
Pres22	dong	pooled	real	77.73 [76.1,79.5]	22.24 [20.4,23.8]	0.02 [0.0,0.1]
Pres22	dong	pooled	marg.null	77.78 [76.0,79.5]	22.12 [20.5,24.0]	0.09 [0.0,0.2]
Pres22	dong	pooled	joint.null	78.17 [76.1,86.5]	21.79 [13.3,23.6]	0.03 [0.0,0.1]
Pres22	dong	sameday	real	99.80 [99.6,99.9]	0.18 [0.1,0.3]	0.02 [0.0,0.1]
Pres22	dong	sameday	marg.null	99.75 [99.6,100.0]	0.22 [0.0,0.4]	0.02 [0.0,0.1]
Pres22	dong	sameday	joint.null	99.58 [99.2,99.8]	0.40 [0.1,0.7]	0.02 [0.0,0.1]
Gen24	const.	early	real	98.74 [97.1,99.8]	0.86 [0.1,2.3]	0.28 [0.0,1.2]
Gen24	const.	early	marg.null	97.87 [93.0,100.0]	1.62 [0.0,6.6]	0.28 [0.0,1.3]
Gen24	const.	early	joint.null	98.83 [97.0,100.0]	0.75 [0.0,2.4]	0.29 [0.0,1.3]
Gen24	const.	early-out	real	98.69 [95.2,100.0]	0.77 [0.0,4.2]	0.31 [0.0,1.3]
Gen24	const.	early-out	marg.null	98.80 [96.6,100.0]	0.73 [0.0,2.8]	0.30 [0.0,1.3]
Gen24	const.	early-out	joint.null	97.84 [94.4,99.6]	1.71 [0.2,5.1]	0.29 [0.0,1.3]
Gen24	const.	early-total	real	99.13 [97.5,100.0]	0.42 [0.0,1.9]	0.29 [0.0,1.2]
Gen24	const.	early-total	marg.null	99.18 [97.9,100.0]	0.38 [0.0,1.4]	0.30 [0.0,1.2]

Table 12 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen24	const.	early-total	joint.null	99.21 [97.9,100.0]	0.36 [0.0,1.5]	0.30 [0.0,1.2]
Gen24	const.	sameday	real	98.88 [96.2,100.0]	0.65 [0.0,3.3]	0.28 [0.0,1.3]
Gen24	const.	sameday	marg.null	99.09 [97.5,100.0]	0.48 [0.0,1.8]	0.29 [0.0,1.4]
Gen24	const.	sameday	joint.null	99.27 [98.1,100.0]	0.29 [0.0,1.2]	0.30 [0.0,1.3]
Gen24	const.	total	real	88.05 [82.1,94.5]	11.52 [4.9,17.2]	0.27 [0.0,1.1]
Gen24	const.	total	marg.null	90.32 [86.2,94.0]	9.24 [5.6,13.4]	0.27 [0.0,1.2]
Gen24	const.	total	joint.null	91.38 [83.0,96.3]	8.16 [3.8,16.9]	0.28 [0.0,1.3]
Gen24	dong	early	real	99.39 [99.1,99.7]	0.57 [0.3,0.9]	0.03 [0.0,0.1]
Gen24	dong	early	marg.null	99.22 [95.7,99.6]	0.74 [0.3,4.3]	0.02 [0.0,0.1]
Gen24	dong	early	joint.null	99.19 [94.7,99.8]	0.77 [0.2,5.3]	0.02 [0.0,0.1]
Gen24	dong	pooled	real	93.71 [92.6,94.8]	6.26 [5.2,7.3]	0.02 [0.0,0.1]
Gen24	dong	pooled	marg.null	92.83 [91.8,93.8]	7.13 [6.2,8.2]	0.03 [0.0,0.1]
Gen24	dong	pooled	joint.null	93.83 [92.7,94.8]	6.13 [5.1,7.3]	0.02 [0.0,0.1]
Gen24	dong	sameday	real	99.82 [99.6,100.0]	0.14 [0.0,0.3]	0.03 [0.0,0.1]
Gen24	dong	sameday	marg.null	99.72 [99.5,99.9]	0.24 [0.1,0.5]	0.02 [0.0,0.1]
Gen24	dong	sameday	joint.null	99.74 [99.5,100.0]	0.22 [0.0,0.4]	0.03 [0.0,0.1]
Pres25	const.	early	real	98.90 [97.2,99.9]	0.70 [0.0,2.3]	0.26 [0.0,1.1]
Pres25	const.	early	marg.null	98.31 [95.2,100.0]	1.25 [0.0,4.4]	0.27 [0.0,1.2]
Pres25	const.	early	joint.null	99.10 [97.6,100.0]	0.49 [0.0,1.9]	0.26 [0.0,1.1]
Pres25	const.	early-out	real	98.75 [96.9,99.9]	0.84 [0.0,2.4]	0.28 [0.0,1.3]
Pres25	const.	early-out	marg.null	98.88 [97.1,100.0]	0.68 [0.0,2.4]	0.28 [0.0,1.2]
Pres25	const.	early-out	joint.null	98.64 [97.0,99.7]	0.97 [0.1,2.4]	0.28 [0.0,1.2]
Pres25	const.	early-total	real	99.35 [98.2,100.0]	0.26 [0.0,1.1]	0.27 [0.0,1.2]
Pres25	const.	early-total	marg.null	99.12 [96.0,100.0]	0.41 [0.0,3.6]	0.28 [0.0,1.2]
Pres25	const.	early-total	joint.null	99.35 [98.2,100.0]	0.27 [0.0,1.1]	0.28 [0.0,1.1]
Pres25	const.	sameday	real	99.02 [97.5,100.0]	0.58 [0.0,2.0]	0.27 [0.0,1.2]
Pres25	const.	sameday	marg.null	98.86 [96.7,100.0]	0.71 [0.0,2.8]	0.27 [0.0,1.2]
Pres25	const.	sameday	joint.null	98.98 [97.3,100.0]	0.62 [0.0,2.1]	0.27 [0.0,1.1]
Pres25	const.	total	real	70.85 [64.0,91.9]	28.77 [7.6,35.4]	0.29 [0.0,1.3]
Pres25	const.	total	marg.null	77.21 [65.7,95.4]	22.06 [4.5,33.3]	0.56 [0.0,2.1]
Pres25	const.	total	joint.null	76.47 [65.9,83.5]	23.13 [16.4,34.0]	0.30 [0.0,1.2]
Pres25	dong	early	real	99.75 [99.5,99.9]	0.22 [0.1,0.4]	0.02 [0.0,0.1]
Pres25	dong	early	marg.null	99.71 [99.1,99.9]	0.26 [0.0,0.9]	0.02 [0.0,0.1]
Pres25	dong	early	joint.null	99.77 [99.6,99.9]	0.20 [0.1,0.4]	0.02 [0.0,0.1]
Pres25	dong	pooled	real	92.72 [78.6,94.1]	7.26 [5.8,21.3]	0.02 [0.0,0.1]
Pres25	dong	pooled	marg.null	80.02 [78.3,89.7]	19.95 [10.2,21.6]	0.03 [0.0,0.1]
Pres25	dong	pooled	joint.null	80.77 [79.5,82.1]	19.21 [17.8,20.5]	0.02 [0.0,0.1]
Pres25	dong	sameday	real	99.27 [98.9,99.7]	0.70 [0.3,1.1]	0.02 [0.0,0.1]
Pres25	dong	sameday	marg.null	98.85 [98.4,99.3]	1.12 [0.7,1.5]	0.02 [0.0,0.1]
Pres25	dong	sameday	joint.null	99.17 [98.7,99.5]	0.80 [0.4,1.2]	0.02 [0.0,0.1]

A.2 Mebane (2023) multimodality diagnostics

Tables 13–16 report $M(\pi_j)$ and $D(\pi_j)$, the multimodality diagnostics proposed by Mebane [2023] and discussed in Section 4.6, for each mixture component $j \in \{1, 2, 3\}$ (no-fraud, incremental fraud, extreme fraud) and every run listed in Tables 5–8. $M(\pi_j)$ is the largest absolute difference between chain-specific posterior means for component j ; $D(\pi_j)$ is the Hartigan dip-test p -value for unimodality on that component’s pooled posterior draws across chains. Cells marked † satisfy $M(\pi_j) > 0.01$ with $D(\pi_j) < 0.05$, the joint threshold Mebane [2023] treats as a substantive multimodality signal rather than an ordinary sampling artifact; as Section 4.6 shows, this flag fires at comparable rates on real and fraud-free null data, so its presence here should not be read as evidence of fraud in any individual row.

Table 13: Mebane (2023) multimodality diagnostics, long-chain configuration, Democratic-coded leader. Each cell reports $M(\pi_j)$ (largest cross-chain gap in chain-specific posterior means) and $D(\pi_j)$ (Hartigan dip-test p -value on pooled draws) for each mixture component j . † marks $M > 0.01$ with $D < 0.05$, the threshold Mebane (2023) treats as a substantive multimodality signal.

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen08	const.	total-no-early	real	0.0003/0.994	0.0004/1.000	0.0004/1.000
Gen08	const.	total-no-early	marg.null	0.0010/0.991	0.0006/0.998	0.0003/0.998
Gen08	const.	total-no-early	joint.null	0.0005/0.994	0.0004/0.998	0.0002/0.999
Gen08	dong	total-no-early	real	0.0007/0.993	0.0007/0.997	0.0000/1.000
Gen08	dong	total-no-early	marg.null	0.0029/0.775	0.0029/0.995	0.0000/0.996
Gen08	dong	total-no-early	joint.null	0.0015/1.000	0.0015/0.996	0.0000/1.000
Gen12	const.	total-no-early	real	0.0007/1.000	0.0005/0.996	0.0004/0.999
Gen12	const.	total-no-early	marg.null	0.0004/0.997	0.0003/0.996	0.0003/1.000
Gen12	const.	total-no-early	joint.null	0.0005/1.000	0.0003/1.000	0.0004/1.000
Gen12	dong	total-no-early	real	0.0040/0.992	0.0041/0.993	0.0001/0.999
Gen12	dong	total-no-early	marg.null	0.0021/0.996	0.0021/0.997	0.0000/0.992
Gen12	dong	total-no-early	joint.null	0.0027/0.999	0.0027/0.784	0.0000/0.963
Pres02	const.	total-no-early	real	0.0411/0.993	0.0414/1.000	0.0005/0.995
Pres02	const.	total-no-early	marg.null	0.0797/0.000 †	0.0783/0.000 †	0.0044/0.999
Pres02	const.	total-no-early	joint.null	0.0361/0.995	0.0361/1.000	0.0007/0.991
Pres02	dong	total-no-early	real	0.0376/0.999	0.0364/1.000	0.0014/0.998
Pres02	dong	total-no-early	marg.null	0.0118/1.000	0.0103/0.981	0.0014/0.996
Pres02	dong	total-no-early	joint.null	0.0182/0.996	0.0179/1.000	0.0003/0.996

Table 13 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres07	const.	total-no-early	real	0.0364/1.000	0.0367/1.000	0.0004/1.000
Pres07	const.	total-no-early	marg.null	0.0721/1.000	0.0724/0.813	0.0005/1.000
Pres07	const.	total-no-early	joint.null	0.0971/0.004 [†]	0.0972/0.001 [†]	0.0001/0.993
Pres07	dong	total-no-early	real	0.0172/1.000	0.0169/0.998	0.0003/1.000
Pres07	dong	total-no-early	marg.null	0.0561/0.000 [†]	0.0561/0.000 [†]	0.0000/1.000
Pres07	dong	total-no-early	joint.null	0.0118/0.995	0.0118/1.000	0.0000/1.000
Pres12	const.	total-no-early	real	0.0597/0.999	0.0597/0.998	0.0045/0.996
Pres12	const.	total-no-early	marg.null	0.0918/0.973	0.0917/0.934	0.0005/0.996
Pres12	const.	total-no-early	joint.null	0.0598/0.998	0.0595/1.000	0.0005/1.000
Pres12	dong	total-no-early	real	0.0448/0.000 [†]	0.0451/0.000 [†]	0.0005/0.971
Pres12	dong	total-no-early	marg.null	0.0162/0.995	0.0154/0.639	0.0020/0.997
Pres12	dong	total-no-early	joint.null	0.0071/0.997	0.0065/1.000	0.0009/0.638
Gen16	const.	early	real	0.0129/0.995	0.0131/1.000	0.0007/1.000
Gen16	const.	early	marg.null	0.0302/0.998	0.0299/0.994	0.0004/0.998
Gen16	const.	early	joint.null	0.0596/0.000 [†]	0.0593/0.000 [†]	0.0007/0.997
Gen16	const.	early-out	real	0.0092/0.998	0.0089/0.947	0.0004/1.000
Gen16	const.	early-out	marg.null	0.0046/0.996	0.0047/0.995	0.0008/0.997
Gen16	const.	early-out	joint.null	0.0041/0.990	0.0039/0.997	0.0004/1.000
Gen16	const.	early-total	real	0.0262/0.998	0.0256/0.995	0.0007/1.000
Gen16	const.	early-total	marg.null	0.0473/0.999	0.0468/0.998	0.0005/0.994
Gen16	const.	early-total	joint.null	0.0211/0.973	0.0213/0.995	0.0002/0.995
Gen16	const.	sameday	real	0.0006/0.807	0.0006/0.995	0.0004/0.999
Gen16	const.	sameday	marg.null	0.0006/0.992	0.0005/0.996	0.0001/1.000
Gen16	const.	sameday	joint.null	0.0004/0.998	0.0002/1.000	0.0003/0.997
Gen16	const.	total	real	0.0083/0.985	0.0083/0.993	0.0001/1.000
Gen16	const.	total	marg.null	0.0220/1.000	0.0219/0.998	0.0008/0.997
Gen16	const.	total	joint.null	0.0087/0.996	0.0082/0.999	0.0005/1.000
Gen16	dong	early	real	0.0114/0.000 [†]	0.0114/0.000 [†]	0.0000/1.000
Gen16	dong	early	marg.null	0.0019/1.000	0.0019/0.997	0.0000/0.996
Gen16	dong	early	joint.null	0.0098/0.000	0.0098/0.000	0.0000/1.000
Gen16	dong	pooled	real	0.0025/0.998	0.0025/0.993	0.0001/0.999
Gen16	dong	pooled	marg.null	0.0022/0.997	0.0022/1.000	0.0000/0.998
Gen16	dong	pooled	joint.null	0.0016/0.997	0.0016/0.992	0.0000/1.000
Gen16	dong	sameday	real	0.0003/0.975	0.0003/1.000	0.0000/0.997
Gen16	dong	sameday	marg.null	0.0006/1.000	0.0006/0.999	0.0000/0.992
Gen16	dong	sameday	joint.null	0.0000/0.936	0.0000/1.000	0.0000/1.000
Pres17	const.	early	real	0.0163/0.998	0.0162/0.998	0.0003/1.000
Pres17	const.	early	marg.null	0.0160/0.970	0.0158/0.992	0.0003/1.000
Pres17	const.	early	joint.null	0.0767/0.000 [†]	0.0765/0.000 [†]	0.0003/1.000
Pres17	const.	early-out	real	0.0672/0.000 [†]	0.0671/0.000 [†]	0.0001/0.997

Table 13 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres17	const.	early-out	marg.null	0.0318/0.994	0.0319/0.495	0.0002/0.998
Pres17	const.	early-out	joint.null	0.0497/0.000 [†]	0.0502/0.000 [†]	0.0005/0.999
Pres17	const.	early-total	real	0.0005/1.000	0.0001/1.000	0.0004/1.000
Pres17	const.	early-total	marg.null	0.0006/1.000	0.0002/0.998	0.0004/1.000
Pres17	const.	early-total	joint.null	0.0038/0.995	0.0039/0.998	0.0002/0.997
Pres17	const.	sameday	real	0.0002/1.000	0.0003/1.000	0.0002/1.000
Pres17	const.	sameday	marg.null	0.0007/1.000	0.0004/0.993	0.0004/1.000
Pres17	const.	sameday	joint.null	0.0005/0.984	0.0004/1.000	0.0001/1.000
Pres17	const.	total	real	0.1199/0.000 [†]	0.1199/0.000 [†]	0.0001/1.000
Pres17	const.	total	marg.null	0.0287/0.996	0.0285/0.997	0.0004/0.998
Pres17	const.	total	joint.null	0.1914/0.000 [†]	0.1918/0.000 [†]	0.0004/0.998
Pres17	dong	early	real	0.0099/0.000	0.0099/0.000	0.0000/0.991
Pres17	dong	early	marg.null	0.0113/0.000 [†]	0.0113/0.000 [†]	0.0000/1.000
Pres17	dong	early	joint.null	0.0025/0.970	0.0025/0.996	0.0000/1.000
Pres17	dong	pooled	real	0.0105/0.997	0.0105/0.991	0.0000/1.000
Pres17	dong	pooled	marg.null	0.0480/0.000 [†]	0.0480/0.000 [†]	0.0000/0.997
Pres17	dong	pooled	joint.null	0.0549/0.000 [†]	0.0549/0.000 [†]	0.0000/1.000
Pres17	dong	sameday	real	0.0006/0.993	0.0006/0.995	0.0000/0.993
Pres17	dong	sameday	marg.null	0.0003/0.966	0.0003/0.992	0.0000/1.000
Pres17	dong	sameday	joint.null	0.0009/0.991	0.0009/0.997	0.0000/0.995
Gen20	const.	early	real	0.0319/0.991	0.0319/0.998	0.0002/1.000
Gen20	const.	early	marg.null	0.0314/1.000	0.0318/1.000	0.0005/1.000
Gen20	const.	early	joint.null	0.0516/0.170	0.0518/0.062	0.0003/0.998
Gen20	const.	early-out	real	0.0197/0.997	0.0197/1.000	0.0002/0.999
Gen20	const.	early-out	marg.null	0.0115/0.996	0.0116/0.997	0.0002/1.000
Gen20	const.	early-out	joint.null	0.0244/0.994	0.0240/0.995	0.0004/0.999
Gen20	const.	early-total	real	0.0156/0.998	0.0156/0.998	0.0002/1.000
Gen20	const.	early-total	marg.null	0.0086/0.950	0.0085/1.000	0.0003/0.998
Gen20	const.	early-total	joint.null	0.0124/0.997	0.0120/0.997	0.0005/1.000
Gen20	const.	sameday	real	0.0040/1.000	0.0040/1.000	0.0002/1.000
Gen20	const.	sameday	marg.null	0.0007/0.997	0.0004/1.000	0.0004/0.999
Gen20	const.	sameday	joint.null	0.0077/0.995	0.0077/1.000	0.0003/1.000
Gen20	const.	total	real	0.0541/1.000	0.0541/0.999	0.0001/1.000
Gen20	const.	total	marg.null	0.0389/0.992	0.0393/0.998	0.0007/0.999
Gen20	const.	total	joint.null	0.0831/0.000 [†]	0.0834/0.000 [†]	0.0005/1.000
Gen20	dong	early	real	0.0136/0.959	0.0136/0.996	0.0000/0.996
Gen20	dong	early	marg.null	0.0043/0.999	0.0042/1.000	0.0000/0.997
Gen20	dong	early	joint.null	0.0482/0.000 [†]	0.0482/0.000 [†]	0.0001/0.997
Gen20	dong	pooled	real	0.0045/0.999	0.0046/0.995	0.0001/0.996
Gen20	dong	pooled	marg.null	0.0037/0.998	0.0037/0.997	0.0000/0.999
Gen20	dong	pooled	joint.null	0.0055/0.996	0.0055/1.000	0.0000/1.000
Gen20	dong	sameday	real	0.0012/0.999	0.0012/0.992	0.0000/0.999
Gen20	dong	sameday	marg.null	0.0006/0.992	0.0006/0.991	0.0000/0.997
Gen20	dong	sameday	joint.null	0.0015/0.994	0.0014/0.989	0.0000/0.992
Pres22	const.	early	real	0.0128/0.998	0.0124/0.995	0.0004/1.000
Pres22	const.	early	marg.null	0.0128/0.999	0.0123/0.999	0.0005/0.997
Pres22	const.	early	joint.null	0.0080/0.997	0.0082/0.998	0.0003/0.998
Pres22	const.	early-out	real	0.0282/0.995	0.0284/1.000	0.0003/1.000
Pres22	const.	early-out	marg.null	0.0243/0.997	0.0242/0.991	0.0010/1.000

Table 13 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres22	const.	early-out	joint.null	0.0721/0.000 [†]	0.0719/0.000 [†]	0.0003/0.992
Pres22	const.	early-total	real	0.0248/0.999	0.0250/0.998	0.0003/1.000
Pres22	const.	early-total	marg.null	0.0165/0.999	0.0162/1.000	0.0004/1.000
Pres22	const.	early-total	joint.null	0.0480/1.000	0.0482/1.000	0.0004/0.997
Pres22	const.	sameday	real	0.0204/0.999	0.0202/1.000	0.0002/1.000
Pres22	const.	sameday	marg.null	0.0041/0.990	0.0044/1.000	0.0003/0.992
Pres22	const.	sameday	joint.null	0.0158/0.997	0.0157/0.994	0.0005/1.000
Pres22	const.	total	real	0.2053/0.000 [†]	0.1969/0.000 [†]	0.0368/0.999
Pres22	const.	total	marg.null	0.2139/0.000 [†]	0.1492/0.000 [†]	0.0647/0.000 [†]
Pres22	const.	total	joint.null	0.1932/0.000 [†]	0.1652/0.000 [†]	0.0289/0.994
Pres22	dong	early	real	0.0865/0.000 [†]	0.0865/0.000 [†]	0.0000/1.000
Pres22	dong	early	marg.null	0.1185/0.000 [†]	0.1185/0.000 [†]	0.0000/0.997
Pres22	dong	early	joint.null	0.0077/0.994	0.0077/0.998	0.0001/1.000
Pres22	dong	pooled	real	0.0901/0.000 [†]	0.0892/0.000 [†]	0.0034/0.999
Pres22	dong	pooled	marg.null	0.1008/0.000 [†]	0.0985/0.000 [†]	0.0031/0.000
Pres22	dong	pooled	joint.null	0.0178/0.998	0.0183/0.998	0.0014/0.998
Pres22	dong	sameday	real	0.0016/0.999	0.0016/0.994	0.0000/0.997
Pres22	dong	sameday	marg.null	0.0022/0.981	0.0022/0.997	0.0000/0.999
Pres22	dong	sameday	joint.null	0.0035/0.993	0.0035/0.987	0.0000/0.999
Gen24	const.	early	real	0.0163/1.000	0.0161/0.995	0.0003/1.000
Gen24	const.	early	marg.null	0.0366/1.000	0.0363/0.998	0.0005/0.999
Gen24	const.	early	joint.null	0.0407/1.000	0.0407/0.999	0.0002/0.999
Gen24	const.	early-out	real	0.0286/0.041 [†]	0.0282/0.000 [†]	0.0004/0.997
Gen24	const.	early-out	marg.null	0.0527/0.995	0.0526/0.991	0.0004/0.999
Gen24	const.	early-out	joint.null	0.0205/0.999	0.0203/0.928	0.0003/1.000
Gen24	const.	early-total	real	0.0408/0.000 [†]	0.0408/0.000 [†]	0.0001/1.000
Gen24	const.	early-total	marg.null	0.0082/0.997	0.0083/0.994	0.0004/1.000
Gen24	const.	early-total	joint.null	0.0410/0.996	0.0408/0.940	0.0004/1.000
Gen24	const.	sameday	real	0.0124/0.995	0.0123/0.999	0.0003/1.000
Gen24	const.	sameday	marg.null	0.0121/1.000	0.0120/1.000	0.0004/1.000
Gen24	const.	sameday	joint.null	0.0206/0.993	0.0207/0.999	0.0003/0.996
Gen24	const.	total	real	0.1181/0.000 [†]	0.1179/0.000 [†]	0.0002/1.000
Gen24	const.	total	marg.null	0.0900/0.000 [†]	0.0898/0.000 [†]	0.0005/1.000
Gen24	const.	total	joint.null	0.1129/0.000 [†]	0.1130/0.000 [†]	0.0001/0.997
Gen24	dong	early	real	0.0322/0.000 [†]	0.0322/0.000 [†]	0.0001/0.998
Gen24	dong	early	marg.null	0.1417/0.000 [†]	0.1417/0.000 [†]	0.0000/1.000
Gen24	dong	early	joint.null	0.0199/0.001 [†]	0.0199/0.001 [†]	0.0000/1.000
Gen24	dong	pooled	real	0.0099/0.994	0.0099/0.996	0.0001/1.000
Gen24	dong	pooled	marg.null	0.0109/0.999	0.0109/0.996	0.0000/1.000
Gen24	dong	pooled	joint.null	0.0140/0.997	0.0139/0.999	0.0001/1.000
Gen24	dong	sameday	real	0.0024/0.994	0.0025/0.998	0.0000/1.000
Gen24	dong	sameday	marg.null	0.0132/0.000 [†]	0.0132/0.000 [†]	0.0000/1.000
Gen24	dong	sameday	joint.null	0.0019/0.992	0.0018/0.995	0.0001/1.000
Pres25	const.	early	real	0.1152/0.000 [†]	0.1148/0.000 [†]	0.0004/0.997
Pres25	const.	early	marg.null	0.0705/0.995	0.0703/0.997	0.0002/1.000
Pres25	const.	early	joint.null	0.0317/0.929	0.0318/0.995	0.0002/1.000
Pres25	const.	early-out	real	0.0626/0.000 [†]	0.0621/0.000 [†]	0.0004/0.999
Pres25	const.	early-out	marg.null	0.0832/0.000 [†]	0.0831/0.000 [†]	0.0002/1.000

Table 13 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres25	const.	early-out	joint.null	0.0910/0.000 [†]	0.0912/0.000 [†]	0.0002/1.000
Pres25	const.	early-total	real	0.0433/1.000	0.0393/0.998	0.0041/1.000
Pres25	const.	early-total	marg.null	0.0237/0.993	0.0197/1.000	0.0040/0.998
Pres25	const.	early-total	joint.null	0.0382/0.995	0.0383/1.000	0.0002/0.988
Pres25	const.	sameday	real	0.0122/0.999	0.0121/0.998	0.0003/1.000
Pres25	const.	sameday	marg.null	0.0159/0.996	0.0156/0.993	0.0004/1.000
Pres25	const.	sameday	joint.null	0.0314/0.996	0.0317/0.995	0.0003/0.995
Pres25	const.	total	real	0.1772/0.000 [†]	0.1483/0.000 [†]	0.0455/0.000 [†]
Pres25	const.	total	marg.null	0.0265/1.000	0.0279/1.000	0.0124/0.997
Pres25	const.	total	joint.null	0.2315/0.000 [†]	0.1660/0.000 [†]	0.0702/0.000 [†]
Pres25	dong	early	real	0.0045/1.000	0.0046/1.000	0.0000/0.996
Pres25	dong	early	marg.null	0.0084/0.999	0.0084/0.999	0.0000/0.992
Pres25	dong	early	joint.null	0.0044/1.000	0.0044/1.000	0.0000/0.998
Pres25	dong	pooled	real	0.1345/0.000 [†]	0.1303/0.000 [†]	0.0042/0.000
Pres25	dong	pooled	marg.null	0.1213/0.000 [†]	0.1176/0.000 [†]	0.0037/0.000
Pres25	dong	pooled	joint.null	0.1346/0.000 [†]	0.1309/0.000 [†]	0.0051/0.000
Pres25	dong	sameday	real	0.0014/0.999	0.0014/0.999	0.0000/0.982
Pres25	dong	sameday	marg.null	0.0009/1.000	0.0010/0.992	0.0001/0.999
Pres25	dong	sameday	joint.null	0.0008/0.996	0.0008/1.000	0.0000/0.995

Table 14: Mebane (2023) multimodality diagnostics, long-chain configuration, conservative-coded leader (partial).

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres12	const.	total-no-early	real	0.0402/0.988	0.0400/0.992	0.0004/0.997
Pres12	const.	total-no-early	marg.null	0.1386/0.000 [†]	0.1386/0.000 [†]	0.0003/0.997
Pres12	const.	total-no-early	joint.null	0.1256/0.000 [†]	0.1258/0.000 [†]	0.0002/0.996
Pres12	dong	total-no-early	real	0.0265/0.995	0.0263/0.923	0.0008/0.996
Pres12	dong	total-no-early	marg.null	0.1012/0.000 [†]	0.1012/0.000 [†]	0.0015/0.000
Pres12	dong	total-no-early	joint.null	0.1096/0.000 [†]	0.1087/0.000 [†]	0.0009/0.005
Gen20	const.	early	real	0.0351/0.995	0.0350/0.996	0.0003/0.995
Gen20	const.	early	marg.null	0.0323/0.999	0.0320/0.996	0.0004/1.000
Gen20	const.	early	joint.null	0.0114/1.000	0.0116/0.994	0.0006/0.997
Gen20	const.	early-out	real	0.0169/0.997	0.0165/0.997	0.0005/0.991
Gen20	const.	early-out	marg.null	0.0116/0.996	0.0118/0.878	0.0004/1.000
Gen20	const.	early-out	joint.null	0.0085/0.995	0.0083/0.994	0.0004/0.997
Gen20	const.	early-total	real	0.0083/0.998	0.0079/0.994	0.0004/0.999
Gen20	const.	early-total	marg.null	0.0118/1.000	0.0117/0.993	0.0003/0.994
Gen20	const.	early-total	joint.null	0.0039/0.999	0.0041/0.995	0.0003/0.993
Gen20	const.	sameday	real	0.0037/0.999	0.0039/0.994	0.0004/1.000
Gen20	const.	sameday	marg.null	0.0037/0.998	0.0038/0.983	0.0005/1.000
Gen20	const.	sameday	joint.null	0.0013/0.997	0.0010/0.997	0.0004/0.997
Gen20	const.	total	real	0.0518/1.000	0.0520/0.996	0.0002/0.990
Gen20	const.	total	marg.null	0.0246/0.985	0.0234/0.987	0.0012/1.000
Gen20	const.	total	joint.null	0.0120/0.993	0.0118/0.999	0.0002/0.997
Gen20	dong	early	real	0.0033/0.993	0.0033/0.998	0.0000/0.997
Gen20	dong	early	marg.null	0.0031/0.997	0.0031/0.991	0.0000/1.000
Gen20	dong	early	joint.null	0.0042/0.995	0.0042/0.999	0.0000/0.991
Gen20	dong	pooled	real	0.0102/0.984	0.0103/0.994	0.0000/0.994
Gen20	dong	pooled	marg.null	0.0106/0.999	0.0106/1.000	0.0000/0.995
Gen20	dong	pooled	joint.null	0.0070/0.994	0.0069/0.997	0.0001/0.995
Gen20	dong	sameday	real	0.0009/0.977	0.0009/0.995	0.0000/1.000
Gen20	dong	sameday	marg.null	0.0011/0.994	0.0010/0.999	0.0000/1.000
Gen20	dong	sameday	joint.null	0.0006/0.993	0.0006/1.000	0.0000/1.000
Gen24	const.	early	real	0.0160/0.997	0.0162/1.000	0.0005/0.994
Gen24	const.	early	marg.null	0.0083/1.000	0.0083/0.997	0.0003/0.996
Gen24	const.	early	joint.null	0.0117/1.000	0.0121/0.990	0.0004/1.000
Gen24	const.	early-out	real	0.0045/0.997	0.0041/0.998	0.0004/0.999
Gen24	const.	early-out	marg.null	0.0208/0.907	0.0205/0.999	0.0003/1.000
Gen24	const.	early-out	joint.null	0.0080/0.996	0.0079/0.999	0.0003/0.997
Gen24	const.	early-total	real	0.0045/1.000	0.0042/1.000	0.0003/0.996
Gen24	const.	early-total	marg.null	0.0041/0.996	0.0041/1.000	0.0001/1.000
Gen24	const.	early-total	joint.null	0.0085/0.999	0.0082/0.998	0.0004/0.998
Gen24	const.	sameday	real	0.0042/1.000	0.0043/0.997	0.0004/1.000

Table 14 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen24	const.	sameday	marg.null	0.0043/1.000	0.0042/0.998	0.0003/1.000
Gen24	const.	sameday	joint.null	0.0046/0.991	0.0045/0.994	0.0002/1.000
Gen24	const.	total	real	0.0821/0.000 [†]	0.0820/0.000 [†]	0.0002/0.996
Gen24	const.	total	marg.null	0.0543/0.998	0.0538/0.997	0.0005/0.998
Gen24	const.	total	joint.null	0.0523/0.846	0.0521/0.630	0.0004/1.000
Gen24	dong	early	real	0.0134/0.000 [†]	0.0134/0.000 [†]	0.0000/1.000
Gen24	dong	early	marg.null	0.0155/0.000 [†]	0.0155/0.000 [†]	0.0000/0.992
Gen24	dong	early	joint.null	0.0116/0.000 [†]	0.0117/0.000 [†]	0.0001/1.000
Gen24	dong	pooled	real	0.0067/0.999	0.0067/1.000	0.0000/1.000
Gen24	dong	pooled	marg.null	0.0157/0.997	0.0157/0.999	0.0000/1.000
Gen24	dong	pooled	joint.null	0.0065/1.000	0.0065/0.992	0.0000/1.000
Gen24	dong	sameday	real	0.0022/0.994	0.0022/0.998	0.0000/1.000
Gen24	dong	sameday	marg.null	0.0008/0.991	0.0008/1.000	0.0001/1.000
Gen24	dong	sameday	joint.null	0.0013/0.993	0.0013/0.998	0.0000/1.000

Table 15: Mebane (2023) multimodality diagnostics, short-chain configuration, Democratic-coded leader. Full coverage, all 11 elections.

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen08	const.	total-no-early	real	0.0014/0.847	0.0014/0.994	0.0007/0.995
Gen08	const.	total-no-early	marg.null	0.0013/0.997	0.0009/0.985	0.0008/0.996
Gen08	const.	total-no-early	joint.null	0.0098/0.494	0.0101/0.998	0.0010/0.994
Gen08	dong	total-no-early	real	0.0011/0.991	0.0011/0.993	0.0000/1.000
Gen08	dong	total-no-early	marg.null	0.0034/0.930	0.0034/0.994	0.0000/0.996
Gen08	dong	total-no-early	joint.null	0.0019/0.994	0.0019/0.998	0.0001/0.991
Gen12	const.	total-no-early	real	0.0008/0.996	0.0005/0.988	0.0012/0.996
Gen12	const.	total-no-early	marg.null	0.0008/0.997	0.0007/0.995	0.0004/0.997
Gen12	const.	total-no-early	joint.null	0.0005/0.981	0.0003/0.995	0.0005/0.999
Gen12	dong	total-no-early	real	0.0032/0.992	0.0032/0.996	0.0001/0.995
Gen12	dong	total-no-early	marg.null	0.0012/0.991	0.0012/0.994	0.0001/0.996
Gen12	dong	total-no-early	joint.null	0.0011/0.992	0.0010/0.886	0.0001/0.987
Pres02	const.	total-no-early	real	0.0570/0.990	0.0563/0.877	0.0009/0.954
Pres02	const.	total-no-early	marg.null	0.0550/0.984	0.0551/0.994	0.0084/0.995
Pres02	const.	total-no-early	joint.null	0.0573/0.913	0.0572/0.882	0.0008/0.994
Pres02	dong	total-no-early	real	0.0502/0.000 [†]	0.0514/0.000 [†]	0.0012/1.000
Pres02	dong	total-no-early	marg.null	0.0557/0.000 [†]	0.0540/0.000 [†]	0.0017/0.000
Pres02	dong	total-no-early	joint.null	0.0082/0.993	0.0079/0.981	0.0006/0.998
Pres07	const.	total-no-early	real	0.0223/0.996	0.0235/0.999	0.0018/0.994
Pres07	const.	total-no-early	marg.null	0.0847/0.992	0.0845/0.915	0.0005/0.993
Pres07	const.	total-no-early	joint.null	0.0537/0.996	0.0536/0.994	0.0012/0.997
Pres07	dong	total-no-early	real	0.0381/0.000 [†]	0.0382/0.000 [†]	0.0001/0.991
Pres07	dong	total-no-early	marg.null	0.0544/0.000 [†]	0.0544/0.000 [†]	0.0000/0.962
Pres07	dong	total-no-early	joint.null	0.0049/0.996	0.0050/0.673	0.0000/0.983
Pres12	const.	total-no-early	real	0.0911/0.253	0.0867/0.916	0.0046/0.992
Pres12	const.	total-no-early	marg.null	0.1128/0.073	0.1083/0.017 [†]	0.0047/0.951

Table 15 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres12	const.	total-no-early	joint.null	0.0278/0.994	0.0297/0.914	0.0021/0.985
Pres12	dong	total-no-early	real	0.0481/0.000 [†]	0.0472/0.000 [†]	0.0009/0.903
Pres12	dong	total-no-early	marg.null	0.0666/0.000 [†]	0.0646/0.000 [†]	0.0020/0.898
Pres12	dong	total-no-early	joint.null	0.0111/0.994	0.0127/0.957	0.0020/1.000
Gen16	const.	early	real	0.0052/0.799	0.0048/0.998	0.0004/0.999
Gen16	const.	early	marg.null	0.0134/0.999	0.0128/1.000	0.0007/0.995
Gen16	const.	early	joint.null	0.0169/0.885	0.0175/0.300	0.0009/0.999
Gen16	const.	early-out	real	0.0039/0.999	0.0042/0.544	0.0004/0.999
Gen16	const.	early-out	marg.null	0.0048/0.994	0.0050/0.997	0.0009/0.998
Gen16	const.	early-out	joint.null	0.0009/0.995	0.0004/1.000	0.0008/0.997
Gen16	const.	early-total	real	0.0198/1.000	0.0215/0.000 [†]	0.0017/0.999
Gen16	const.	early-total	marg.null	0.0242/0.663	0.0241/0.878	0.0007/0.960
Gen16	const.	early-total	joint.null	0.0514/0.999	0.0513/0.993	0.0003/0.991
Gen16	const.	sameday	real	0.0012/0.995	0.0009/0.997	0.0006/0.993
Gen16	const.	sameday	marg.null	0.0004/0.994	0.0006/0.993	0.0007/0.996
Gen16	const.	sameday	joint.null	0.0014/0.993	0.0015/0.911	0.0018/0.993
Gen16	const.	total	real	0.0049/0.891	0.0046/0.993	0.0009/0.997
Gen16	const.	total	marg.null	0.0091/1.000	0.0088/0.993	0.0006/1.000
Gen16	const.	total	joint.null	0.0083/0.993	0.0085/0.995	0.0004/0.984
Gen16	dong	early	real	0.0030/1.000	0.0030/0.995	0.0000/0.991
Gen16	dong	early	marg.null	0.0103/0.000 [†]	0.0102/0.000 [†]	0.0001/0.997
Gen16	dong	early	joint.null	0.0078/0.000	0.0078/0.000	0.0000/1.000
Gen16	dong	pooled	real	0.0017/0.992	0.0016/0.996	0.0001/0.998
Gen16	dong	pooled	marg.null	0.0028/0.903	0.0028/0.992	0.0001/0.997
Gen16	dong	pooled	joint.null	0.0016/0.942	0.0016/0.998	0.0001/0.996
Gen16	dong	sameday	real	0.0001/0.996	0.0000/1.000	0.0001/0.997
Gen16	dong	sameday	marg.null	0.0003/0.990	0.0003/0.997	0.0000/0.997
Gen16	dong	sameday	joint.null	0.0001/0.988	0.0001/0.997	0.0001/0.998
Pres17	const.	early	real	0.0118/0.998	0.0117/0.995	0.0004/0.991
Pres17	const.	early	marg.null	0.0319/0.993	0.0323/0.543	0.0008/0.993
Pres17	const.	early	joint.null	0.0369/0.606	0.0372/0.000 [†]	0.0006/0.912
Pres17	const.	early-out	real	0.0484/1.000	0.0487/0.997	0.0006/0.999
Pres17	const.	early-out	marg.null	0.0313/0.981	0.0313/0.179	0.0003/0.997
Pres17	const.	early-out	joint.null	0.0617/0.702	0.0618/0.989	0.0006/0.995
Pres17	const.	early-total	real	0.0569/0.000 [†]	0.0565/0.000 [†]	0.0004/0.991
Pres17	const.	early-total	marg.null	0.0009/0.998	0.0012/0.999	0.0006/1.000
Pres17	const.	early-total	joint.null	0.0082/0.906	0.0083/0.899	0.0007/0.999
Pres17	const.	sameday	real	0.0008/0.810	0.0008/0.995	0.0003/1.000
Pres17	const.	sameday	marg.null	0.0019/1.000	0.0011/0.997	0.0009/0.995
Pres17	const.	sameday	joint.null	0.0011/0.994	0.0011/0.983	0.0004/0.992
Pres17	const.	total	real	0.1562/0.000 [†]	0.1562/0.000 [†]	0.0004/0.993
Pres17	const.	total	marg.null	0.1442/0.000 [†]	0.1438/0.000 [†]	0.0007/0.995
Pres17	const.	total	joint.null	0.1795/0.000 [†]	0.1801/0.000 [†]	0.0009/0.995
Pres17	dong	early	real	0.0182/0.000 [†]	0.0181/0.000 [†]	0.0000/0.994
Pres17	dong	early	marg.null	0.0097/0.303	0.0096/0.067	0.0000/0.998

Table 15 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres17	dong	early	joint.null	0.0021/0.996	0.0020/0.991	0.0001/0.983
Pres17	dong	pooled	real	0.0542/0.000 [†]	0.0542/0.000 [†]	0.0000/0.999
Pres17	dong	pooled	marg.null	0.0708/0.000 [†]	0.0708/0.000 [†]	0.0001/1.000
Pres17	dong	pooled	joint.null	0.0415/0.000 [†]	0.0415/0.000 [†]	0.0000/0.997
Pres17	dong	sameday	real	0.0006/0.977	0.0006/0.995	0.0001/0.995
Pres17	dong	sameday	marg.null	0.0003/0.994	0.0004/0.996	0.0000/1.000
Pres17	dong	sameday	joint.null	0.0034/0.009	0.0034/0.000	0.0000/0.993
Gen20	const.	early	real	0.0289/0.977	0.0289/0.683	0.0010/0.998
Gen20	const.	early	marg.null	0.0082/0.986	0.0078/0.995	0.0009/0.994
Gen20	const.	early	joint.null	0.0385/0.992	0.0385/0.978	0.0001/0.998
Gen20	const.	early-out	real	0.0248/0.480	0.0246/0.000 [†]	0.0010/1.000
Gen20	const.	early-out	marg.null	0.0272/0.939	0.0267/0.487	0.0005/0.993
Gen20	const.	early-out	joint.null	0.0112/0.996	0.0112/0.816	0.0009/0.996
Gen20	const.	early-total	real	0.0531/0.045 [†]	0.0529/0.002 [†]	0.0005/0.996
Gen20	const.	early-total	marg.null	0.0281/0.881	0.0280/0.994	0.0006/0.990
Gen20	const.	early-total	joint.null	0.0314/0.997	0.0319/0.997	0.0005/0.993
Gen20	const.	sameday	real	0.0044/0.994	0.0046/0.988	0.0005/0.667
Gen20	const.	sameday	marg.null	0.0044/0.992	0.0047/1.000	0.0008/0.965
Gen20	const.	sameday	joint.null	0.0048/0.995	0.0045/0.966	0.0004/0.997
Gen20	const.	total	real	0.0226/0.995	0.0228/0.992	0.0008/0.806
Gen20	const.	total	marg.null	0.0675/0.998	0.0672/0.983	0.0005/0.997
Gen20	const.	total	joint.null	0.1020/0.991	0.1021/0.931	0.0007/0.995
Gen20	dong	early	real	0.1530/0.000 [†]	0.1529/0.000 [†]	0.0000/0.996
Gen20	dong	early	marg.null	0.1397/0.000 [†]	0.1397/0.000 [†]	0.0000/0.961
Gen20	dong	early	joint.null	0.0523/0.000 [†]	0.0523/0.000 [†]	0.0000/0.999
Gen20	dong	pooled	real	0.0017/0.999	0.0018/0.996	0.0001/0.995
Gen20	dong	pooled	marg.null	0.0042/0.992	0.0042/0.998	0.0001/0.859
Gen20	dong	pooled	joint.null	0.0068/0.926	0.0068/0.776	0.0001/0.992
Gen20	dong	sameday	real	0.0005/0.876	0.0005/0.996	0.0000/0.998
Gen20	dong	sameday	marg.null	0.0019/0.968	0.0019/0.998	0.0000/0.998
Gen20	dong	sameday	joint.null	0.0020/0.991	0.0020/0.911	0.0001/0.994
Pres22	const.	early	real	0.0600/0.113	0.0595/0.001 [†]	0.0004/0.998
Pres22	const.	early	marg.null	0.0068/0.902	0.0076/0.995	0.0009/0.990
Pres22	const.	early	joint.null	0.0087/0.980	0.0085/0.994	0.0009/0.999
Pres22	const.	early-out	real	0.0293/0.993	0.0286/0.993	0.0007/0.824
Pres22	const.	early-out	marg.null	0.0532/0.000 [†]	0.0527/0.000 [†]	0.0006/0.870
Pres22	const.	early-out	joint.null	0.0696/0.000 [†]	0.0700/0.000 [†]	0.0007/0.950
Pres22	const.	early-total	real	0.0128/0.991	0.0134/0.996	0.0006/0.997
Pres22	const.	early-total	marg.null	0.0176/0.999	0.0176/0.993	0.0003/0.955
Pres22	const.	early-total	joint.null	0.0294/0.991	0.0292/0.999	0.0004/0.999
Pres22	const.	sameday	real	0.0083/1.000	0.0084/0.995	0.0003/0.982
Pres22	const.	sameday	marg.null	0.0120/1.000	0.0126/0.993	0.0006/0.996
Pres22	const.	sameday	joint.null	0.0106/0.999	0.0109/0.995	0.0003/0.997
Pres22	const.	total	real	0.1832/0.000 [†]	0.1662/0.308	0.0436/0.991
Pres22	const.	total	marg.null	0.1160/0.014 [†]	0.0992/0.000 [†]	0.0449/0.999
Pres22	const.	total	joint.null	0.0736/0.994	0.0434/0.993	0.0592/0.000 [†]
Pres22	dong	early	real	0.0808/0.000 [†]	0.0808/0.000 [†]	0.0001/0.993
Pres22	dong	early	marg.null	0.0669/0.000 [†]	0.0669/0.000 [†]	0.0001/0.997

Table 15 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres22	dong	early	joint.null	0.0662/0.000 [†]	0.0662/0.000 [†]	0.0001/0.997
Pres22	dong	pooled	real	0.0948/0.000 [†]	0.0936/0.000 [†]	0.0012/0.999
Pres22	dong	pooled	marg.null	0.0770/0.000 [†]	0.0738/0.000 [†]	0.0032/0.024
Pres22	dong	pooled	joint.null	0.1062/0.000 [†]	0.1062/0.000 [†]	0.0015/1.000
Pres22	dong	sameday	real	0.0013/0.994	0.0013/0.997	0.0000/0.974
Pres22	dong	sameday	marg.null	0.0022/0.990	0.0023/0.997	0.0004/0.958
Pres22	dong	sameday	joint.null	0.0036/0.989	0.0037/0.993	0.0001/0.989
Gen24	const.	early	real	0.0208/0.946	0.0208/0.986	0.0002/0.992
Gen24	const.	early	marg.null	0.0586/0.331	0.0592/0.158	0.0009/1.000
Gen24	const.	early	joint.null	0.0170/0.991	0.0166/0.963	0.0008/0.995
Gen24	const.	early-out	real	0.0690/0.802	0.0691/0.907	0.0007/0.989
Gen24	const.	early-out	marg.null	0.0568/0.000 [†]	0.0570/0.000 [†]	0.0005/1.000
Gen24	const.	early-out	joint.null	0.0042/0.990	0.0042/0.997	0.0003/0.993
Gen24	const.	early-total	real	0.0089/0.959	0.0088/0.995	0.0003/0.909
Gen24	const.	early-total	marg.null	0.0198/0.998	0.0199/0.798	0.0005/0.994
Gen24	const.	early-total	joint.null	0.0052/0.957	0.0047/1.000	0.0008/0.993
Gen24	const.	sameday	real	0.0053/0.993	0.0049/0.807	0.0009/0.995
Gen24	const.	sameday	marg.null	0.0166/0.966	0.0171/0.290	0.0005/0.994
Gen24	const.	sameday	joint.null	0.0081/0.993	0.0081/0.985	0.0005/0.996
Gen24	const.	total	real	0.0432/0.990	0.0474/0.991	0.0043/0.952
Gen24	const.	total	marg.null	0.0459/0.981	0.0500/0.993	0.0042/0.994
Gen24	const.	total	joint.null	0.1107/0.000 [†]	0.1110/0.000 [†]	0.0003/0.991
Gen24	dong	early	real	0.0803/0.000 [†]	0.0802/0.000 [†]	0.0001/0.871
Gen24	dong	early	marg.null	0.0907/0.000 [†]	0.0907/0.000 [†]	0.0001/0.994
Gen24	dong	early	joint.null	0.0064/0.996	0.0064/0.845	0.0001/0.998
Gen24	dong	pooled	real	0.0053/0.997	0.0053/0.999	0.0000/1.000
Gen24	dong	pooled	marg.null	0.0088/0.993	0.0089/0.996	0.0001/0.999
Gen24	dong	pooled	joint.null	0.0161/0.992	0.0162/1.000	0.0000/0.995
Gen24	dong	sameday	real	0.0016/0.992	0.0016/1.000	0.0000/0.998
Gen24	dong	sameday	marg.null	0.0025/0.994	0.0025/0.990	0.0001/0.988
Gen24	dong	sameday	joint.null	0.0035/1.000	0.0036/0.983	0.0001/0.992
Pres25	const.	early	real	0.0144/0.997	0.0146/0.982	0.0003/0.996
Pres25	const.	early	marg.null	0.0362/0.995	0.0362/0.992	0.0009/0.998
Pres25	const.	early	joint.null	0.0167/0.899	0.0163/0.973	0.0007/0.999
Pres25	const.	early-out	real	0.0641/0.000 [†]	0.0638/0.000 [†]	0.0004/0.772
Pres25	const.	early-out	marg.null	0.0672/0.000 [†]	0.0675/0.000 [†]	0.0008/0.995
Pres25	const.	early-out	joint.null	0.1100/0.114	0.1106/0.053	0.0006/1.000
Pres25	const.	early-total	real	0.0346/0.975	0.0350/0.992	0.0004/0.994
Pres25	const.	early-total	marg.null	0.0261/1.000	0.0295/0.800	0.0042/1.000
Pres25	const.	early-total	joint.null	0.0423/0.973	0.0409/0.993	0.0014/0.996
Pres25	const.	sameday	real	0.0140/0.991	0.0149/0.984	0.0009/0.992
Pres25	const.	sameday	marg.null	0.0074/0.973	0.0079/1.000	0.0006/0.996
Pres25	const.	sameday	joint.null	0.0085/0.997	0.0083/0.992	0.0004/0.998
Pres25	const.	total	real	0.0373/0.970	0.0327/0.992	0.0181/0.951
Pres25	const.	total	marg.null	0.2119/0.000 [†]	0.1891/0.000 [†]	0.0339/0.986
Pres25	const.	total	joint.null	0.0938/0.990	0.0793/0.994	0.0455/0.770
Pres25	dong	early	real	0.0993/0.000 [†]	0.0993/0.000 [†]	0.0000/0.994
Pres25	dong	early	marg.null	0.0187/1.000	0.0188/1.000	0.0000/0.998

Table 15 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres25	dong	early	joint.null	0.1193/0.000 [†]	0.1181/0.000 [†]	0.0012/0.996
Pres25	dong	pooled	real	0.1418/0.000 [†]	0.1373/0.000 [†]	0.0446/0.000 [†]
Pres25	dong	pooled	marg.null	0.1319/0.000 [†]	0.1268/0.000 [†]	0.0059/0.000
Pres25	dong	pooled	joint.null	0.1086/0.000 [†]	0.1052/0.000 [†]	0.0033/0.000
Pres25	dong	sameday	real	0.0015/0.930	0.0015/0.987	0.0000/0.966
Pres25	dong	sameday	marg.null	0.0022/0.974	0.0023/0.972	0.0001/0.993
Pres25	dong	sameday	joint.null	0.0020/0.998	0.0020/0.982	0.0001/0.996

Table 16: Mebane (2023) multimodality diagnostics, short-chain configuration, conservative-coded leader. Full coverage, all 11 elections.

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen08	const.	total-no-early	real	0.0012/0.998	0.0009/0.996	0.0006/0.998
Gen08	const.	total-no-early	marg.null	0.0007/0.994	0.0009/1.000	0.0011/1.000
Gen08	const.	total-no-early	joint.null	0.0009/0.980	0.0007/0.991	0.0004/0.995
Gen08	dong	total-no-early	real	0.0007/0.979	0.0007/1.000	0.0000/0.993
Gen08	dong	total-no-early	marg.null	0.0016/0.990	0.0016/0.898	0.0000/0.998
Gen08	dong	total-no-early	joint.null	0.0022/0.991	0.0022/0.962	0.0000/0.996
Gen12	const.	total-no-early	real	0.0066/0.984	0.0060/0.989	0.0006/0.928
Gen12	const.	total-no-early	marg.null	0.0056/0.982	0.0053/0.980	0.0013/0.995
Gen12	const.	total-no-early	joint.null	0.0270/0.991	0.0264/0.995	0.0007/0.998
Gen12	dong	total-no-early	real	0.0053/0.969	0.0054/0.979	0.0000/0.995
Gen12	dong	total-no-early	marg.null	0.0047/0.993	0.0048/0.995	0.0001/1.000
Gen12	dong	total-no-early	joint.null	0.0014/0.983	0.0014/0.993	0.0000/1.000
Pres02	const.	total-no-early	real	0.0447/0.996	0.0450/1.000	0.0006/0.997
Pres02	const.	total-no-early	marg.null	0.0187/0.999	0.0184/0.999	0.0004/0.997
Pres02	const.	total-no-early	joint.null	0.0581/0.984	0.0586/0.994	0.0008/0.995
Pres02	dong	total-no-early	real	0.0076/0.999	0.0076/0.996	0.0000/0.994
Pres02	dong	total-no-early	marg.null	0.0182/0.999	0.0182/1.000	0.0001/0.609
Pres02	dong	total-no-early	joint.null	0.0106/0.991	0.0106/0.995	0.0000/1.000
Pres07	const.	total-no-early	real	0.0141/0.998	0.0137/0.991	0.0005/0.999
Pres07	const.	total-no-early	marg.null	0.0238/0.926	0.0239/0.725	0.0006/1.000
Pres07	const.	total-no-early	joint.null	0.0325/0.991	0.0327/0.922	0.0011/0.997
Pres07	dong	total-no-early	real	0.0096/0.992	0.0096/0.995	0.0000/0.999
Pres07	dong	total-no-early	marg.null	0.0084/1.000	0.0084/0.995	0.0000/1.000
Pres07	dong	total-no-early	joint.null	0.0078/0.995	0.0078/0.999	0.0001/0.995
Pres12	const.	total-no-early	real	0.1800/0.000 [†]	0.1800/0.000 [†]	0.0005/0.996
Pres12	const.	total-no-early	marg.null	0.0820/0.995	0.0782/0.998	0.0120/0.993

Table 16 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres12	const.	total-no-early	joint.null	0.1504/0.000 [†]	0.1508/0.000 [†]	0.0012/0.999
Pres12	dong	total-no-early	real	0.0590/0.000 [†]	0.0598/0.000 [†]	0.0011/0.989
Pres12	dong	total-no-early	marg.null	0.0108/0.989	0.0099/0.996	0.0012/0.000
Pres12	dong	total-no-early	joint.null	0.0433/0.291	0.0436/0.332	0.0003/0.996
Gen16	const.	early	real	0.0173/0.997	0.0175/0.998	0.0002/0.998
Gen16	const.	early	marg.null	0.0471/0.952	0.0477/0.993	0.0006/1.000
Gen16	const.	early	joint.null	0.0045/0.811	0.0045/0.996	0.0006/0.995
Gen16	const.	early-out	real	0.0008/0.929	0.0006/1.000	0.0004/1.000
Gen16	const.	early-out	marg.null	0.0019/0.996	0.0010/0.997	0.0015/0.979
Gen16	const.	early-out	joint.null	0.0013/0.976	0.0008/0.993	0.0011/0.995
Gen16	const.	early-total	real	0.0226/0.859	0.0225/0.000 [†]	0.0004/0.992
Gen16	const.	early-total	marg.null	0.0176/0.996	0.0175/1.000	0.0005/0.997
Gen16	const.	early-total	joint.null	0.0219/1.000	0.0217/0.976	0.0007/0.994
Gen16	const.	sameday	real	0.0007/0.995	0.0005/0.987	0.0002/0.993
Gen16	const.	sameday	marg.null	0.0011/0.920	0.0007/0.982	0.0008/0.993
Gen16	const.	sameday	joint.null	0.0008/0.998	0.0003/0.994	0.0006/0.991
Gen16	const.	total	real	0.0085/0.974	0.0082/0.692	0.0005/0.996
Gen16	const.	total	marg.null	0.0039/0.999	0.0038/1.000	0.0010/0.967
Gen16	const.	total	joint.null	0.0010/0.985	0.0005/0.997	0.0007/1.000
Gen16	dong	early	real	0.0074/0.000	0.0073/0.000	0.0001/0.999
Gen16	dong	early	marg.null	0.0246/0.000 [†]	0.0247/0.000 [†]	0.0001/0.997
Gen16	dong	early	joint.null	0.0026/0.987	0.0026/0.783	0.0000/0.997
Gen16	dong	pooled	real	0.0051/0.991	0.0050/0.995	0.0001/0.999
Gen16	dong	pooled	marg.null	0.0027/0.996	0.0027/0.995	0.0001/0.996
Gen16	dong	pooled	joint.null	0.0068/0.909	0.0069/0.488	0.0001/0.995
Gen16	dong	sameday	real	0.0004/0.996	0.0003/0.992	0.0001/1.000
Gen16	dong	sameday	marg.null	0.0020/0.026	0.0020/0.000	0.0001/0.998
Gen16	dong	sameday	joint.null	0.0010/1.000	0.0010/0.994	0.0000/1.000
Pres17	const.	early	real	0.0203/0.804	0.0208/0.998	0.0010/0.996
Pres17	const.	early	marg.null	0.0445/0.188	0.0447/0.000 [†]	0.0009/0.991
Pres17	const.	early	joint.null	0.0071/0.991	0.0076/0.972	0.0007/0.997
Pres17	const.	early-out	real	0.0006/0.993	0.0008/0.991	0.0007/0.996
Pres17	const.	early-out	marg.null	0.0038/0.993	0.0042/0.991	0.0006/0.996
Pres17	const.	early-out	joint.null	0.0120/0.987	0.0124/0.997	0.0009/0.995
Pres17	const.	early-total	real	0.0013/0.991	0.0006/0.997	0.0007/1.000
Pres17	const.	early-total	marg.null	0.0042/0.983	0.0043/0.998	0.0004/0.972
Pres17	const.	early-total	joint.null	0.0123/0.996	0.0126/0.995	0.0008/0.994
Pres17	const.	sameday	real	0.0013/0.983	0.0012/0.998	0.0002/0.982
Pres17	const.	sameday	marg.null	0.0202/1.000	0.0201/0.995	0.0007/1.000
Pres17	const.	sameday	joint.null	0.0009/0.991	0.0005/0.994	0.0007/0.996
Pres17	const.	total	real	0.0508/0.992	0.0507/0.997	0.0010/1.000
Pres17	const.	total	marg.null	0.1629/0.000 [†]	0.1621/0.000 [†]	0.0010/0.998
Pres17	const.	total	joint.null	0.1157/0.000 [†]	0.1157/0.000 [†]	0.0003/0.999
Pres17	dong	early	real	0.0071/0.661	0.0069/0.003	0.0003/0.996
Pres17	dong	early	marg.null	0.0138/0.000 [†]	0.0138/0.000 [†]	0.0000/0.998

Table 16 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres17	dong	early	joint.null	0.0057/0.000	0.0057/0.000	0.0000/0.988
Pres17	dong	pooled	real	0.1363/0.000 [†]	0.1363/0.000 [†]	0.0001/0.996
Pres17	dong	pooled	marg.null	0.0646/0.000 [†]	0.0647/0.000 [†]	0.0000/0.991
Pres17	dong	pooled	joint.null	0.1032/0.000 [†]	0.1033/0.000 [†]	0.0001/0.997
Pres17	dong	sameday	real	0.0502/0.000 [†]	0.0502/0.000 [†]	0.0001/0.997
Pres17	dong	sameday	marg.null	0.0463/0.000 [†]	0.0463/0.000 [†]	0.0000/0.997
Pres17	dong	sameday	joint.null	0.0234/0.000 [†]	0.0234/0.000 [†]	0.0000/0.998
Gen20	const.	early	real	0.0317/0.996	0.0319/1.000	0.0009/0.992
Gen20	const.	early	marg.null	0.0194/0.996	0.0198/0.968	0.0004/0.963
Gen20	const.	early	joint.null	0.0138/0.994	0.0133/0.996	0.0008/0.984
Gen20	const.	early-out	real	0.0042/0.996	0.0038/0.999	0.0005/0.978
Gen20	const.	early-out	marg.null	0.0003/0.993	0.0002/1.000	0.0002/0.999
Gen20	const.	early-out	joint.null	0.0087/0.990	0.0086/0.999	0.0004/1.000
Gen20	const.	early-total	real	0.0034/0.999	0.0037/0.991	0.0008/1.000
Gen20	const.	early-total	marg.null	0.0076/0.448	0.0072/0.999	0.0006/0.999
Gen20	const.	early-total	joint.null	0.0118/0.998	0.0122/0.067	0.0005/0.985
Gen20	const.	sameday	real	0.0005/0.999	0.0008/0.996	0.0005/0.998
Gen20	const.	sameday	marg.null	0.0003/0.999	0.0003/0.988	0.0005/0.938
Gen20	const.	sameday	joint.null	0.0005/0.992	0.0003/0.998	0.0005/0.991
Gen20	const.	total	real	0.0430/0.997	0.0429/0.992	0.0003/0.984
Gen20	const.	total	marg.null	0.0393/1.000	0.0385/0.966	0.0007/0.990
Gen20	const.	total	joint.null	0.0196/0.994	0.0193/1.000	0.0008/0.924
Gen20	dong	early	real	0.0296/0.000 [†]	0.0296/0.000 [†]	0.0000/0.982
Gen20	dong	early	marg.null	0.0338/0.000 [†]	0.0338/0.000 [†]	0.0001/0.991
Gen20	dong	early	joint.null	0.0197/0.000 [†]	0.0197/0.000 [†]	0.0001/0.995
Gen20	dong	pooled	real	0.0016/0.679	0.0015/0.453	0.0001/0.996
Gen20	dong	pooled	marg.null	0.0156/0.995	0.0156/0.991	0.0000/1.000
Gen20	dong	pooled	joint.null	0.0054/0.997	0.0054/0.997	0.0000/1.000
Gen20	dong	sameday	real	0.0012/0.990	0.0012/0.568	0.0000/1.000
Gen20	dong	sameday	marg.null	0.0012/0.990	0.0012/0.856	0.0000/0.998
Gen20	dong	sameday	joint.null	0.0012/0.973	0.0012/0.997	0.0000/1.000
Pres22	const.	early	real	0.0181/0.889	0.0187/0.618	0.0010/0.993
Pres22	const.	early	marg.null	0.0366/0.990	0.0372/0.995	0.0009/0.993
Pres22	const.	early	joint.null	0.0044/0.980	0.0047/1.000	0.0006/0.998
Pres22	const.	early-out	real	0.0282/0.995	0.0278/0.960	0.0006/0.996
Pres22	const.	early-out	marg.null	0.0278/0.323	0.0276/0.008 [†]	0.0008/0.985
Pres22	const.	early-out	joint.null	0.0596/0.000 [†]	0.0599/0.000 [†]	0.0003/0.994
Pres22	const.	early-total	real	0.0129/0.995	0.0125/0.972	0.0006/1.000
Pres22	const.	early-total	marg.null	0.0125/0.992	0.0118/0.998	0.0009/1.000
Pres22	const.	early-total	joint.null	0.0044/0.998	0.0040/0.998	0.0004/0.997
Pres22	const.	sameday	real	0.0079/1.000	0.0086/0.997	0.0007/0.996
Pres22	const.	sameday	marg.null	0.0086/0.905	0.0083/0.999	0.0005/0.997
Pres22	const.	sameday	joint.null	0.0010/0.990	0.0007/0.993	0.0007/0.987
Pres22	const.	total	real	0.2202/0.000 [†]	0.2166/0.000 [†]	0.0078/0.612
Pres22	const.	total	marg.null	0.1885/0.000 [†]	0.1900/0.000 [†]	0.0078/0.955
Pres22	const.	total	joint.null	0.0676/1.000	0.0838/0.994	0.0162/0.818
Pres22	dong	early	real	0.0025/0.997	0.0025/0.998	0.0000/0.999
Pres22	dong	early	marg.null	0.0032/0.925	0.0033/0.994	0.0001/1.000

Table 16 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres22	dong	early	joint.null	0.0021/0.991	0.0021/0.903	0.0000/1.000
Pres22	dong	pooled	real	0.0126/0.992	0.0129/0.997	0.0003/0.992
Pres22	dong	pooled	marg.null	0.0147/0.782	0.0147/0.934	0.0003/0.973
Pres22	dong	pooled	joint.null	0.0858/0.000 [†]	0.0855/0.000 [†]	0.0002/0.992
Pres22	dong	sameday	real	0.0003/0.995	0.0003/0.995	0.0001/0.995
Pres22	dong	sameday	marg.null	0.0020/0.595	0.0020/0.945	0.0001/1.000
Pres22	dong	sameday	joint.null	0.0031/0.974	0.0032/0.984	0.0001/0.993
Gen24	const.	early	real	0.0047/0.993	0.0041/0.982	0.0006/0.987
Gen24	const.	early	marg.null	0.0497/0.998	0.0491/0.994	0.0006/0.996
Gen24	const.	early	joint.null	0.0125/0.993	0.0122/0.997	0.0004/0.997
Gen24	const.	early-out	real	0.0294/0.995	0.0287/1.000	0.0009/0.997
Gen24	const.	early-out	marg.null	0.0161/0.994	0.0159/0.998	0.0003/0.993
Gen24	const.	early-out	joint.null	0.0285/0.998	0.0284/0.995	0.0004/1.000
Gen24	const.	early-total	real	0.0094/0.999	0.0095/1.000	0.0004/0.999
Gen24	const.	early-total	marg.null	0.0040/0.985	0.0037/0.997	0.0005/0.997
Gen24	const.	early-total	joint.null	0.0044/0.976	0.0047/1.000	0.0009/0.994
Gen24	const.	sameday	real	0.0214/0.994	0.0212/0.997	0.0011/1.000
Gen24	const.	sameday	marg.null	0.0046/1.000	0.0043/0.997	0.0009/0.997
Gen24	const.	sameday	joint.null	0.0006/1.000	0.0004/0.992	0.0004/0.998
Gen24	const.	total	real	0.0726/0.997	0.0720/0.995	0.0008/0.994
Gen24	const.	total	marg.null	0.0214/0.997	0.0213/0.996	0.0002/0.992
Gen24	const.	total	joint.null	0.0787/0.996	0.0796/0.996	0.0011/0.999
Gen24	dong	early	real	0.0014/0.993	0.0014/0.984	0.0000/0.990
Gen24	dong	early	marg.null	0.0338/0.000 [†]	0.0339/0.000 [†]	0.0001/0.923
Gen24	dong	early	joint.null	0.0448/0.000 [†]	0.0448/0.000 [†]	0.0000/0.995
Gen24	dong	pooled	real	0.0077/0.988	0.0078/0.998	0.0001/1.000
Gen24	dong	pooled	marg.null	0.0050/0.954	0.0050/0.842	0.0000/0.944
Gen24	dong	pooled	joint.null	0.0087/0.997	0.0087/0.993	0.0000/1.000
Gen24	dong	sameday	real	0.0010/0.867	0.0010/0.901	0.0001/0.951
Gen24	dong	sameday	marg.null	0.0019/0.997	0.0019/0.932	0.0000/0.995
Gen24	dong	sameday	joint.null	0.0018/0.997	0.0017/0.997	0.0001/0.998
Pres25	const.	early	real	0.0123/0.992	0.0122/0.988	0.0004/1.000
Pres25	const.	early	marg.null	0.0305/0.943	0.0315/0.773	0.0010/0.993
Pres25	const.	early	joint.null	0.0079/0.974	0.0083/0.996	0.0011/0.981
Pres25	const.	early-out	real	0.0109/0.974	0.0113/0.576	0.0009/0.993
Pres25	const.	early-out	marg.null	0.0123/0.982	0.0118/0.997	0.0005/0.990
Pres25	const.	early-out	joint.null	0.0045/0.922	0.0042/0.982	0.0005/0.995
Pres25	const.	early-total	real	0.0005/0.993	0.0003/0.988	0.0006/0.961
Pres25	const.	early-total	marg.null	0.0240/0.993	0.0242/0.983	0.0004/0.992
Pres25	const.	early-total	joint.null	0.0006/1.000	0.0006/0.994	0.0004/0.997
Pres25	const.	sameday	real	0.0076/0.999	0.0080/0.997	0.0006/0.997
Pres25	const.	sameday	marg.null	0.0172/0.995	0.0167/0.995	0.0005/0.952
Pres25	const.	sameday	joint.null	0.0087/0.977	0.0085/0.871	0.0006/0.993
Pres25	const.	total	real	0.2056/0.000 [†]	0.2054/0.000 [†]	0.0002/0.997
Pres25	const.	total	marg.null	0.2361/0.000 [†]	0.2266/0.000 [†]	0.0095/0.993
Pres25	const.	total	joint.null	0.1022/0.994	0.1021/0.992	0.0007/1.000
Pres25	dong	early	real	0.0017/1.000	0.0017/0.954	0.0000/0.998
Pres25	dong	early	marg.null	0.0062/0.094	0.0062/0.006	0.0000/0.780
Pres25	dong	early	joint.null	0.0012/0.994	0.0012/0.994	0.0001/0.994

Table 16 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres25	dong	pooled	real	0.1404/0.000 [†]	0.1404/0.000 [†]	0.0001/0.969
Pres25	dong	pooled	marg.null	0.0965/0.000 [†]	0.0965/0.000 [†]	0.0003/1.000
Pres25	dong	pooled	joint.null	0.0043/0.992	0.0043/0.991	0.0000/0.983
Pres25	dong	sameday	real	0.0043/0.960	0.0043/0.990	0.0000/0.995
Pres25	dong	sameday	marg.null	0.0033/1.000	0.0033/1.000	0.0000/0.993
Pres25	dong	sameday	joint.null	0.0041/0.992	0.0041/0.991	0.0001/0.993

A.3 Model-consistent self-null: complete results

Table 17 lists all 57 completed cells of the model-consistent self-null grid discussed in Section 4.3: every (election, level, channel) cell in the short-chain, Democratic-coded configuration, with population mixture weights fixed at $\pi = (1, 0, 0)$ by construction using the fitted qbl model’s own class-1 hyperparameters. One cell, pres21/dong/early, is omitted because its second-stage fit crashed on an invalid simulated draw (a unit for which the simulated random effects implied more votes than eligible voters, a low-probability edge case of the Gaussian random-effect simulation rather than a systematic issue; every other cell in the grid completed without incident). π_1 is the model’s own posterior no-fraud probability on data drawn entirely from its class-1 process; under an unbiased test this would recover values near 100% throughout, which it manifestly does not.

Table 17: Model-consistent self-null results: π forced to $(1, 0, 0)$ using the qbl model’s own fitted class-1 (no-fraud) data-generating equations, short-chain configuration, Democratic-coded leader. N is the unit count; “Flagged” is the number of units receiving a non-clean MAP classification. One cell (pres21/dong/early) failed with an invalid simulated draw and is omitted (Appendix A.3).

Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
Gen08	const.	total-no-early	198	99.14	0.71	0/198	1.01
Gen08	dong	total-no-early	2733	99.61	0.38	2/2733	1.26*
Gen12	const.	total-no-early	209	98.94	0.90	0/209	1.12*
Gen12	dong	total-no-early	2923	99.34	0.65	1/2923	1.11*
Pres02	const.	total-no-early	244	84.96	14.87	15/244	2.01*
Pres02	dong	total-no-early	3515	84.48	15.51	229/3515	1.21*
Pres07	const.	total-no-early	248	98.16	1.66	2/248	2.21*
Pres07	dong	total-no-early	3576	97.87	2.12	22/3576	1.27*
Pres12	const.	total-no-early	251	86.28	13.60	9/251	1.46*
Pres12	dong	total-no-early	3479	85.14	14.74	153/3479	9.41*
Gen16	const.	early	234	97.00	2.86	0/234	1.80*
Gen16	const.	early-out	234	98.66	1.22	0/234	1.08*
Gen16	const.	early-total	234	97.81	2.04	0/234	1.51*
Gen16	const.	sameday	234	99.28	0.58	0/234	1.03
Gen16	const.	total	234	99.00	0.86	0/234	1.48*

Table 17 – continued

Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
Gen16	dong	early	3135	99.40	0.58	0/3135	3.44*
Gen16	dong	pooled	3135	99.76	0.23	0/3135	1.66*
Gen16	dong	sameday	3135	99.69	0.31	0/3135	4.97*
Pres17	const.	early	250	97.19	2.71	1/250	1.51*
Pres17	const.	early-out	250	97.31	2.59	0/250	1.74*
Pres17	const.	early-total	250	98.77	1.04	0/250	2.66*
Pres17	const.	sameday	250	99.32	0.54	0/250	1.02
Pres17	const.	total	250	82.07	17.78	5/250	1.26*
Pres17	dong	early	3491	99.42	0.57	2/3491	12.12*
Pres17	dong	pooled	3491	89.62	10.37	99/3491	1.52*
Pres17	dong	sameday	3491	99.92	0.07	0/3491	1.01
Gen20	const.	early	253	95.67	4.20	0/253	1.45*
Gen20	const.	early-out	253	97.32	2.55	0/253	1.47*
Gen20	const.	early-total	253	96.02	3.84	2/253	1.69*
Gen20	const.	sameday	253	99.36	0.52	0/253	1.01
Gen20	const.	total	253	90.10	9.79	2/253	1.80*
Gen20	dong	early	3480	98.65	1.34	1/3480	1.70*
Gen20	dong	pooled	3480	95.96	4.03	31/3480	1.27*
Gen20	dong	sameday	3480	99.94	0.05	0/3480	1.20*
Pres22	const.	early	250	97.79	2.08	0/250	1.46*
Pres22	const.	early-out	250	95.90	4.01	1/250	1.29*
Pres22	const.	early-total	250	98.51	1.36	0/250	1.21*
Pres22	const.	sameday	250	98.98	0.92	0/250	1.26*
Pres22	const.	total	248	86.97	13.10	6/248	3.04*
Pres22	dong	early	3509	99.32	0.67	1/3509	1.06*
Pres22	dong	pooled	3509	83.04	16.95	230/3509	1.50*
Pres22	dong	sameday	3509	99.67	0.32	0/3509	1.49*
Gen24	const.	early	246	95.29	4.59	0/246	1.84*
Gen24	const.	early-out	246	95.66	4.21	1/246	1.47*
Gen24	const.	early-total	246	98.41	1.45	0/246	1.06*
Gen24	const.	sameday	246	99.30	0.57	0/246	1.02
Gen24	const.	total	246	87.99	11.87	10/246	1.20*
Gen24	dong	early	2739	98.19	1.80	0/2739	1.92*
Gen24	dong	pooled	2739	93.81	6.19	46/2739	1.95*
Gen24	dong	sameday	2739	99.76	0.23	0/2739	1.39*
Pres25	const.	early	253	94.95	4.92	1/253	1.56*
Pres25	const.	early-out	253	95.08	4.79	0/253	2.29*
Pres25	const.	early-total	253	97.98	1.89	0/253	1.13*
Pres25	const.	sameday	253	99.00	0.85	0/253	1.69*
Pres25	const.	total	242	75.13	24.45	15/242	4.60*
Pres25	dong	pooled	3553	80.22	19.77	302/3553	1.67*
Pres25	dong	sameday	3553	99.65	0.34	0/3553	1.39*

B Code and data-processing documentation

This appendix documents the analysis workflow referenced throughout the paper: what each script does, the software it requires, and the commands needed to reproduce a model run from a raw election-result CSV through to a posterior summary. The full codebase is not reproduced inline because, in total, it comprises several thousand lines of R and Python; the complete, version-controlled source is provided as part of the submission.

B.1 Workflow overview

The analysis workflow has two layers. A Python layer (i) loads and cleans a raw National Election Commission result file into the unit-level (w, a, N, prov) table required by `eforensics` (leader votes, abstentions, eligible voters, and a province grouping code; Section 3), (ii) optionally replaces that table with one of the three fraud-free null constructions of Section 3.3, and (iii) invokes an R layer that calls the `eforensics` package’s `eforensics()` function via JAGS and writes the posterior summary back out as CSV. Orchestration across the full grid of elections, levels, channels, sources, and leader codings is handled by shell scripts that launch many single-cell Python invocations concurrently and log each cell’s output separately, so that a long-running batch can be interrupted and resumed without repeating completed cells.

B.2 Software requirements

R (≥ 4.0 recommended), with the packages `eforensics` [Ferrari et al., 2019] (installed from Walter Mebane’s own fork, `UMeforensics/eforensics_public` on GitHub, rather than any CRAN mirror, to guarantee the exact model implementation referenced throughout this paper), `coda` (posterior summary and Gelman–Rubin diagnostics), and, optionally, `dipTest` (Hartigan dip test for the Mebane multimodality diagnostic of Section 4.6; runs degrade gracefully with the corresponding diagnostic reported as unavailable if this package is absent). **JAGS** ≥ 4.3 must be installed separately and on the system path, since `eforensics` calls it as an external MCMC sampler rather than bundling it. **Python** ≥ 3.8 , with `numpy` and `pandas` for the data-preparation and null-generation layer, and, for the post-processing scripts that build this paper’s tables and figures, `matplotlib`. No Python package is required to have any dependency on R; the two communicate only through CSV files written to a temporary directory and an `Rscript` subprocess call.

B.3 Script reference

mebane_crosscheck.R. Fits the `qbl` model to a single election/level/channel/source cell. Takes an input CSV with columns `leader_votes`, `total_votes`, `eligible`, `prov_code`; writes a one-row summary CSV (mixture weights, credible intervals, $\hat{R}$, the Mebane multimodality diagnostics, and, where the installed package exposes it, total fraudulent-vote estimates F_t/F_w) and a per-unit MAP-classification CSV. Usage: `Rscript mebane_crosscheck.R <input.csv> <output.csv> [n_iter] [n_chains] [burn_in] [use_parcomp] [n_adapt]`.

mebane_crosscheck_selfnull_fit.R. Identical to `mebane_crosscheck.R`, but additionally monitors and writes out the posterior-median class-1 (no-fraud) hyperparameters needed to construct the model-consistent self-null of Section 4.3: the raw province design-matrix coefficients on the turnout and vote-share equations, the corresponding random-intercept means, and the random-intercept variances. Writes two additional output files, `*_hyperparams.csv` and `*_provlevels.csv` (the latter records the province factor-level ordering needed to map the coefficients back onto the original province codes). Usage: `Rscript mebane_crosscheck_selfnull_fit.R <input.csv> <output.csv> [n_iter] [n_chains] [burn_in] [use_parcomp]`.

mebane_r_pipeline_bidirectional.py. The main Python data layer. Contains the per-election configuration (candidate name patterns, result-file names, early-voting status), the Korean-language CSV parser that builds dong- and constituency-level unit tables from a raw result file, the marginal- and joint-null generators, and the driver function that writes a unit table to a temporary CSV, calls `mebane_crosscheck.R` via `Rscript`, and reads the result back in. `leader_side` (`'dem'` or `'con'`) controls which party or candidate is coded as the model’s leader (Section 2). Usage: `python3 mebane_r_pipeline_bidirectional.py <elec_id> <level> <channel> <source> [leader_side] [r_n_iter] [r_n_chains] [r_`

burn_in] [use_parcomp] [r_n_adapt], where source is one of real, marginal_null, joint_null.

mebane_self_null_pipeline.py. Implements the two-stage self-null procedure of Section 3.3: it calls mebane_crosscheck_selfnull_fit.R on the real data to harvest class-1 hyperparameters, simulates a synthetic dataset from those hyperparameters with every unit forced into the no-fraud class, and refits mebane_crosscheck.R to that synthetic dataset. Depends on mebane_r_pipeline_bidirectional.py for data loading and on mebane_crosscheck_selfnull_fit.R/mebane_crosscheck.R for the two model fits. Usage: `python3 mebane_self_null_pipeline.py <elec_id> <level> <channel> <leader_side> [r_n_iter] [r_n_chains] [r_burn_in] [use_parcomp] [seed]`.

run_mebane_r_all_parallel_bidirectional.sh and **run_mebane_self_null_parallel.sh.** Bash drivers that loop over every (election, level, channel, source, leader_side) combination in the study grid, launching mebane_r_pipeline_bidirectional.py or mebane_self_null_pipeline.py respectively as concurrent background subprocesses (bounded by a configurable MAX_PARALLEL), and writing each cell’s console output to its own log file plus a short summary snippet. A cell with an existing non-empty log is skipped unless FORCE_RERUN=1 is set, making both scripts safe to interrupt and resume. Environment variables documented in each script’s header (R_N_ITER, R_N_CHAINS, R_BURN_IN, LEADER_SIDES, MAX_PARALLEL, SEED_BASE, MEBANE_PROVINCE_GROUPING, among others) control the MCMC settings and scope of a given run without editing the script itself.

make_appendix_table.py, **make_appendix_table_components.py,** **make_appendix_table_mebane_md.py,** and **make_appendix_table_selfnull.py.** Each reads an aggregated results CSV (appendix_full_results.csv, appendix_components_results.csv, appendix_mebane_md_results.csv, or appendix_selfnull_results.csv, one row per completed cell) and writes the corresponding LaTeX longtable included in Appendix A.

make_figures.py. Reads the same aggregated results CSVs and generates every figure in the main text (Figures 2–4) as PDF files.

B.4 Reproducing a single cell end to end

1. Install R, JAGS, and the R packages listed above; install eforensics from UMeForensics/eforensics_public rather than CRAN.
2. Download the relevant election-result CSV file(s) from the data.go.kr pages listed under Data availability, and place them in the working directory under the filenames referenced in the per-election configuration (e.g. 21st_election_result.csv for the 2020 general election).
3. Run, e.g., `python3 mebane_r_pipeline_bidirectional.py 21 dong early real dem 1000 4 300 FALSE 1000` to fit the real-data cell for the 21st general election, dong level, in-precinct early-voting channel, Democratic-coded leader, at the short-chain configuration’s settings; substitute marginal_null or joint_null for real to fit the corresponding null instead, or use mebane_self_null_pipeline.py for the model-consistent self-null.
4. For the complete grid used in this paper, run `./run_mebane_r_all_parallel_bidirectional.sh` (real/marginal-null/joint-null grid) and, separately, `./run_mebane_self_null_parallel.sh` (self-null grid); both accept an optional list of election IDs and a level as positional arguments to restrict scope, and MCMC settings, concurrency, and leader-coding scope via the environment variables documented in their headers.

5. Aggregate the resulting per-cell summary CSVs into the four `appendix_*_results.csv` files (one row per completed cell, with the columns referenced throughout Appendix A), then run the `make_appendix_table_*.py` and `make_figures.py` scripts described above from a directory containing those aggregated CSVs to regenerate the tables and figures in this paper.

References

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